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Gupta

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(54) **PORT MANAGEMENT FOR RELIABLE DATA SERVICE (RDS) PROTOCOL**

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PCT Pub. Date: **Jul. 23, 2020**

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H04W 76/10 (2018.01)
H04W 76/12 (2018.01)

(52) **U.S. Cl.**
CPC ***H04W 76/12*** (2018.02)

(58) **Field of Classification Search**
None
See application file for complete search history.

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WO WO-2018144249 A1 * 8/2018 .... H04W 16/14

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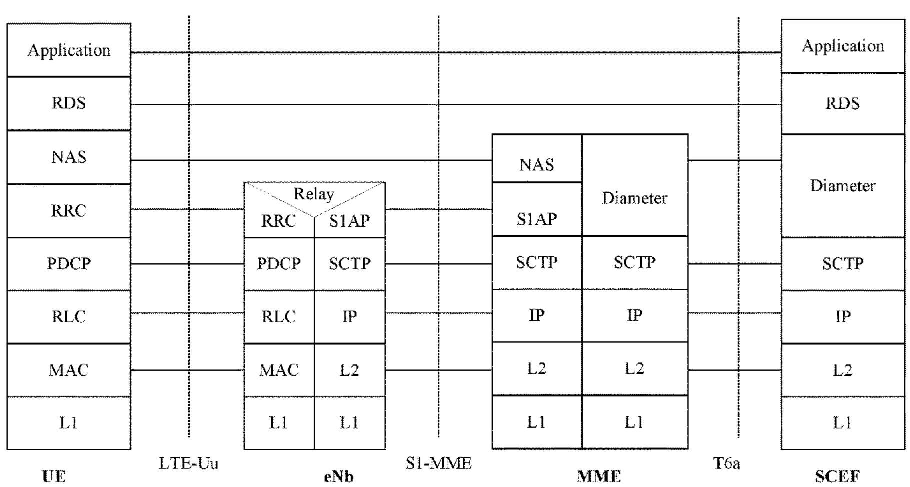
(74) *Attorney, Agent, or Firm* — Dorsey & Whitney LLP

(57) **ABSTRACT**

Systems, apparatuses, methods, and computer-readable media are provided for use in a wireless network for dynamic port discovery. Some embodiments are directed to an originator device that includes processor circuitry and radio front circuitry. The processor circuitry can be configured to generate a command frame for dynamic port discovery for communicating with a receiver device using a Reliable Data Service (RDS) protocol. The processor circuitry can be further configured to transmit, using the radio front end circuitry, the command frame to the receiver device and receive, using the radio front end circuitry, a response frame from the receiver device. The processor circuitry can be further configured to assign, using the received response frame, a source port number and a destination port number to an application identifier. The application identifier can be associated with an application on the originator device communicating with a corresponding application on the receiver device.

14 Claims, 23 Drawing Sheets



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3GPP TS 24.250 V15.1.0, 3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; Protocol for Reliable Data Service between UE and SCEF; Stage 3 (Release 15), Dec. 2018; 22 pages.
International Search Report and Written Opinion directed to related International Application No. PCT/US2020/013504, issued on Apr. 14, 2020, 11 pages.

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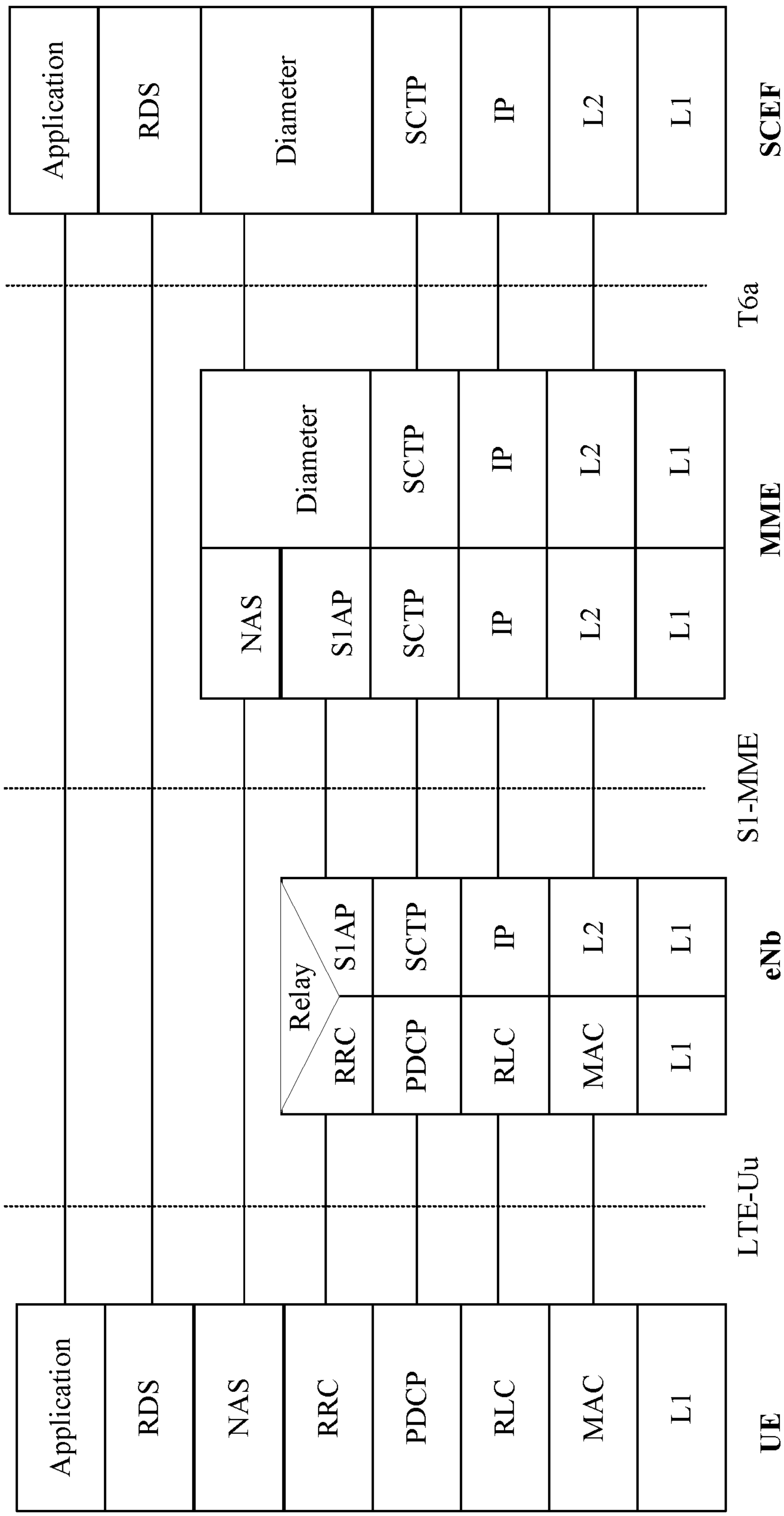

Application
RDS
NAS
RRC
PDCP
RLC
MAC
L1
UE
LTE-Uu
Relay
RRC
PDCP
RLC
MAC
L1
S1AP
SCTP
IP
L2
L1
eNb
S1-MME
NAS
S1AP
SCTP
IP
L2
L1
Diameter
SCTP
IP
L2
L1
MME
T6a
Application
RDS
Diameter
SCTP
IP
L2
L1
SCEF

Figure 1

Bits								
8	7	6	5	4	3	2	1	Octet
Type								1
Length								2
High-order octet								3
…								…
Low-order octet								n

Figure 2

8	7	6	5	4	3	2	1	
Port 7	Port 6	Port 5	Port 4	Port 3	Port 2	Port 1	Port 0	octet 1
Port 15	Port 14	Port 13	Port 12	Port 11	Port 10	Port 9	Port 8	octet 2

Figure 3

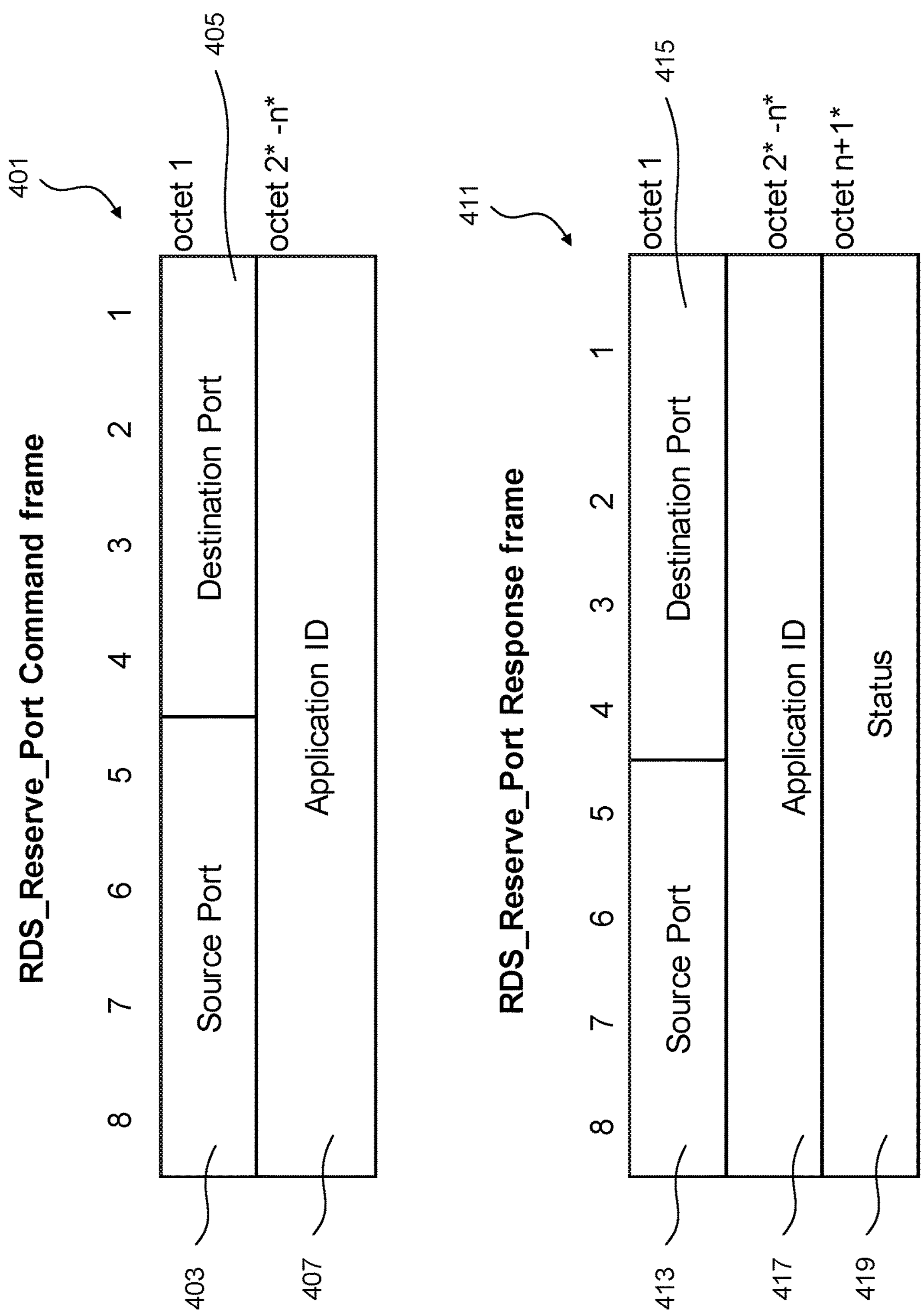

RDS_Reserve_Port Command frame
401
1
2
3
4
5
6
7
8
octet 1
octet 2* -n*
Destination Port
Source Port
Application ID
405
403
407
RDS_Reserve_Port Response frame
411
octet 1
octet 2* -n*
octet n+1*
Destination Port
Source Port
Application ID
Status
415
413
417
419

Figure 4

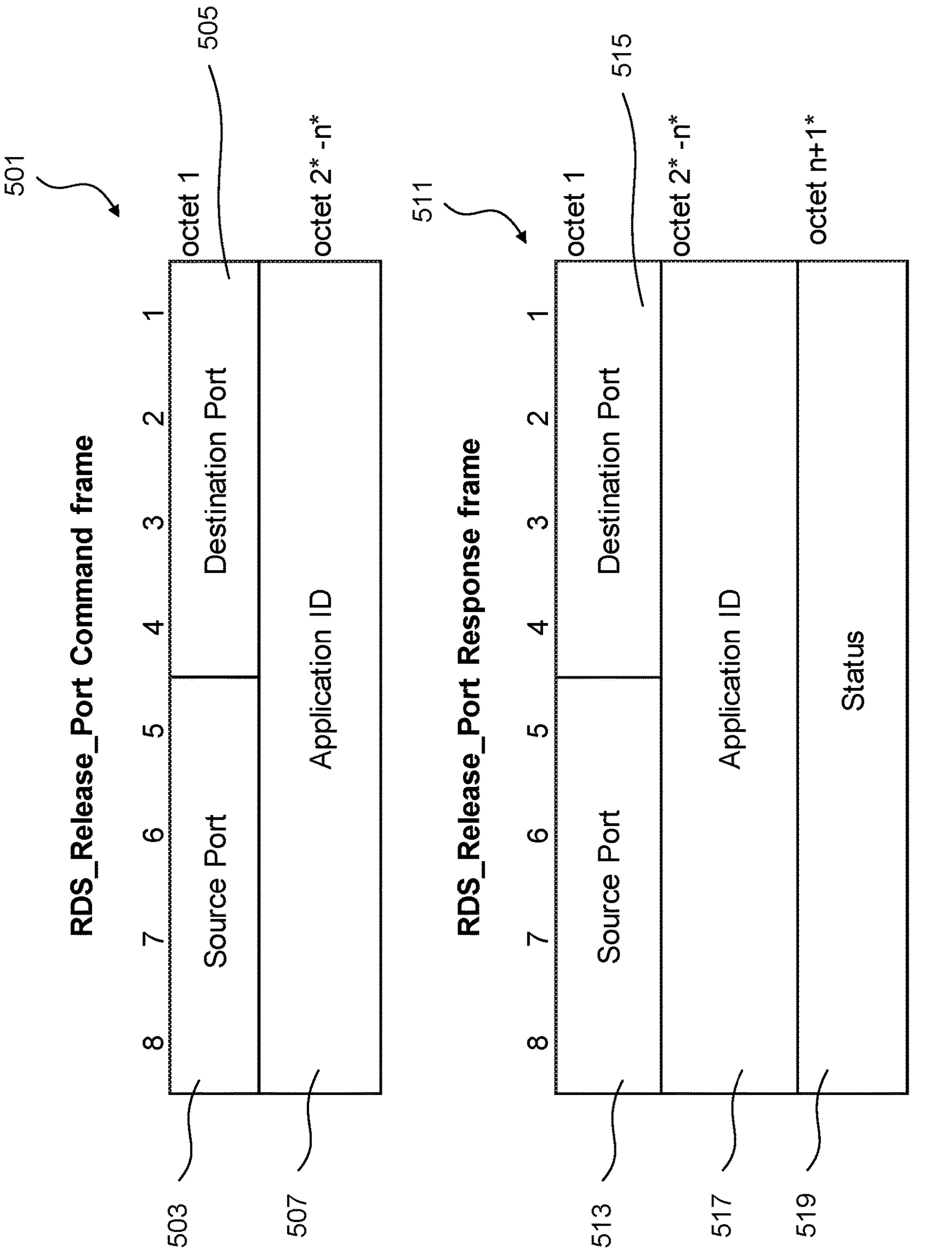

RDS_Release_Port Command frame
501
8 7 6 5 4 3 2 1
Source Port
Destination Port
octet 1
Application ID
octet 2* -n*
503
505
507
RDS_Release_Port Response frame
511
8 7 6 5 4 3 2 1
Source Port
Destination Port
octet 1
Application ID
octet 2* -n*
Status
octet n+1*
513
515
517
519

Figure 5

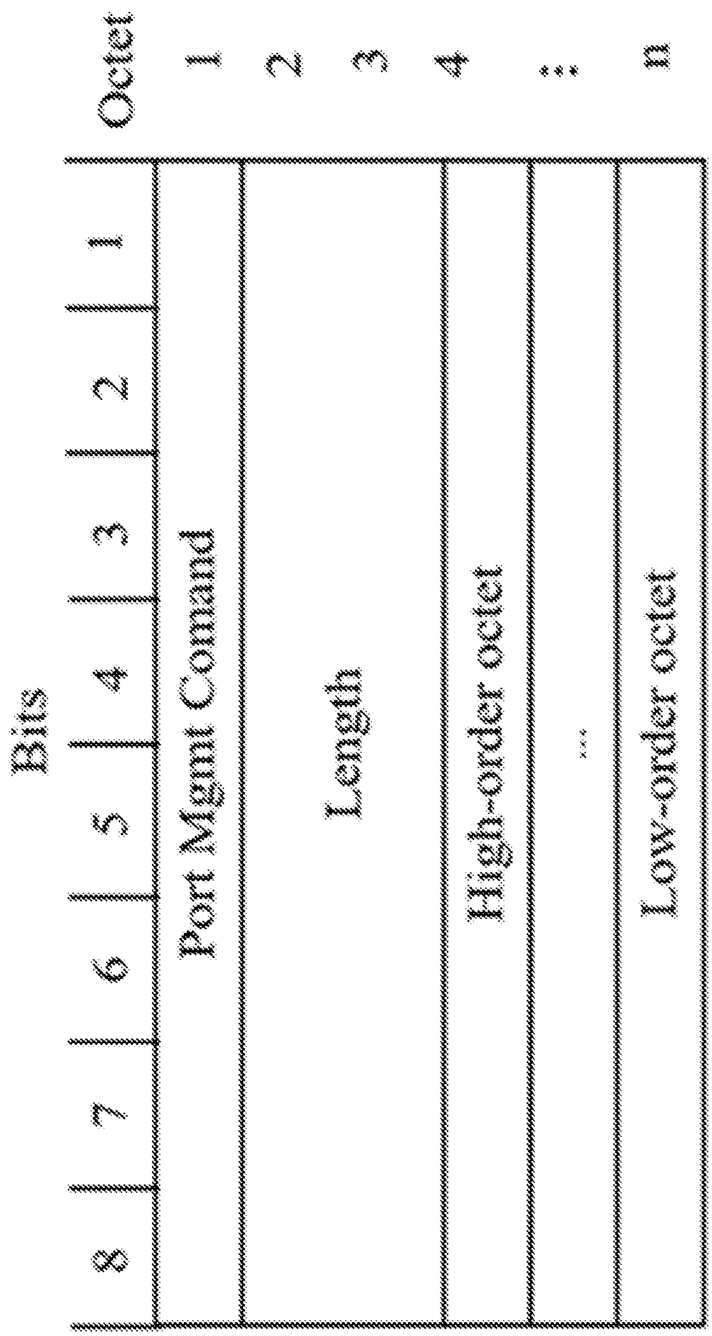

Bits
8
7
6
5
4
3
2
1
Octet
Port Mgmt Comand
1
Length
2
3
High-order octet
4
...
...
Low-order octet
n

Figure 6

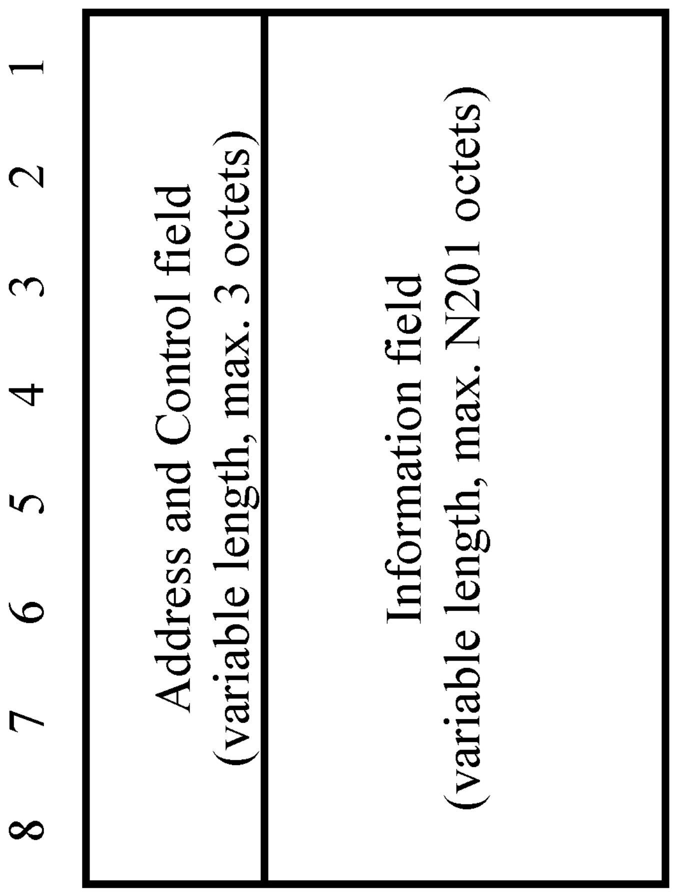

8 7 6 5 4 3 2 1
Address and Control field
(variable length, max. 3 octets)
Information field
(variable length, max. N201 octets)

Figure 7

Address and Control Field Bits

Format	8	7	6	5	4	3	2	1	Octet
I Format	PD	0	A	X	ADS	N(S)			1
	N(R)			R1	R2	R3	S1	S2	2
	Source Port				Destination Port				3
S Format	PD	1	1	0	ADS	A	X	X	1
	N(R)			R1	R2	R3	S1	S2	2
	Source Port				Destination Port				3
UI Format	PD	1	0	X	ADS	N(U)			1
	Source Port				Destination Port				2
U Format	PD	1	1	1	ADS	CR	X	X	1
	X	X	X	X	M4	M3	M2	M1	2
	Source Port				Destination Port				3

Figure 8

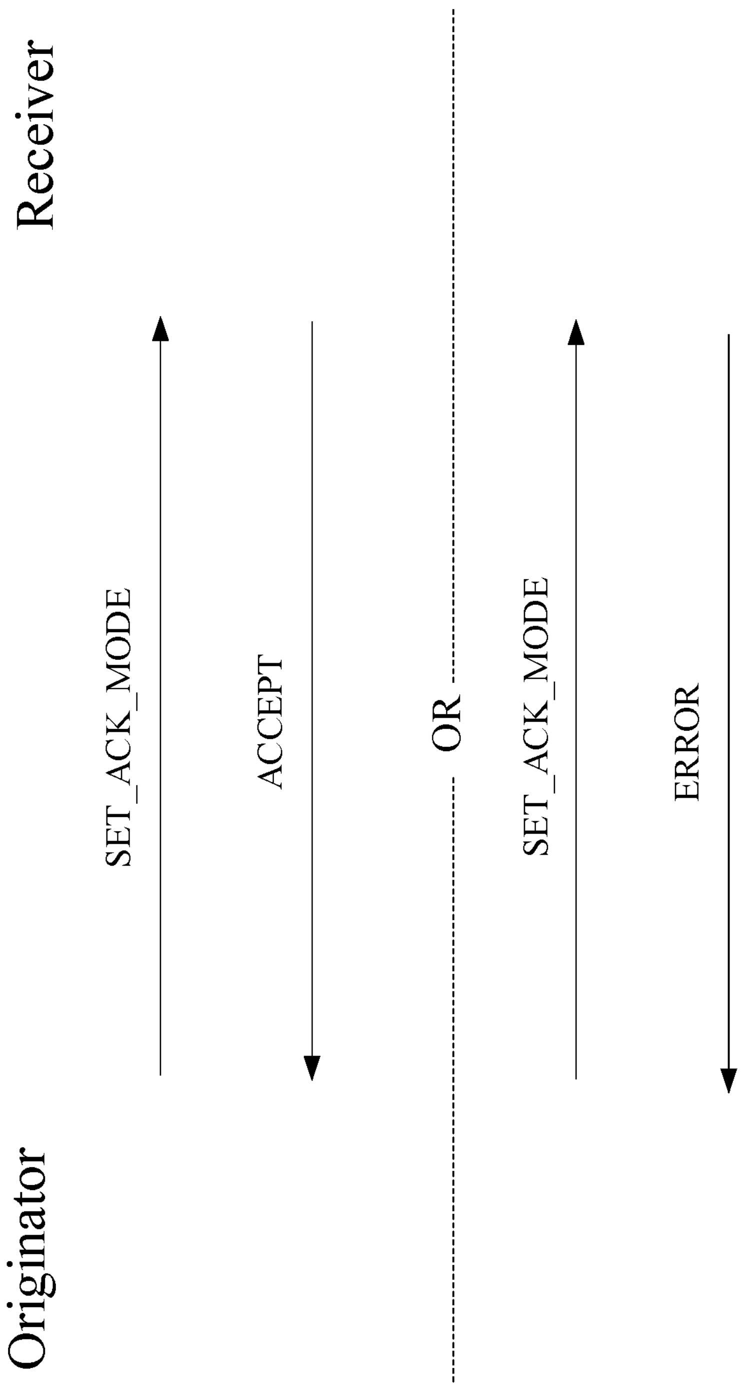

Receiver
Originator
SET_ACK_MODE
ACCEPT
OR
SET_ACK_MODE
ERROR

Figure 9

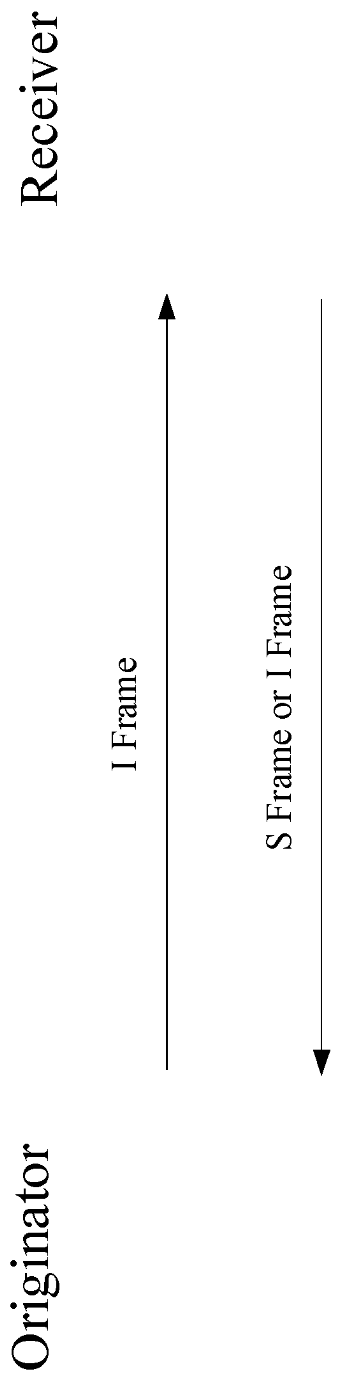


Figure 10

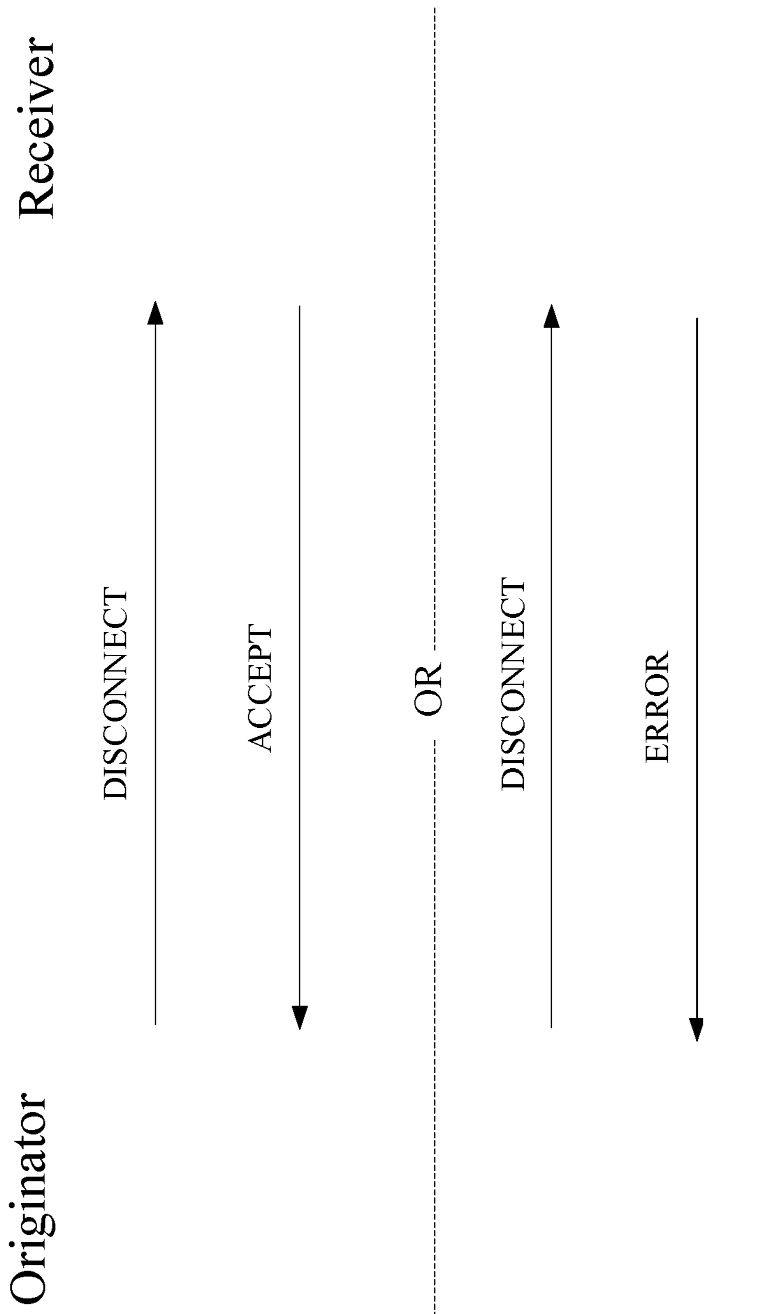

Receiver
Originator
DISCONNECT
ACCEPT
OR
DISCONNECT
ERROR

Figure 11

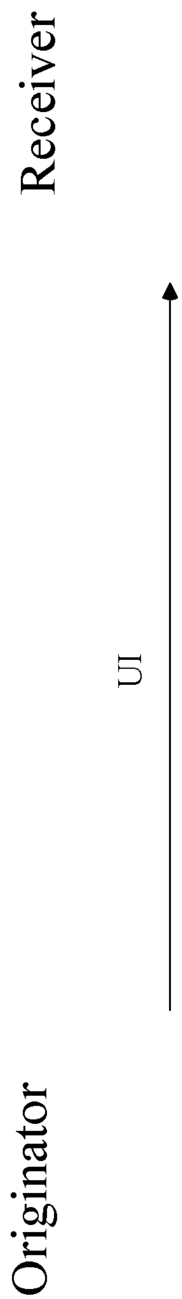


Figure 12

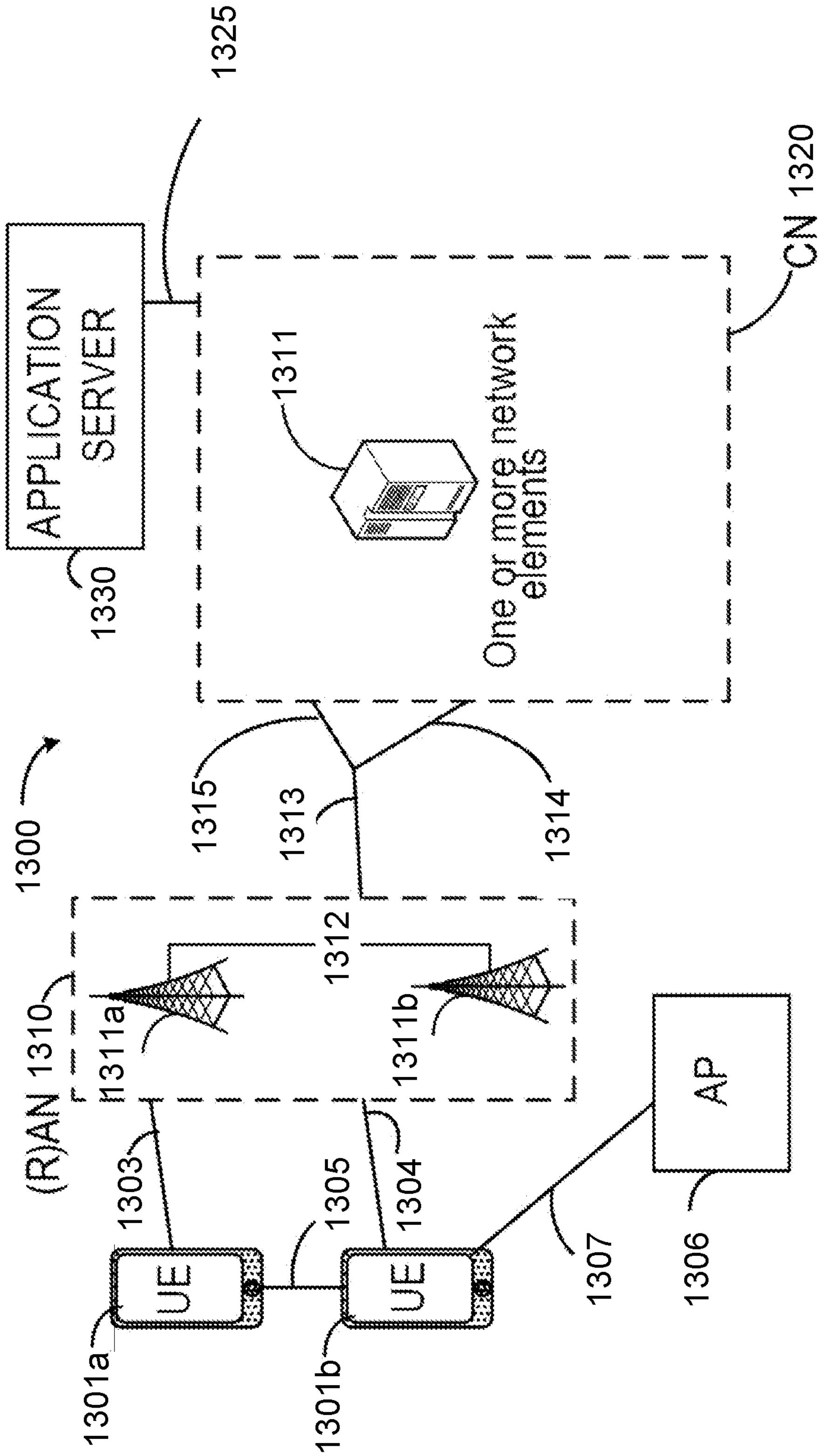

1300
APPLICATION SERVER
1330
1325
CN 1320
1311
One or more network elements
1315
1313
1314
(R)AN 1310
1311a
1312
1311b
1303
1305
1304
UE
UE
1301a
1301b
1307
AP
1306

Figure 13

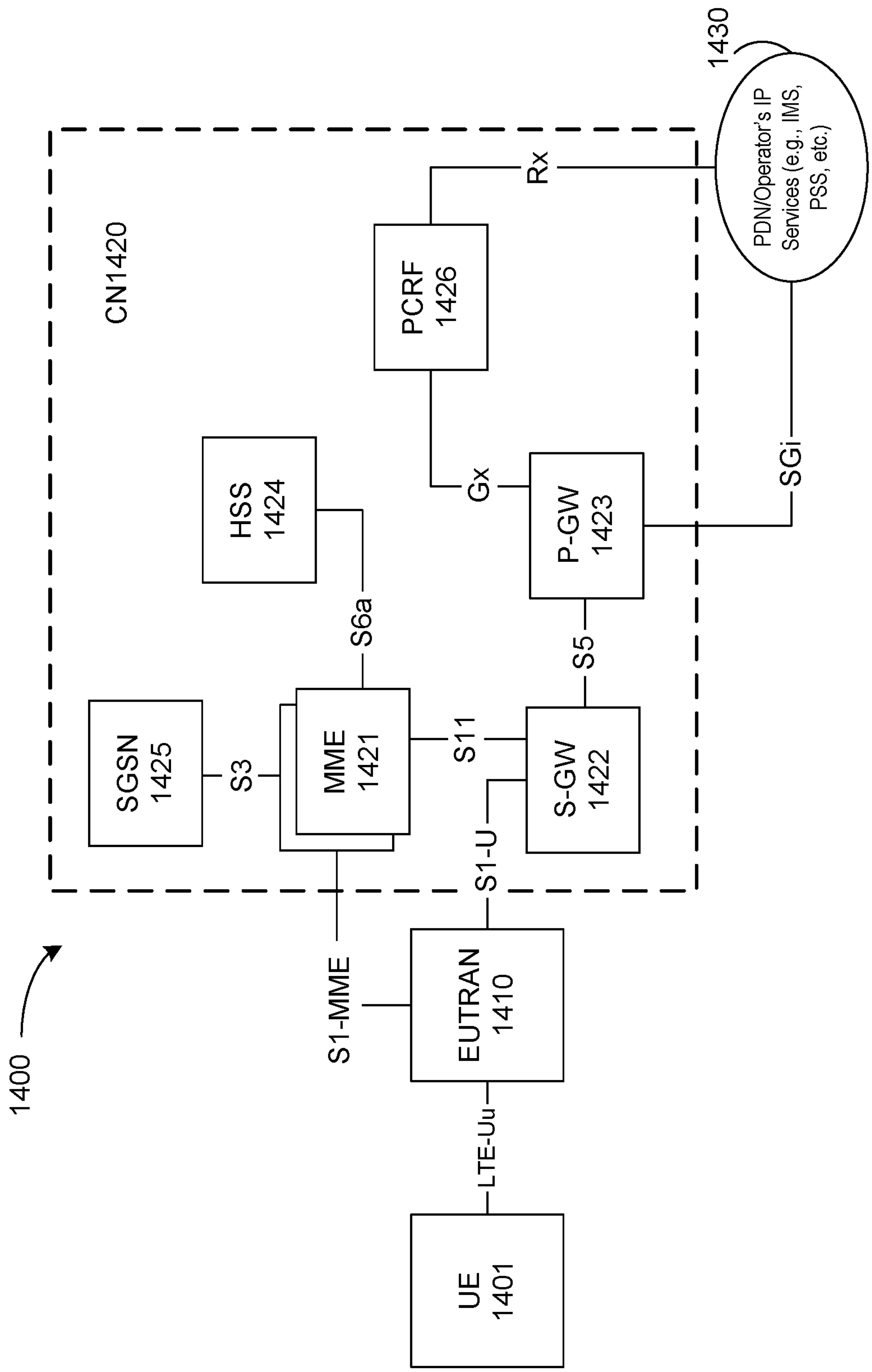

1400
CN1420
SGSN
1425
MME
1421
HSS
1424
S-GW
1422
P-GW
1423
PCRF
1426
S3
S6a
S11
S5
Gx
Rx
SGi
S1-U
S1-MME
EUTRAN
1410
LTE-Uu
UE
1401
1430
PDN/Operator's IP
Services (e.g., IMS,
PSS, etc.)

Figure 14

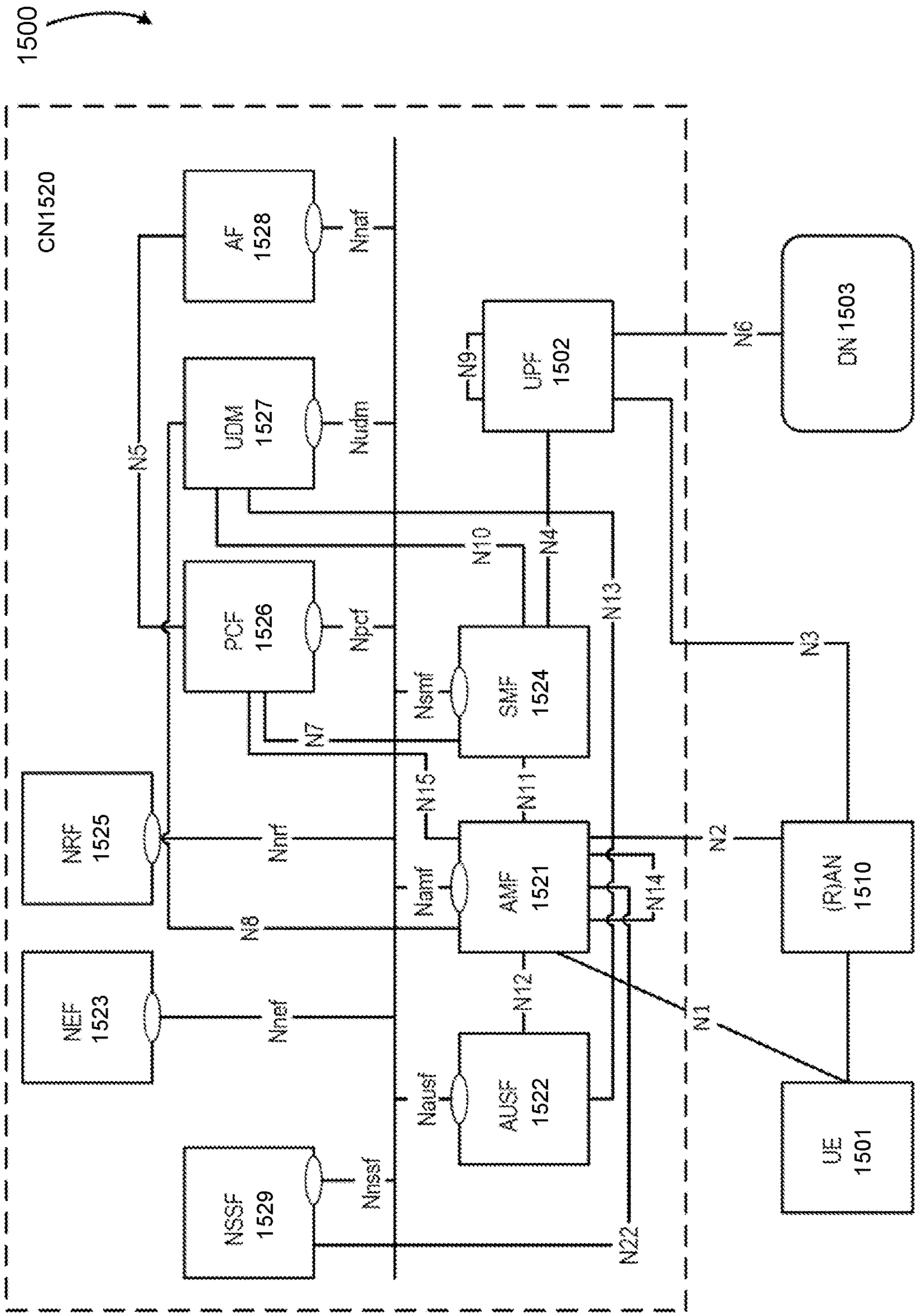

1500
CN1520
NSSF
1529
NEF
1523
NRF
1525
AUSF
1522
AMF
1521
SMF
1524
PCF
1526
UDM
1527
AF
1528
UPF
1502
DN 1503
(R)AN
1510
UE
1501
Nnssf
Nnef
Nnrf
Npcf
Nudm
Nnaf
Nausf
Namf
Nsmf
N1
N2
N3
N4
N5
N6
N7
N8
N9
N10
N11
N12
N13
N14
N15
N22

Figure 15

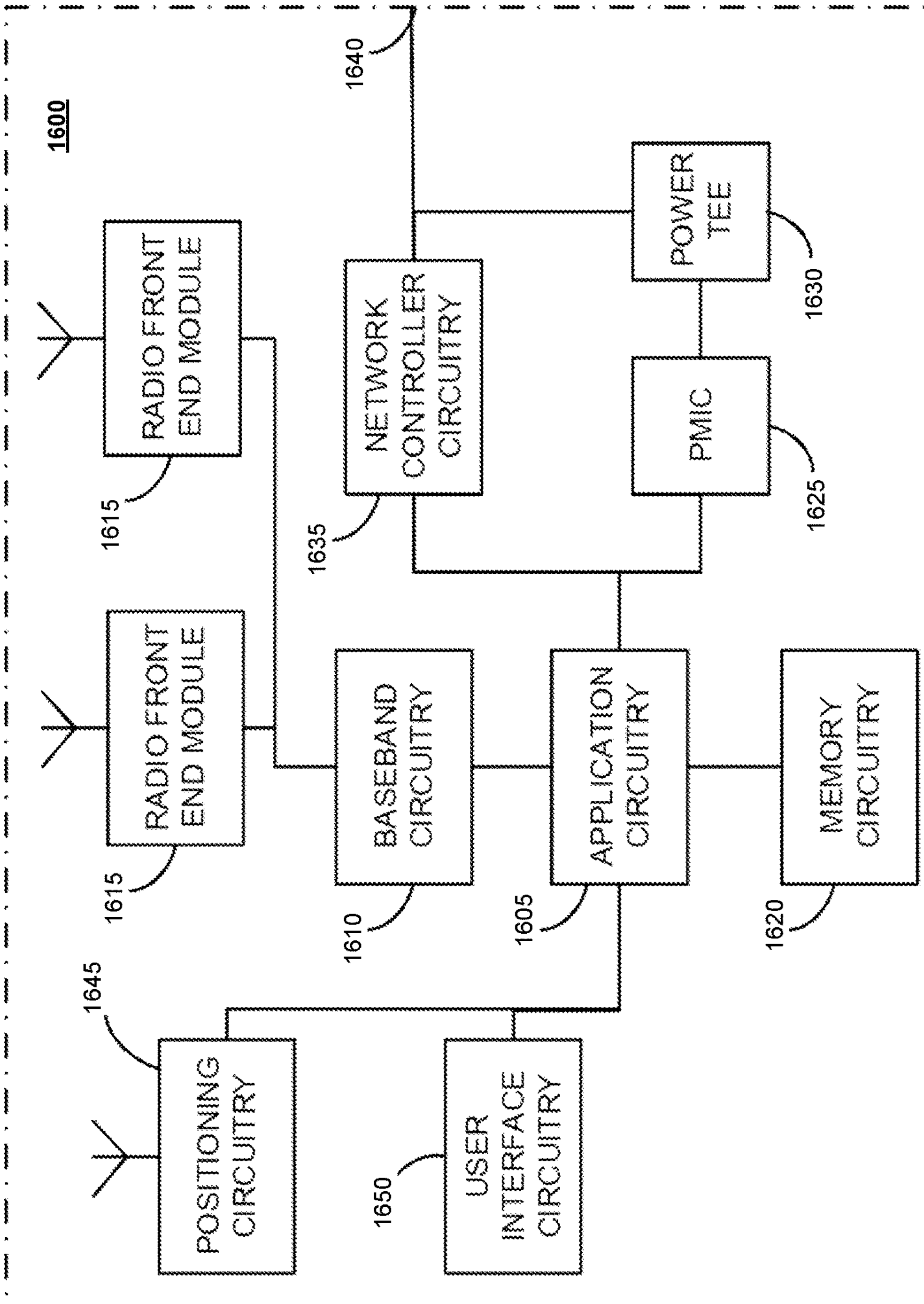

1600
RADIO FRONT END MODULE
1615
RADIO FRONT END MODULE
1615
POSITIONING CIRCUITRY
1645
BASEBAND CIRCUITRY
1610
USER INTERFACE CIRCUITRY
1650
NETWORK CONTROLLER CIRCUITRY
1635
1640
APPLICATION CIRCUITRY
1605
POWER TEE
1630
PMIC
1625
MEMORY CIRCUITRY
1620

Figure 16

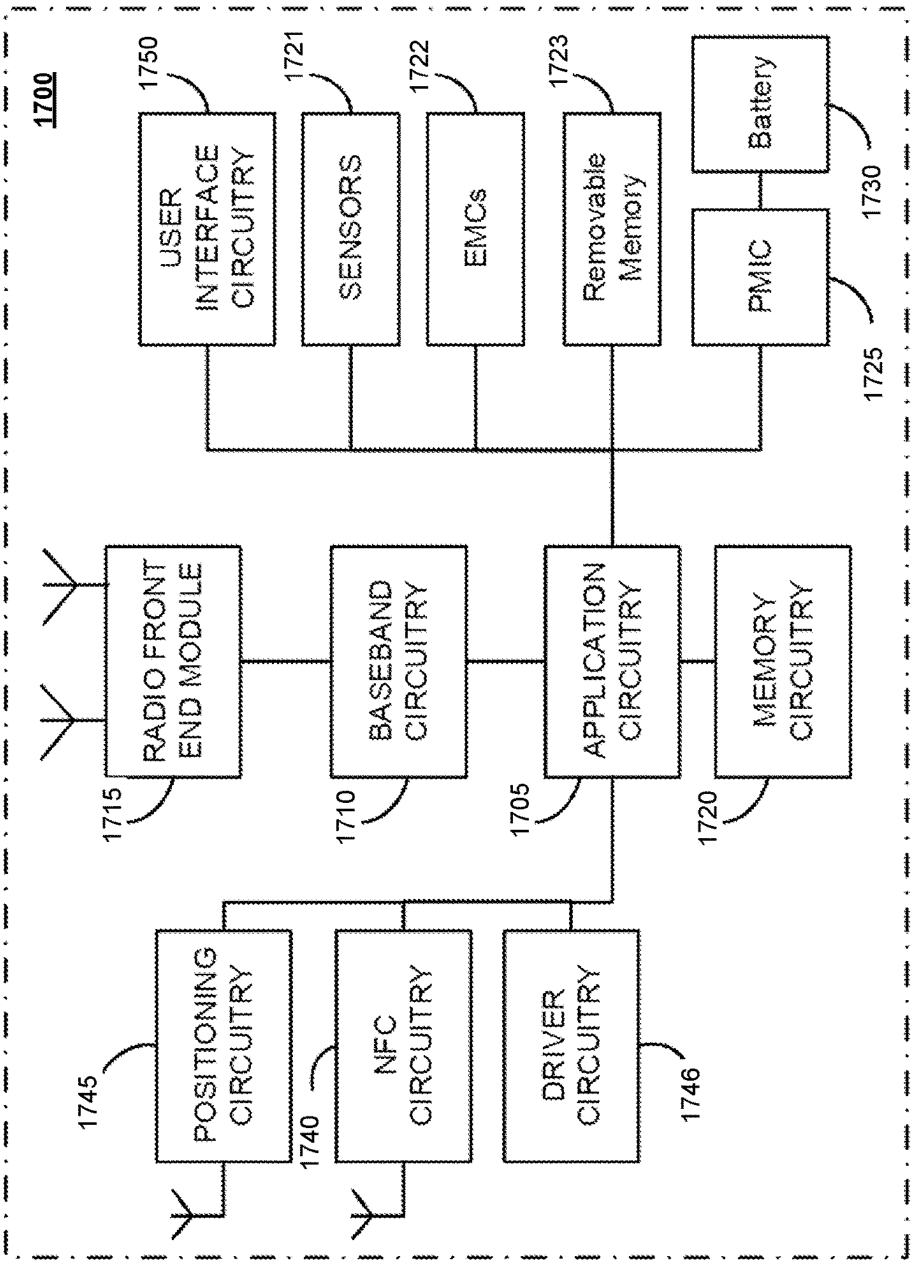

1700
USER INTERFACE CIRCUITRY
1750
SENSORS
1721
EMCs
1722
Removable Memory
1723
Battery
1730
PMIC
1725
RADIO FRONT END MODULE
1715
BASEBAND CIRCUITRY
1710
APPLICATION CIRCUITRY
1705
MEMORY CIRCUITRY
1720
POSITIONING CIRCUITRY
1745
NFC CIRCUITRY
1740
DRIVER CIRCUITRY
1746

Figure 17

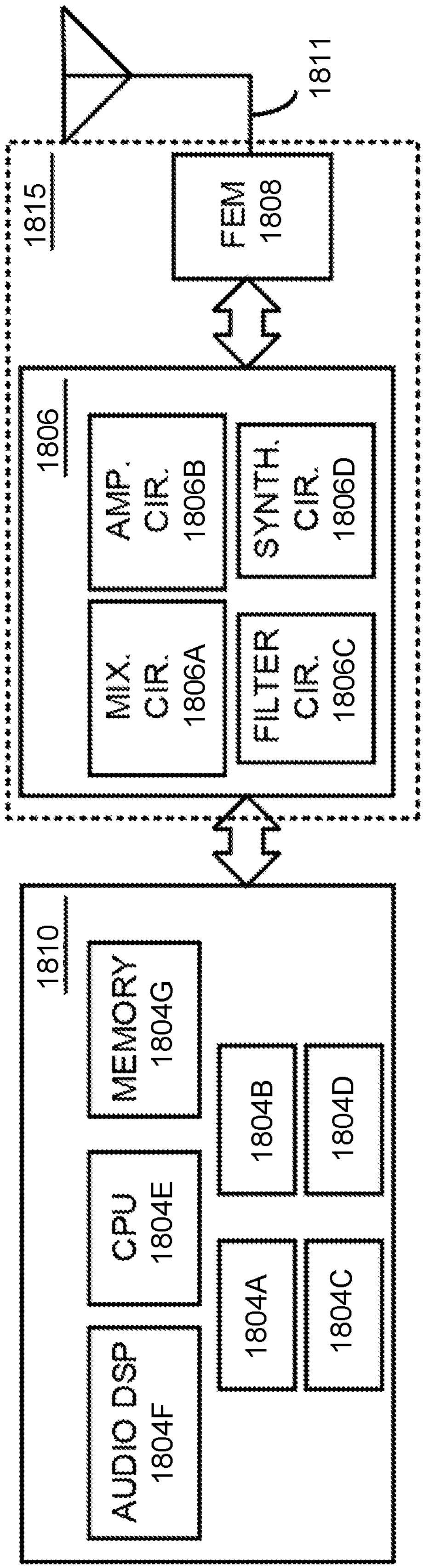
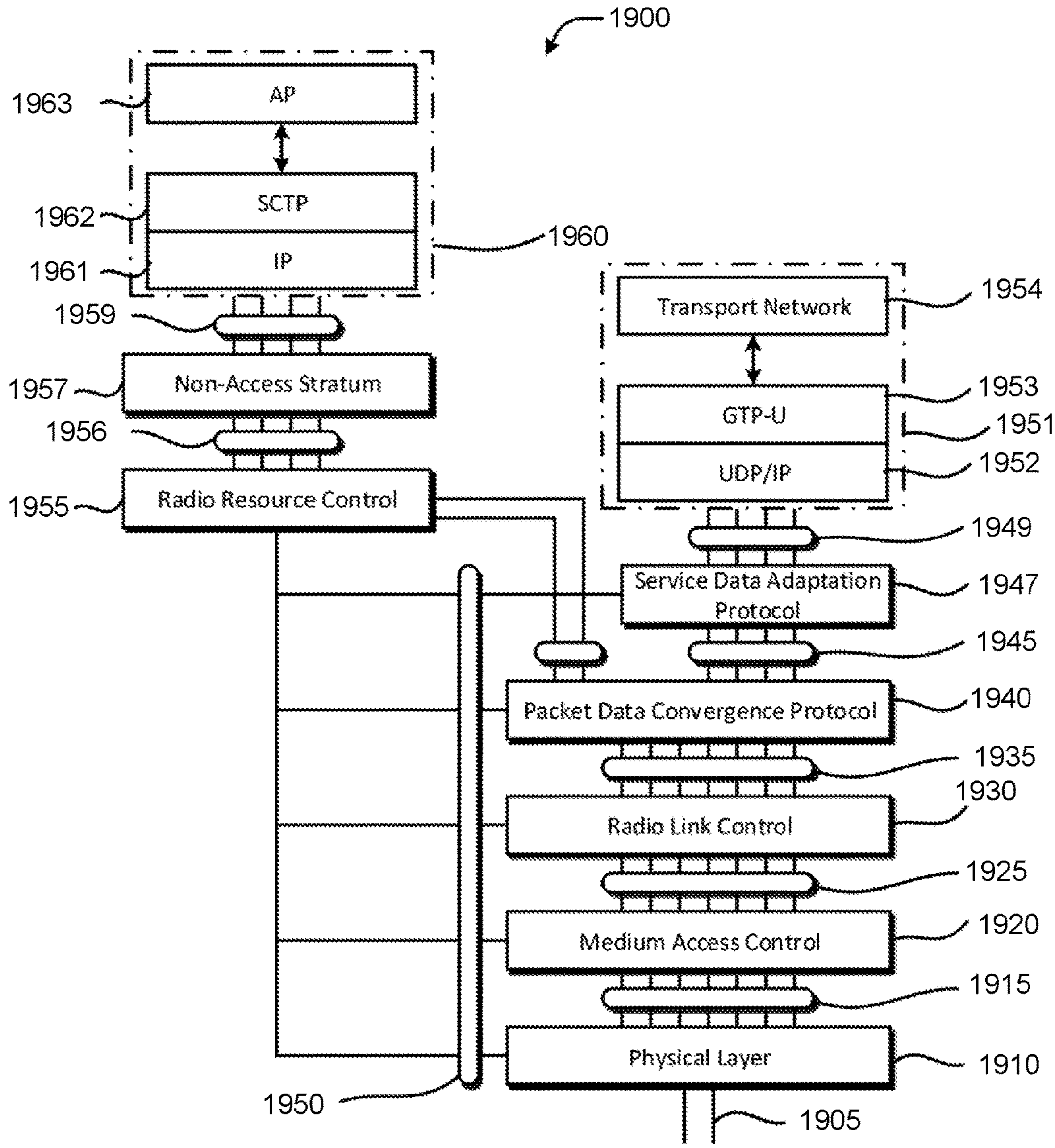

1811
1815
FEM
1808
1806
AMP.
CIR.
1806B
SYNTH.
CIR.
1806D
MIX.
CIR.
1806A
FILTER
CIR.
1806C
1810
MEMORY
1804G
CPU
1804E
AUDIO DSP
1804F
1804B
1804D
1804A
1804C

Figure 18



1900
AP
1963
SCTP
1962
IP
1961
1960
1959
Non-Access Stratum
1957
1956
Radio Resource Control
1955
Transport Network
1954
GTP-U
1953
1951
UDP/IP
1952
1949
Service Data Adaptation Protocol
1947
1945
Packet Data Convergence Protocol
1940
1935
Radio Link Control
1930
1925
Medium Access Control
1920
1915
Physical Layer
1910
1950
1905

Figure 19

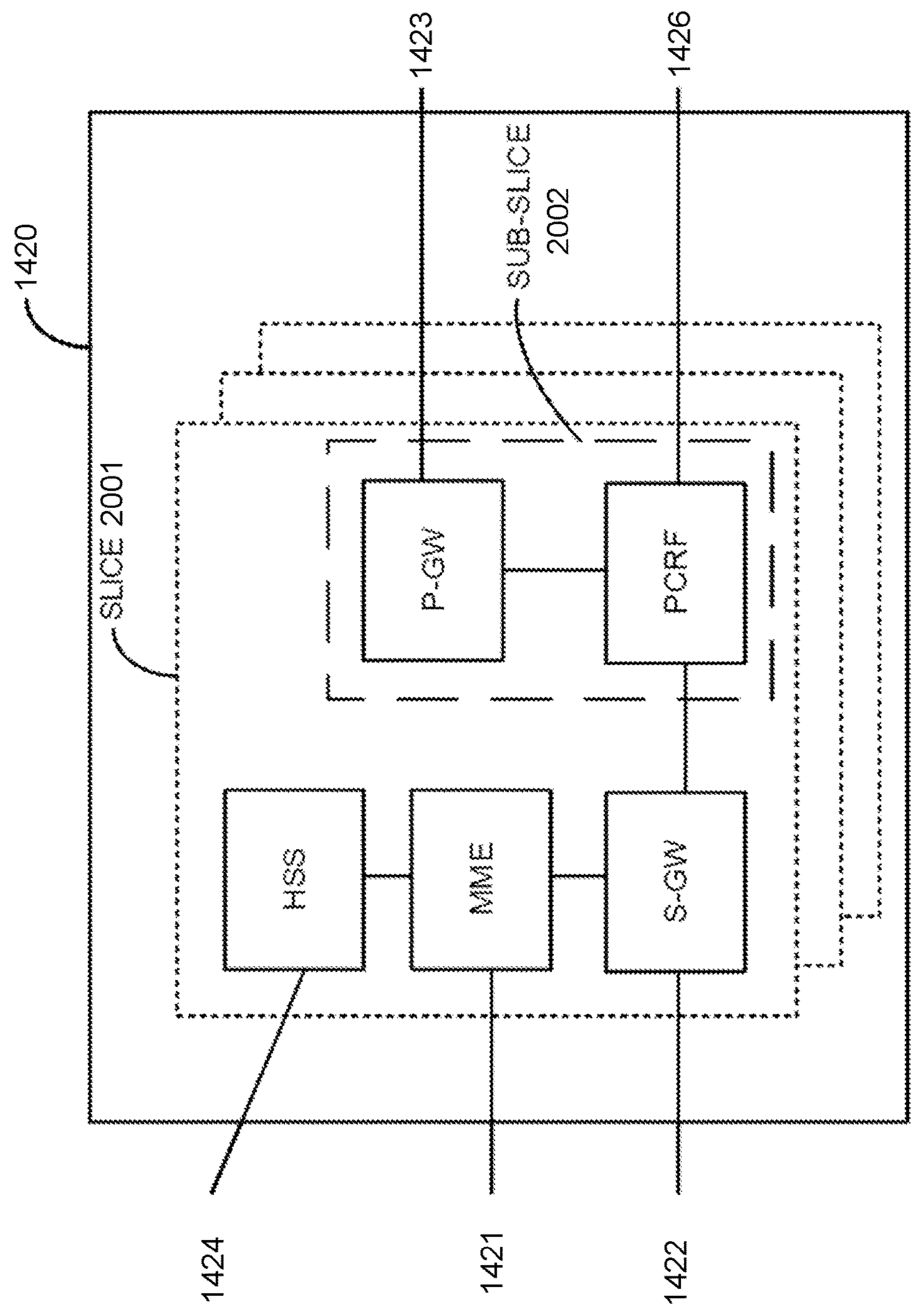

1420
SLICE 2001
SUB-SLICE 2002
HSS
MME
S-GW
P-GW
PCRF
1423
1426
1424
1421
1422

Figure 20

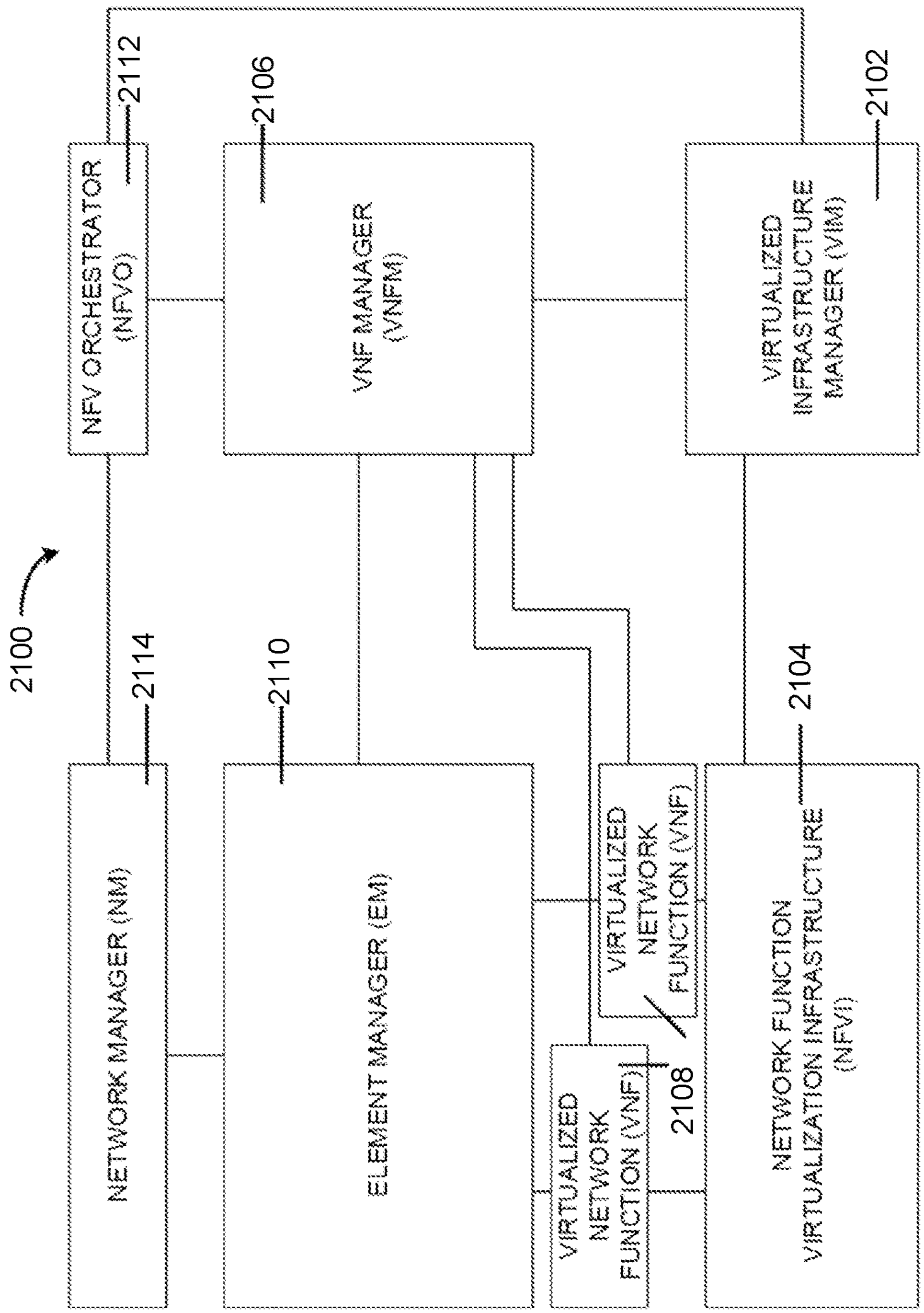

2100
NFV ORCHESTRATOR (NFVO)
2112
VNF MANAGER (VNFM)
2106
VIRTUALIZED INFRASTRUCTURE MANAGER (VIM)
2102
NETWORK MANAGER (NM)
2114
ELEMENT MANAGER (EM)
2110
VIRTUALIZED NETWORK FUNCTION (VNF)
VIRTUALIZED NETWORK FUNCTION (VNF)
2108
NETWORK FUNCTION VIRTUALIZATION INFRASTRUCTURE (NFVI)
2104

Figure 21

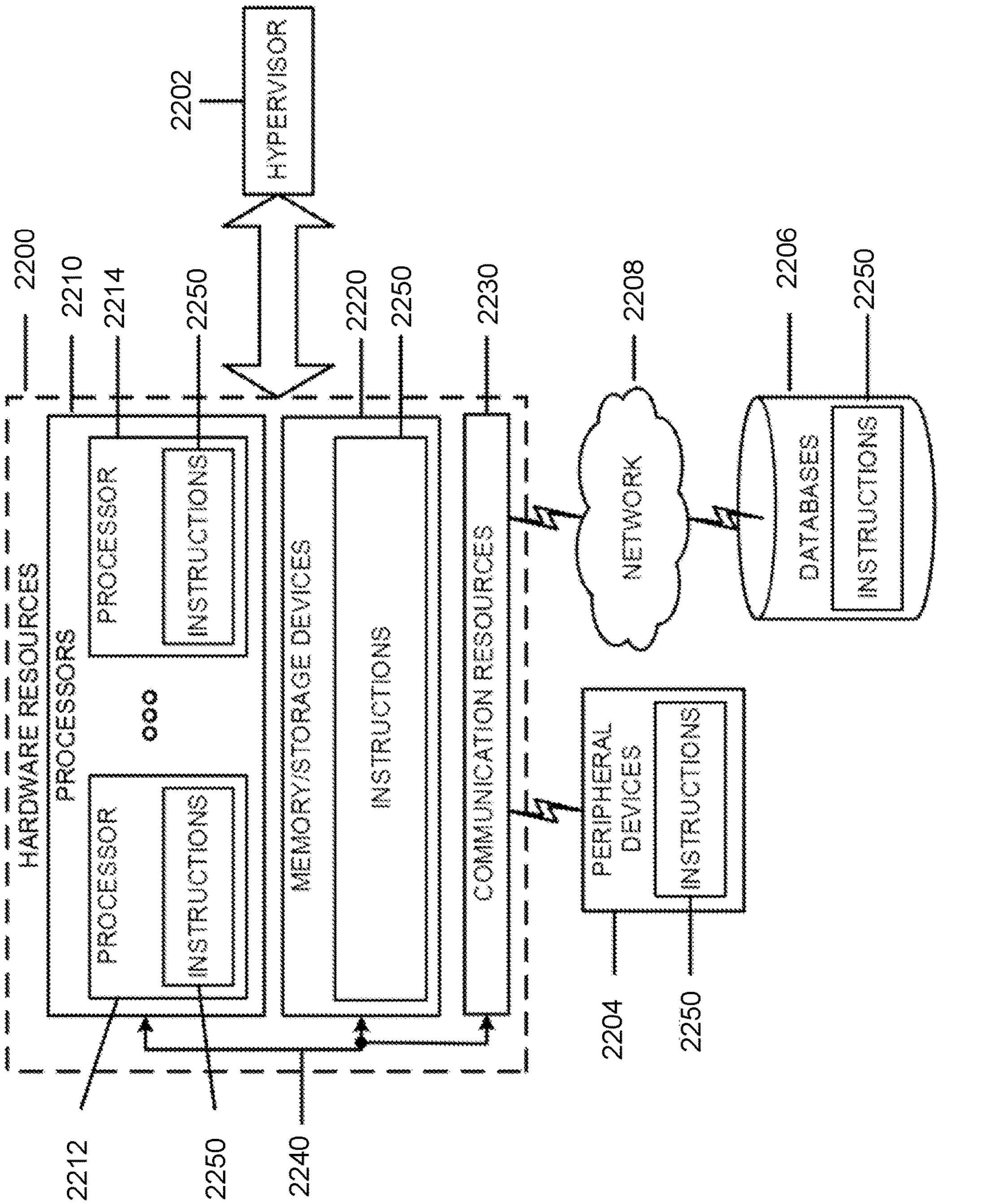

HYPERVISOR
2202
2200
2210
2214
2250
2220
2250
2230
2208
2206
2250
HARDWARE RESOURCES
PROCESSORS
PROCESSOR
INSTRUCTIONS
PROCESSOR
INSTRUCTIONS
MEMORY/STORAGE DEVICES
INSTRUCTIONS
COMMUNICATION RESOURCES
NETWORK
DATABASES
INSTRUCTIONS
PERIPHERAL DEVICES
INSTRUCTIONS
2204
2250
2212
2250
2240

Figure 22

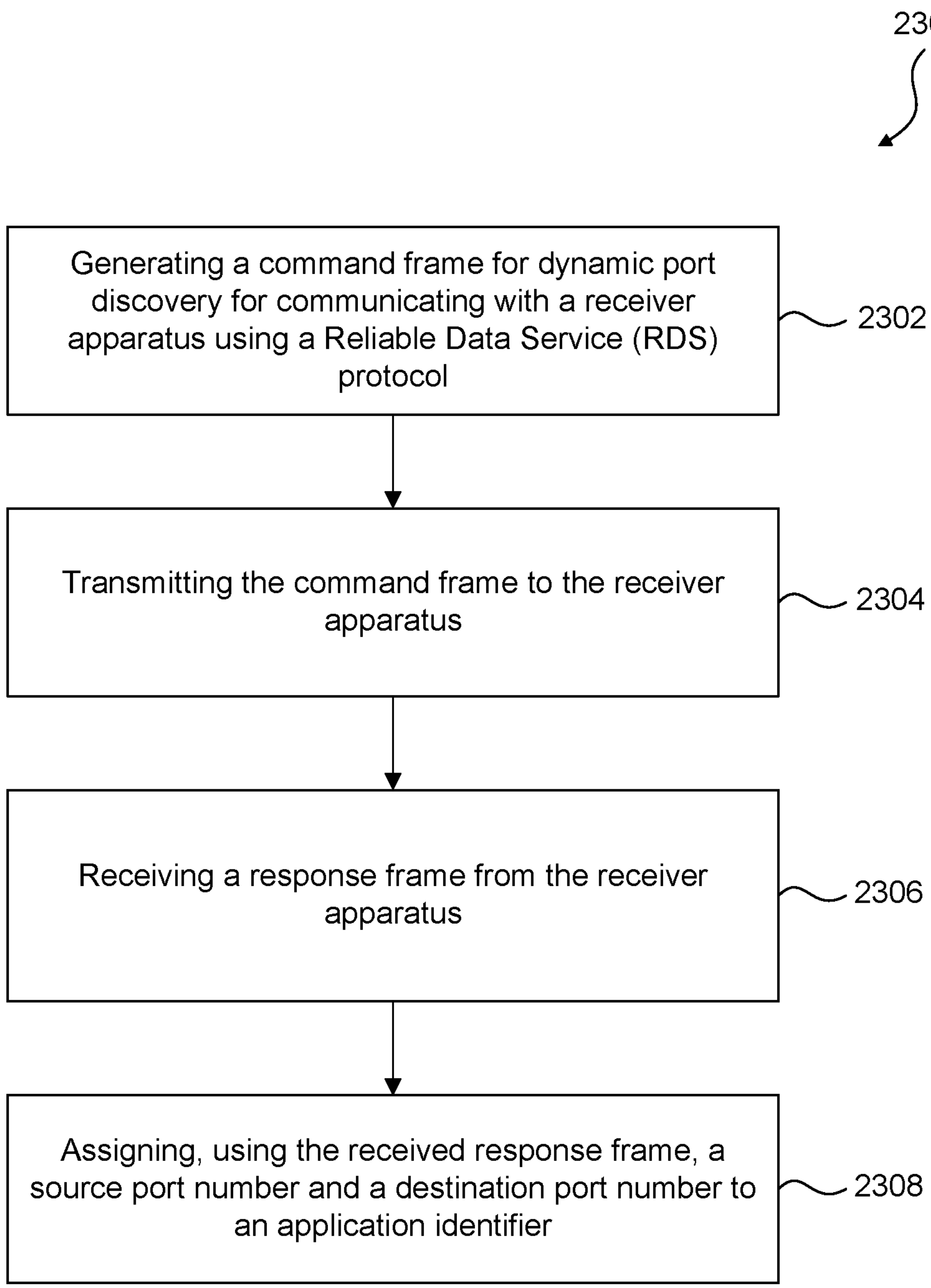

2300
Generating a command frame for dynamic port discovery for communicating with a receiver apparatus using a Reliable Data Service (RDS) protocol
2302
Transmitting the command frame to the receiver apparatus
2304
Receiving a response frame from the receiver apparatus
2306
Assigning, using the received response frame, a source port number and a destination port number to an application identifier
2308

Figure 23

PORT MANAGEMENT FOR RELIABLE DATA SERVICE (RDS) PROTOCOL

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a U.S. National Phase of International Application No. PCT/US2020/013504, filed Jan. 14, 2020, which claims the benefit under 35 U.S.C. § 119 (e) of U.S. Provisional Application No. 62/792,089, filed Jan. 14, 2019, both of which are hereby incorporated by reference in their entireties.

BACKGROUND

Various embodiments generally may relate to the field of wireless communications.

SUMMARY

Some embodiments of this disclosure include systems, apparatuses, methods, and computer-readable media for use in a wireless network for dynamic port discovery.

Some embodiments are directed to an originator device. The originator device includes processor circuitry and radio front end circuitry. The processor circuitry can be configured to generate a command frame for dynamic port discovery for communicating with a receiver device using a Reliable Data Service (RDS) protocol. The processor circuitry can be further configured to transmit, using the radio front end circuitry, the command frame to the receiver device and receive, using the radio front end circuitry, a response frame from the receiver device. The processor circuitry can be further configured to assign, using the received response frame, a source port number and a destination port number to an application identifier. The application identifier can be associated with an application on the originator device communicating with a corresponding application on the receiver device.

Some embodiments are directed to an originator device. The originator device includes a memory that stores instructions and a processor. The processor, upon executing the instructions, can be configured to generate a command frame for dynamic port discovery for communicating with a receiver device using a Reliable Data Service (RDS) protocol. The processor can be also configured to transmit the command frame to the receiver device and receive a response frame from the receiver device. The processor can be further configured to assign, using the received response frame, a source port number and a destination port number to an application identifier. The application identifier can be associated with an application on the originator device communicating with a corresponding application on the receiver device.

Some embodiments are directed to a method including generating, at an originator device, a command frame for dynamic port discovery for communicating with a receiver device using a Reliable Data Service (RDS) protocol. The method can also include transmitting the command frame to the receiver device and receiving a response frame from the receiver device. The method can also include assigning, using the received response frame, a source port number and a destination port number to an application identifier. The application identifier can be associated with an application on the originator device communicating with a corresponding application on the receiver device.

BRIEF DESCRIPTION OF THE DRAWINGS/FIGURES

FIG. **1** depicts an example of protocol layering for reliable data transfer between UE and SCEF via E-UTRAN, in accordance with some embodiments.

FIG. **2** depicts an example SET_PARAMETERS field format, in accordance with some embodiments.

FIG. **3** depicts an example bitmap of port numbers, in accordance with some embodiments.

FIG. **4** depicts an example RDS_Reserve_Port Command frame and an example RDS_Reserve_Port Response frame, in accordance with some embodiments.

FIG. **5** depicts an example RDS_Release_Port Command frame and an example RDS_Release_Port Response frame, in accordance with some embodiments.

FIG. **6** depicts an example RDS_PORT_MGMT field format, in accordance with some embodiments.

FIG. **7** depicts an example RDS frame format, in accordance with some embodiments.

FIG. **8** depicts an example Address and Control field format, in accordance with some embodiments.

FIG. **9** depicts an example establishment of acknowledged transfer procedure, in accordance with some embodiments.

FIG. **10** depicts an example acknowledged information transfer procedure, in accordance with some embodiments.

FIG. **11** depicts an example termination of acknowledged transfer procedure, in accordance with some embodiments.

FIG. **12** depicts an example unacknowledged information transfer procedure, in accordance with some embodiments.

FIG. **13** depicts an architecture of a system of a network, in accordance with some embodiments.

FIG. **14** depicts an architecture of a system including a first core network, in accordance with some embodiments.

FIG. **15** depicts an architecture of a system including a second core network in accordance with some embodiments.

FIG. **16** depicts an example of infrastructure equipment, in accordance with various embodiments.

FIG. **17** depicts example components of a computer platform, in accordance with various embodiments.

FIG. **18** depicts example components of baseband circuitry and radio frequency circuitry, in accordance with various embodiments.

FIG. **19** is an illustration of various protocol functions that may be used for various protocol stacks, in accordance with various embodiments.

FIG. **20** illustrates components of a core network in accordance with various embodiments.

FIG. **21** is a block diagram illustrating components, according to some example embodiments, of a system to support NFV.

FIG. **22** depicts a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein.

FIG. **23** depicts an example flowchart for practicing the various embodiments discussed herein, for example, for dynamic port discovery.

The features and advantages of the embodiments will become more apparent from the detailed description set forth below when taken in conjunction with the drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements. The drawing in which an element first appears is indicated by the leftmost digit(s) in the corresponding reference number.

DETAILED DESCRIPTION

The following detailed description refers to the accompanying drawings. The same reference numbers may be used in different drawings to identify the same or similar elements. In the following description, for purposes of explanation and not limitation, specific details are set forth such as particular structures, architectures, interfaces, techniques, etc. in order to provide a thorough understanding of the various aspects of various embodiments. However, it will be apparent to those skilled in the art having the benefit of the present disclosure that the various aspects of the various embodiments may be practiced in other examples that depart from these specific details. In certain instances, descriptions of well-known devices, circuits, and methods are omitted so as not to obscure the description of the various embodiments with unnecessary detail. For the purposes of the present document, the phrase "A or B" means (A), (B), or (A and B).

The Reliable Data Service (RDS) protocol is used to perform non-IP data transfer between a UE (e.g., UE **1301** of FIG. **13**) and Service Capability Exposure Function (SCEF)/PDN-GW. RDS supports peer-to-peer data transfers and provides reliable data delivery between the UE and the SCEF or P-GW. The data is transferred via a PDN connection between the UE and SCEF or P-GW. The UE can connect to multiple SCEFs or P-GWs. A UE can connect to multiple SCS/AS via the SCEF or P-GW. RDS supports multiple applications on the UE to simultaneously conduct data transfers with their peer entities on the SCEF or P-GW using a single PDN connection between the UE and SCEF or P-GW. RDS supports both acknowledged and unacknowledged data transfers. RDS supports variable-length frames and allows detection and elimination of duplicate frames at the receiving endpoint.

FIG. **1** shows a reference model of the protocol layering for reliable data transfer between the UE and SCEF. The RDS operates above the NAS and Diameter layers in the reference architecture. RDS establishes a peer-to-peer logical link between the UE and SCEF or P-GW. The logical link is identified by a pair of port numbers and EPS bearer ID. Each port number is used to identify an application on the UE side or at the network (SCEF or P-GW) side and is carried in the address field of each frame. The source port number identifies the application on the originator and the destination port number identifies the application on the receiver. When a single application on the originator conducts data transfer with a single application on the receiver, the source port number and destination port number need not be used. Each RDS frame consists of a header and an information field of variable length. The header contains information about port numbers and the frame number that is used to identify the frame and provide reliable transmission. The information field contains the payload to be transferred between the UE and SCEF or P-GW.

The UE establishes a PDN connection with the SCEF or P-GW either during Attach or through UE requested PDN connectivity. The UE uses the EPS bearer ID to select the bearer to transfer RDS PDUs to the SCEF or P-GW. The EPS bearer ID identifies the destination (at the UE or at the SCEF or P-GW) and is not carried in the frame as it is already included in the NAS ESM message header. Once the UE and network successfully negotiate to use RDS for a particular PDN connection, the PDN connection transfers data only using RDS protocol.

RDS supports both single and multiple applications within the UE. RDS also provides functionality for flow control and sequence control to maintain the sequential order of frames across the logical link.

In acknowledged operation the information is transmitted in order in numbered Information (I) frames. The I frames are acknowledged at the RDS layer. Error recovery and reordering mechanisms based on retransmission of acknowledged I frames are specified. Several I frames can be acknowledged at the same time. Flow control is implemented via a sliding window mechanism. In unacknowledged operation the information is transmitted in numbered Unconfirmed Information (UI) frames. The UI frames are not acknowledged at the RDS layer. Error recovery and reordering mechanisms are not defined. Duplicate UI frames are discarded. Flow control procedures are not defined.

The RDS protocol is currently being used in 4G and is also likely to be used in 5G systems for CIoT type applications. Currently, port number assignment and management is not included in RDS protocol and is done outside it by proprietary mechanisms, some other OTT applications or through static configuration.

Embodiments herein extend the RDS protocol so that dynamic port discovery can be added to it. The purpose of port discovery is to allow originators (UE or SCEF) to dynamically discover destination port numbers and other associated applications that use RDS transfer over the EPS bearer ID. For UEs with limited subscription to only non-IP APN(s), this RDS based discovery application may be the only way to dynamically discover these port numbers.

RDS may be used for CIoT systems and applications, for example, data transfer for applications like smart metering, power grid management and other IoT type applications. Dynamic port discovery makes the protocol more flexible and usable.

RDS Port Discovery

Source Port Number (Source Port)

When a UE application starts to use the PDN connection to transmit RDS frames, the UE and the SCEF or P-GW establish which source port number will be used for the application on the UE side for MO traffic and which destination port number will be used for the application intended to receive the frames on the SCEF or P-GW side. Similarly for MT traffic when an application in the network starts to use the PDN connection to transmit RDS frames, the UE and the SCEF or P-GW establish which source port number will be used for the application on the SCEF or P-GW side and which destination port number will be used for the application intended to receive the frames on the UE side. How the applications on the originator side and their peer entities on receiver side synchronize port numbers is outside the scope of this specification.

One way around this is that port numbers are hard coded and assigned a priori. Having these small ranges of port numbers hard coded to specific IoT applications might not be suitable for flexible deployments as it requires firmware upgrade at the UE side each time a port number is added, deleted or updated. For UEs that support both IP and non-IP protocol stack and have subscription to both IP and non-IP APNs, there is a way to dynamically discover these port numbers outside the RDS protocol. For example, a web discovery server running over http/https protocol might be used to dynamically discover/allocate these port numbers. For UEs that does not have subscription to IP based APN, there is not a way to dynamically discover these port numbers but to use RDS protocol.

In various embodiments, the RDS protocol is extended so that dynamic port discovery can be added to it. The purpose of port discovery is to allow originators (UE or SCEF) to dynamically discover destination port numbers and other associated applications that use RDS transfer over the EPS bearer ID. For UEs with limited subscription to only non-IP APN(s), this RDS based discovery application may be the only way to dynamically discover these port numbers.

In a first embodiment, the SET_PARAMETERs command/response is extended for port discovery. In a second embodiment, a new command RDS_PORT_MGMT is defined for port discovery.

The highlighted yellow text is new text proposed to TS 24.250.

Embodiment 1

SET_PARAMETERS Command/Response

The SET_PARAMETERS command and response is used to negotiate values of parameters between originator and receiver in both acknowledged and unacknowledged mode of transfer. These parameters include the version of the RDS.

If the originator wants to negotiate the value of parameters, the originator sends a SET_PARAMETERS command including the set of parameters along with their values to the receiver. The receiver sends a SET_PARAMETERS response, either confirming these parameter values by returning the requested values, or proposing different ones in their place. Both, the originator and the receiver uses the negotiated values after the completion of the negotiation process.

Table 1 lists the negotiable RDS layer parameters. FIG. **2** shows the SET_PARAMETERS field format. A parameter item consists of Type and Length octets followed by the value of that parameter. The Length octet indicates the number of octets that the value actually occupies.

TABLE 1

RDS layer parameters

Parameter Name	Type	Length	Format (87654321)	Range
RDS_Version	0	1	bbbbbbbb	0 through 255
RDS_Query_Port_List	1	2	Bitmap	See below
RDS_Reserve_Port	2			See below
RDS_Release_Port	3			See below

RDS_Query Port List

The RDS_Query_Port_List parameter allows the originator to query the receiver as to which port numbers are assigned and which are free on the receiver side. The originator passes a bitmap of Source port numbers (0-15) to receiver. If a port is assigned on originator side, the bit position is set to 1 and if a port is free then the corresponding bit position is set to 0. The receiver sends a corresponding bitmap of Destination ports on receiver side to originator. In some variants the Application ID associated with reserved port numbers may also be included after the bitmap of port numbers. An example bitmap of port numbers, in accordance with some embodiments, is illustrated in FIG. **3**.

RDS_Reserve_Port

The RDS_Reserve_Port command allows the originator to assign a combination of Source port and Destination port numbers to a specific application. The originator sends the Source Port number, Destination Port number and Application Id to the receiver in the command frame. The receiver returns the Source Port number, Destination Port number, Application Id and Status to the originator in response frame. The Status field indicates if the receiver was able to successfully assign the Source and Destination port numbers for the specified application.

The application identifier Application ID is specified as follows:

a) "OS Id+OS App Id" where OS Id contains a Universally Unique IDentifier (UUID) as specified in IETF RFC 4122 ("A Universally Unique IDentifier (UUID) URN Namespace") of 16 bytes and OS app id contains an OS specific application identifier of N bytes; or

b) Only "OS App Id" (example: URN) of 4 bytes that is more universally recognized by IoT applications without subscription to IP-based APNs.

FIG. **4** depicts an example RDS_Reserve_Port Command frame **401** and an example RDS_Reserve_Port Response frame **411**, in accordance with some embodiments. According to some embodiments, RDS_Reserve_Port Command frame **401** can include Source Port number **403**, Destination Port number **405**, and Application Identifier **407**. According to some embodiments, RDS_Reserve_Port Response frame **411** can include Source Port number **413**, Destination Port number **415**, Application Identifier **417**, and Status **419**. Status **419** in the RDS_Reserve_Port Response frame **411** can be as follows:

Status:

0 Successful

1 Destination Port already assigned

2 Application ID not valid

RDS_Release Port

The RDS_Release_Port command allows the originator to release a combination of Source port and Destination port numbers which have been previously assigned to a specific application. The originator sends the Source Port number, Destination Port number and Application Id to the receiver in the command frame. The receiver returns the Source Port number, Destination Port number, Application Id and Status to the originator in response frame. The Status field indicates if the receiver was able to successfully release the Source and Destination port numbers from the specified application.

FIG. **5** depicts an example RDS_Release_Port Command frame **501** and an example RDS_Release_Port Response frame **511**, in accordance with some embodiments. According to some embodiments, RDS_Release_Port Command frame **501** can include Source Port number **503**, Destination Port number **505**, and Application Identifier **507**. According to some embodiments, RDS_Release_Port Response frame **511** can include Source Port number **513**, Destination Port number **515**, Application Identifier **517**, and Status **519**. Status **519** in the RDS_Release_Port Response frame **511** can be as follows:

Status:

0 Successful

1 Port Busy

2 Application ID not valid

Embodiment 2

Commands and Responses

General

The following commands and responses are used by the UE side and SCEF side as shown in table 2. Each link connection supports the appropriate set of commands and responses for the type of operation desired.

TABLE 2

Commands and responses								
			Encoding					
Frame	Commands	Responses	S1	S2	M4	M3	M2	M1
S Frame or I Frame	SACK	SACK	1	1	—	—	—	—
U Frame	ERROR	ERROR	—	—	0	0	0	1
U Frame	DISCONNECT	—	—	—	0	1	0	0
U Frame	—	ACCEPT	—	—	0	1	1	0
U Frame	SET_ACK_MODE	—	—	—	0	1	1	1
U Frame	SET_PARAMETERS	SET_PARAMETERS	—	—	1	0	1	1
U Frame	RDS_PORT_MGMT	RDS_PORT_MGMT	—	—	1	0	1	0

RDS_PORT_MGMT Command/Response

The RDS_PORT_MGMT command and response is used to discover which port numbers are free and which port numbers have associated applications with them.

Applications can then use this command to reserve or release a port number at originator and receiver.

If the originator wants to discover the list of all port numbers already reserved, or reserve a new port number or release an already assigned port number, the originator sends a RDS_PORT_MGMT command including the set of parameters along with their values to the receiver. The receiver sends a RDS_PORT_MGMT response, either returning all the information requested by the originator or reserving or releasing a port number as requested by the originator. The receiver also includes the status of operation in the response.

FIG. **6** shows the RDS_PORT_MGMT field format. A parameter item consists of Port Mgmt Command and Length octets followed by other parameters depending on type of port management command. The Length octet indicates the number of octets that the frame actually occupies.

The following Port Mgmt Commands are supported.

RDS_Query Port_List	1
RDS_Reserve_Port	2
RDS_Release_Port	3

The frame formats for command and response are similar to that described in Option 1.

RDS Aspects

Frame Structure and Format of Fields

General

The peer-to-peer transfers using RDS conforms to the frame format as shown in FIG. **7**. The frame header consists of the Address and Control field and is a minimum of 1 octet and a maximum of 3 octets long. The Information field is of variable length.

Address and Control Field

Address and Control Field Format

The Address and Control field identifies the type of frame and consists of minimum of 1 octet and maximum of 3 octets. The following types of control field frames are specified:

confirmed information transfer (I frame);

supervisory functions (S frame);

unconfirmed information transfer (UI frame); and

control functions (U frame).

The address and control field format for RDS is shown in FIG. **8**. The description of address and control field bits is shown in Table 3.

TABLE 3

Address and Control field bits description	
Control field bits	Description
A	Acknowledgement request bit
M_n	Unnumbered function bit
N(R)	Receive sequence number
N(S)	Send sequence number
N(U)	Unconfirmed sequence number
S_n	Supervisory function bit
R_n	Selective acknowledgement bitmap bit
PD	Protocol Discriminator bit
C/R	Command/Response bit
ADS	Address bit
Source Port	Source port number
Destination Port	Destination port number
X	Spare bit

Protocol Discriminator Bit (PD)

The PD bit indicates whether a frame is an RDS frame or belongs to a different protocol. RDS frames have the PD bit set to 0. If a frame with the PD bit set to 1 is received, then it is treated as an invalid frame.

Address Bit (ADS)

The ADS bit controls if the Source Port and Destination Port are included in the frame format. When a single application on UE side conducts data transfer with a single application on SCEF or P-GW side, the source and destination port numbers need not be used. The source and destination port numbers enable multiple applications on the UE side to simultaneously communicate with their peer entities on the SCEF or P-GW side using the same PDN connection. Source Port and Destination Port are included in the frame format only if the ADS bit is set to 1.

Source Port Number (Source Port)

When a UE application starts to use the PDN connection to transmit RDS frames, the UE and the SCEF or P-GW establish which source port number will be used for the application on the UE side for MO traffic and which destination port number will be used for the application intended to receive the frames on the SCEF or P-GW side. Similarly for MT traffic when an application in the network starts to use the PDN connection to transmit RDS frames, the UE and the SCEF or P-GW establish which source port number will be used for the application on the SCEF or P-GW side and which destination port number will be used for the application intended to receive the frames on the UE side. How the applications on the originator side and their peer entities on receiver side synchronize port numbers is outside the scope of this specification.

Source Port is included only if the ADS bit is 1. Source Port has values from 0 to 15.

Destination Port Number (Destination Port)

The Destination Port is used to identify the destination application on the receiver that is receiving the frame.

Destination Port is included only if the ADS bit is 1. Destination Port has values from 0 to 15.

Information Transfer Frame—I

The I frame is used to perform an information transfer between peer entities with acknowledgement. Each I frame has a send sequence number N(S), a receive sequence number N(R) and an acknowledgement request bit A, that may be set to 0 or 1. The use of N(S), N(R), and A is defined below.

Each I frame also contains supervisory function bits S(n) and the Selective Acknowledgement bitmap R(n) which are defined below.

Supervisory Frame—S

The S frame is used to perform supervisory control functions such as acknowledge I frames. The supervisory frame has supervisory function bits S(n) that are used for encoding commands and responses which perform the supervisory control functions. Each supervisory frame has a receive sequence number N(R) and an acknowledgement request bit A, that may be set to 0 or 1. In acknowledged operation, all I and S frames contain R(n), the Selective Acknowledgement bitmap. The commands and responses are defined below and the use of S(n), N(R), A and R(n) is defined below.

Unconfirmed Information Frame—UI

The UI frame is used to perform an information transfer between peer entities without acknowledgement. The UI frames contain N(U), the unconfirmed sequence number of transmitted UI frames. No verification of sequence numbers is performed for UI frames.

Unnumbered Frame—U

The U frame is used to provide additional link control functions. Each U frame has Unnumbered function bits M (n) that are used to encode link control commands and response. The U frame has Command/Response bit (C/R) that identifies a U frame as either a command or a response. The U fame contains no sequence number. The commands and responses are defined below and the use of M (n) is defined below.

Command/Response Bit (C/R)

The C/R bit identifies a frame as either a command or a response. The UE side sends commands with the C/R bit set to 0, and responses with the C/R bit set to 1. The SCEF or P-GW side does the opposite; i.e., commands are sent with C/R set to 1, and responses are sent with C/R set to 0. The combinations for the SCEF or P-GW side and UE side are shown in table 4.

TABLE 4

C/R field bit usage

Type	Direction	C/R value
Command	SCEF or P-GW side to UE side	1
Command	UE side to SCEF or P-GW side	0
Response	SCEF or P-GW side to UE side	0
Response	UE side to SCEF or P-GW side	1

Control Field Parameters and Associated State Variables

The various parameters associated with the control field frames are described below.

Acknowledgement Request Bit (a)

All I and S frames contain the Acknowledgement Request (A) bit.

The A bit set to 1 is used to solicit an acknowledgement (i.e., an I frame or S frame) from the receiver. The A bit set to 0 is used to indicate that the receiver is not requested to send an acknowledgement.

Acknowledged Operation Variables and Parameters

Send State Variable V(s)

In acknowledged operation, each originator has an associated send state variable V(S) when using I frames. V(S) denotes the sequence number of the next in-sequence I frame to be transmitted. V(S) can take on the value 0 through (MAX SEQUENCE NUMBER−1). The value of V(S) is incremented by 1 with each successive I frame transmission, and will not exceed acknowledge state variable V(A) by more than the maximum number of outstanding I frames k. The value of k may be in the range 1≤k≤(MAX SEQUENCE NUMBER/2−1). V(S) will not be incremented when an I frame is retransmitted.

Acknowledge State Variable V(a)

In acknowledged operation, each peer originator has an associated acknowledge state variable V(A) when using I frame and supervisory frame commands and responses. V(A) identifies the first I frame in the transmit window, so that V(A)−1 equals N(S) of the last in-sequence acknowledged I frame. V(A) can take on the value 0 through (MAX SEQUENCE NUMBER−1). The value of V(A) is updated by the valid N(R) values received from its peer. A valid N(R) value is in the range V(A)≤N(R)≤V(S).

Send Sequence Number N(s)

In acknowledged operation, only I frames contain N(S), the send sequence number of transmitted I frames. At the time that an in-sequence I frame is designated for transmission, the value of N(S) is set equal to the value of the send state variable V(S).

Receive State Variable V(R)

In acknowledged operation, each receiver has an associated receive state variable V(R) when using I frame and supervisory frame commands and responses. V(R) denotes the sequence number of the next in-sequence I frame expected to be received. V(R) can take on the value 0 through (MAX SEQUENCE NUMBER−1). The value of V(R) is incremented by one with the receipt of an error-free, in-sequence I frame whose send sequence number N(S) equals V(R).

Receive Sequence Number N(R)

In acknowledged operation, all I frames and S frames contain N(R), the expected send sequence number of the next in-sequence received I frame. At the time that a frame of the above types is designated for transmission, the value of N(R) is set equal to the value of the receive state variable V(R). N(R) indicates that the receiver transmitting the N(R) has correctly received all I frames numbered up to and including N(R)−1.

Selective Acknowledgement (SACK) Bitmap R(n)

In acknowledged operation, all I frames and S frames contain R(n), the SACK bitmap. At the time that a S frame is designated for transmission, the value of each bit R(n) in the bitmap is set to 0 or 1 depending on whether I frame number N(R)+n has been received or not. R(n)=1 indicates that the receiver transmitting the S frame has correctly received I frame number N(R)+n. R(n)=0 indicates that the receiver transmitting the S frame has not correctly received I frame number N(R)+n. The SACK bitmap contains (k) bits.

Unacknowledged Operation Variables and Parameters

Unconfirmed Send State Variable V(U)

Each peer entity has an associated unconfirmed send state variable V(U) when using UI frame commands. V(U) denotes the sequence number of the next UI frame to be transmitted. V(U) can take on the value 0 through 7. The value of V(U) is incremented by 1 with each successive UI frame transmission.

Unconfirmed Sequence Number N(U)

Only UI frames contain N(U), the unconfirmed sequence number of transmitted UI frames. At the time that a UI frame is designated for transmission, the value of N(U) is set equal to the value of the unconfirmed send state variable V(U).

Unconfirmed Receive State Variable V(UR)

Each peer entity has an associated unconfirmed receive state variable V(UR) when using UI frame commands. V(UR) denotes the sequence number of the next in-sequence UI frame expected to be received. V(UR) can take on the value 0 through 7.

Other Parameters and Variables

The only other parameter defined for unacknowledged operation is the number of octets (N**201**) in the information field of the UI frame.

Procedures

Types of RDS Procedures

The following RDS protocol procedures are defined:

Establishment of acknowledged transfer;

Acknowledged information transfer;

Termination of acknowledged transfer; and

Unacknowledged information transfer.

Establishment of Acknowledged Transfer Procedure

General

The purpose of the establishment of acknowledged transfer procedure is for the originator to establish acknowledged transmission of information with the receiver. All frames other than U and UI frames received during the establishment of acknowledged transfer procedure is ignored.

Establishment of Acknowledged Transfer Procedure Initiation

FIG. **9** depicts an example establishment of acknowledged transfer procedure, in accordance with some embodiments. The originator initiates the establishment of acknowledged transfer procedure when upper layers indicate information is to be transmitted using acknowledged operation. The originator and the receiver identify the source and destination port numbers before initiating establishment of acknowledged transfer procedure.

The originator initiates the establishment of acknowledged transfer procedure by transmitting a SET_ACK_MODE command to receiver. When a single application on the originator conducts data transfer with a single application on the receiver, the Source Port and Destination Port numbers need not be used; otherwise the originator sets the Source Port to the port number of the source application on the originator and the Destination Port to the port number of the destination application on the receiver. The originator clears all exception conditions, discard all queued I frames, reset the retransmission counter and timer T**200** is set.

If a logical link between the UE and SCEF identified without port numbers exists and the originator needs to initiate establishment of an additional acknowledged transfer procedure, the additional logical link between the UE and SCEF is identified with port numbers while the first logical link can remain without port numbers.

Establishment of Acknowledged Transfer Procedure Accepted by Receiver

Upon receiving a SET_ACK_MODE command, the receiver checks if the Destination Port number contained in the SET_ACK_MODE command corresponds to an application on the receiver.

If the check is successful and if the application accepts to enter acknowledged transfer mode, the receiver sends an ACCEPT response to the originator. When a single application on the originator conducts data transfer with a single application on the receiver, the Source Port and Destination Port numbers need not be used; otherwise the receiver sets the Source Port to the port number of the application on the receiver and the Destination Port to the port number of the application on the originator. The receiver resets timer T**200** if active, clear all exception conditions and set V(S), V(R) and V(A) to 0.

Establishment of Acknowledged Transfer Procedure Completed by Originator

Upon receipt of the ACCEPT response and if the Destination Port number corresponds to the application which initiated the establishment of acknowledged transfer procedure, the originator enters acknowledged mode transfer. The originator resets timer T**200** if active, clear all exception conditions and set V(S), V(R) and V(A) to 0 and the establishment of acknowledged transfer procedure is successfully completed.

Establishment of Acknowledged Transfer Procedure not Accepted by Receiver

Upon receiving a SET_ACK_MODE command if the Destination Port number contained in the SET_ACK_MODE command,

- is not supported by the receiver; or
- corresponds to an application that is not able to perform acknowledged transfer mode, then the receiver sends an ERROR response to the originator.

Acknowledged Information Transfer Procedure

General

The purpose of the acknowledged information transfer procedure is for the originator to transfer I frames to receiver and receive acknowledgements for these frames from receiver.

The originator stores the history of the transmitted I frames, and remembers the I-frame transmission sequence. The history is used to decide which I frames to retransmit. Due to retransmission, the history is not necessarily an in-order sequence.

A frame within the receive window is either:

- received: the frame has been correctly received; or
- not received: the frame has not been correctly received.

A frame within the transmit window is either:

- not yet transmitted: the frame has not yet been transmitted;
- transmitted: the frame has been (re-) transmitted, but the originator does not know if the frame has been received by the receiver;
- acknowledged: the frame has been acknowledged by the receiver; or
- marked for retransmission: the originator has decided to retransmit this I frame.

I frames is transmitted in ascending N(S) order. When I frames are retransmitted, the frame with the lowest N(S) is retransmitted first. This is used by the receiver to detect lost frames.

Transmitting I Frames and Requesting Acknowledgements

FIG. **10** depicts an example acknowledged information transfer procedure, in accordance with some embodiments. If the originator has received information to be transmitted from upper layers, the information is inserted in an I frame. The control field parameters N(S) and N(R) is assigned the values V(S) and V(R), respectively. V(S) is incremented by 1 at the end of the transmission of the I frame.

The originator requests an acknowledgement from the receiver by transmitting an I or S frame with the A bit set to 1. The originator may request an acknowledgement at any time. An acknowledgement is requested when:

the last I frame in a sequence of one or more I frames is transmitted; or

$V(S)=V(A)+k$ as a result of the transmission of the I frame.

The originator transmits a frame in the following order of priority:

If there are any I frames marked for retransmission then the originator increments by 1 the retransmission count variable for the I frame with the lowest send sequence number N(S). If the retransmission count variable exceeds the value of N**200**, then the originator initiates the Establishment of acknowledged operation procedure as described above. If the retransmission count variable does not exceed the value of N**200**, then the originator retransmits the I frame.

If the originator has a new I frame to transmit, if $V(S)<V(A)+k$ (where k is the maximum number of outstanding I frames) then the new I frame is transmitted.

If the originator has an acknowledgement to transmit then the originator transmits an S frame.

When requesting an acknowledgement, the originator sets timer T**201** and associate the timer with the I frame currently being transmitted, or, if the A bit is transmitted in an S frame, with the I frame last transmitted.

Receiving I Frames and Sending Acknowledgements

When the receiver receives a valid I frame whose N(S) is equal to the current V(R), the receiver:

passes the contents of the Information field to the appropriate upper layer entity;

increments its V(R) by 1; and

responds with a I or S frame containing the SACK bitmap, if the A bit of the received I frame was set to 1.

When the receiver receives a valid I frame whose N(S) is not in the range $V(R)\leq N(S)<V(R)+k$, the receiver discards the frame as a duplicate.

When the receiver receives a valid I frame where $V(R)<N(S)<V(R)+k$, then the receiver stores the I frame until all frames from V(R) to $N(S)-1$ inclusive are correctly received. Once all the frames are correctly received the receiver then:

passes the contents of the Information field to the appropriate upper layer entity; and

sets its $V(R)=N(S)+1$.

Whenever the receiver detects an error in the sequence of received I frames, it transmits an I or S frame.

If the receiver receives an I frame with a higher N(S) than the N(S) of the previously received I frame, and if there are I frames missing between these two N(S) values, then the receiver assumes that the missing I frames have been lost. If the receiver receives an I frame with a lower N(S) than the N(S) of the previously received I frame, it can assume that its peer originator has (re-) started retransmission due to the reception of an acknowledgement.

Receiving Acknowledgements

On receipt of a valid I or S frame, the originator, if N(R) is valid, treats the N(R) contained in this frame as an acknowledgement for all the I frames it has transmitted with an N(S) up to and including the received $N(R)-1$. A valid N(R) value is one that is in the range $V(A)\leq N(R)\leq V(S)$. If N(R) is not valid, then the received SACK bitmap is disregarded.

V(A) is then set to N(R).

On receipt of a valid I or S frame containing the SACK bitmap, the originator considers all I frames with the corresponding bit set to 1 in the SACK bitmap as acknowledged.

If timer T**201** is active and associated with an acknowledged I frame, then timer T**201** is reset.

The originator determines which I frames to retransmit by analyzing it's I frame transmission sequence history and the acknowledgements received. An unacknowledged I frame that was transmitted prior to an acknowledged I frame is considered lost and is marked for retransmission. Acknowledged I frames is removed from the I frame transmission sequence history.

Termination of Acknowledged Transfer Procedure

General

The purpose of the termination of acknowledged transfer procedure is to terminate the acknowledged transmission of information between the UE side and the SCEF or P-GW side. All frames other than U and UI frames received during the termination of acknowledged transfer procedure is ignored and all queued I frames is discarded.

Termination of Acknowledged Transfer Procedure Initiation

FIG. **11** depicts an example termination of acknowledged transfer procedure, in accordance with some embodiments. The originator or receiver initiates the termination of acknowledged transfer procedure when upper layers indicate termination of acknowledged operation.

The originator initiates the termination of acknowledged transfer procedure by transmitting a DISCONNECT command. When a single application on the originator conducts data transfer with a single application on the receiver, the Source Port and Destination Port numbers need not be used; otherwise the originator sets the Source Port to the port number of the source application on the originator and the Destination Port to the port number of the destination application on the receiver. The originator resets the retransmission counter and timer T**200** is set.

Termination of Acknowledged Transfer Procedure Accepted by Receiver

Upon receiving a DISCONNECT command, the receiver checks if the Destination Port number contained in the DISCONNECT command corresponds to an application on the receiver.

If the check is successful the receiver sends an ACCEPT response to the originator. When a single application on the originator conducts data transfer with a single application on the receiver, the Source Port and Destination Port numbers need not be used; otherwise the receiver sets the Source Port to the port number of the application on the receiver and the Destination Port to the port number of the application on the originator.

Termination of Acknowledged Transfer Procedure Completed by Originator

Upon receipt of the ACCEPT response and if the Destination Port number corresponds to the application which initiated the termination of acknowledged transfer procedure, the originator terminates acknowledged transfer mode. The originator resets timer T**200** if active, and the termination of acknowledged transfer procedure is successfully completed.

Termination of Acknowledged Transfer Procedure not Accepted by Receiver

Upon receiving a DISCONNECT command if the Destination Port number contained in the DISCONNECT command

is not supported by the receiver; or

corresponds to an application that is not in acknowledged transfer mode, then the receiver sends an ERROR response to the originator.

Unacknowledged Information Transfer Procedure

General

The purpose of the unacknowledged information transfer procedure is for the originator to perform unacknowledged transmission of information to the receiver. No error recovery mechanisms are defined for unacknowledged operation.

Unacknowledged Information Transfer Procedure Initiation

FIG. **12** depicts an example unacknowledged information transfer procedure, in accordance with some embodiments. The originator initiates the unacknowledged information transfer procedure when information from upper layers is to be transmitted using unacknowledged operation. The originator and the receiver negotiate the use of source and destination port numbers before initiating unacknowledged information transfer.

The originator initiates the unacknowledged information transfer procedure by transmitting a UI frame to receiver. When a single application on the originator conducts data transfer with a single application on the receiver, the Source Port and Destination Port numbers need not be used; otherwise the originator sets the Source Port to the port number of the source application on the originator and the Destination Port to the port number of the destination application on the receiver. The originator sets the unconfirmed sequence number N(U) in UI frame to the value of unconfirmed send state variable V(U).

Unacknowledged Information Transfer Procedure Accepted by Receiver

Upon receiving a UI frame the receiver passes the contents of the Information field to the appropriate upper layer corresponding to the Destination Port. The receiver sets the unconfirmed receive state variable V(UR) to N(U)+1.

Unacknowledged Information Transfer Procedure Completed by Originator

Upon transmission of the UI frame the unacknowledged information transfer procedure is completed by the originator.

Unacknowledged Information Transfer Procedure not Accepted by Receiver

Upon receiving a UI frame,

if the Destination Port number is not in use by the receiver; or

if N(U) of the received UI frame is in the range (V(UR)−k')≤N(U)<V(UR) and if a UI frame with the same N(U) has already been received,

then the UI frame is discarded by the receiver without any further action. The value of k' may be in the range 1<k'<MAX SEQUENCE NUMBER/2.

Abnormal Cases

Expiry of Timer T**200**

Timer T**200** is set when a U frame with any of the following commands is transmitted.

SET_ACK_MODE;

DISCONNECT; and

SET_PARAMETERS.

If timer T**200** expires before a response to the sent command is received then the originator retransmits the command, and resets and start timer T**200** and increment the retransmission counter. After retransmission of the command N**200** times, the originator aborts the operation and notify the upper layers.

Expiry of Timer T**201**

On expiry of timer T**201**, the originator increments by 1 the retransmission count variable for the I frame associated with timer T**201**. If the value of the retransmission count variable does not exceed N**200**, the originator resets and start timer T**201**, and retransmit the I frame with the A bit set to 1. If the value of the retransmission count variable exceeds N**200**, the originator sends the ERROR command to the receiver and initiate the establishment of acknowledged transfer procedure to re-establish the acknowledged transfer mode with the receiver.

List of RDS Parameters

The following parameters are applicable for Reliable Data Service (RDS).

RDS version number (Version): The RDS version number (Version) is an RDS layer parameter. The default version number is 0.

Retransmission timers (T**200** and T**201**): The default value of timers T**200** and T**201** is 250 seconds.

Maximum number of retransmissions (N**200**): The default value of N**200** is 3.

Maximum number of outstanding I frames (k): k is the maximum number of sequentially-numbered I frames that may be outstanding (i.e. unacknowledged) at any given time. k is also denoted as window size. The default value of k is 3. The value of MAX SEQUENCE NUMBER is 8.

Maximum length of Information Field (N**201**): The default value of N**201** is 1520 octets.

Systems and Implementations

FIG. **13** illustrates an example architecture of a system **1300** of a network, in accordance with various embodiments. The following description is provided for an example system **1300** that operates in conjunction with the LTE system standards and 5G or NR system standards as provided by 3GPP technical specifications. However, the example embodiments are not limited in this regard and the described embodiments may apply to other networks that benefit from the principles described herein, such as future 3GPP systems (e.g., Sixth Generation (6G)) systems, IEEE 802.16 protocols (e.g., WMAN, WiMAX, etc.), or the like.

As shown by FIG. **13**, the system **1300** includes UE **1301***a* and UE **1301***b* (collectively referred to as "UEs **1301**" or "UE **1301**"). In this example, UEs **1301** are illustrated as smartphones (e.g., handheld touchscreen mobile computing devices connectable to one or more cellular networks), but may also comprise any mobile or non-mobile computing device, such as consumer electronics devices, cellular phones, smartphones, feature phones, tablet computers, wearable computer devices, personal digital assistants (PDAs), pagers, wireless handsets, desktop computers, laptop computers, in-vehicle infotainment (IVI), in-car entertainment (ICE) devices, an Instrument Cluster (IC), head-up display (HUD) devices, onboard diagnostic (OBD) devices, dashtop mobile equipment (DME), mobile data terminals (MDTs), Electronic Engine Management System (EEMS), electronic/engine control units (ECUs), electronic/engine control modules (ECMs), embedded systems, microcontrollers, control modules, engine management systems (EMS), networked or "smart" appliances, MTC devices, M2M, IoT devices, and/or the like.

In some embodiments, any of the UEs **1301** may be IoT UEs, which may comprise a network access layer designed for low-power IoT applications utilizing short-lived UE connections. An IoT UE can utilize technologies such as

M2M or MTC for exchanging data with an MTC server or device via a PLMN, ProSe or D2D communication, sensor networks, or IoT networks. The M2M or MTC exchange of data may be a machine-initiated exchange of data. An IoT network describes interconnecting IoT UEs, which may include uniquely identifiable embedded computing devices (within the Internet infrastructure), with short-lived connections. The IoT UEs may execute background applications (e.g., keep-alive messages, status updates, etc.) to facilitate the connections of the IoT network.

The UEs **1301** may be configured to connect, for example, communicatively couple, with an or RAN **1310**. In embodiments, the RAN **1310** may be an NG RAN or a 5G RAN, an E-UTRAN, or a legacy RAN, such as a UTRAN or GERAN. As used herein, the term "NG RAN" or the like may refer to a RAN **1310** that operates in an NR or 5G system **1300**, and the term "E-UTRAN" or the like may refer to a RAN **1310** that operates in an LTE or 4G system **1300**. The UEs **1301** utilize connections (or channels) **1303** and **1304**, respectively, each of which comprises a physical communications interface or layer (discussed in further detail below).

In this example, the connections **1303** and **1304** are illustrated as an air interface to enable communicative coupling, and can be consistent with cellular communications protocols, such as a GSM protocol, a CDMA network protocol, a PTT protocol, a POC protocol, a UMTS protocol, a 3GPP LTE protocol, a 5G protocol, a NR protocol, and/or any of the other communications protocols discussed herein. In embodiments, the UEs **1301** may directly exchange communication data via a ProSe interface **1305**. The ProSe interface **1305** may alternatively be referred to as a SL interface **1305** and may comprise one or more logical channels, including but not limited to a PSCCH, a PSSCH, a PSDCH, and a PSBCH.

The UE **1301***b* is shown to be configured to access an AP **1306** (also referred to as "WLAN node **1306**," "WLAN **1306**," "WLAN Termination **1306**," "WT **1306**" or the like) via connection **1307**. The connection **1307** can comprise a local wireless connection, such as a connection consistent with any IEEE 802.11 protocol, wherein the AP **1306** would comprise a wireless fidelity (Wi-Fi®) router. In this example, the AP **1306** is shown to be connected to the Internet without connecting to the core network of the wireless system (described in further detail below). In various embodiments, the UE **1301***b*, RAN **1310**, and AP **1306** may be configured to utilize LWA operation and/or LWIP operation. The LWA operation may involve the UE **1301***b* in RRC CONNECTED being configured by a RAN node **1311***a-b* to utilize radio resources of LTE and WLAN. LWIP operation may involve the UE **1301***b* using WLAN radio resources (e.g., connection **1307**) via IPsec protocol tunneling to authenticate and encrypt packets (e.g., IP packets) sent over the connection **1307**. IPsec tunneling may include encapsulating the entirety of original IP packets and adding a new packet header, thereby protecting the original header of the IP packets.

The RAN **1310** can include one or more AN nodes or RAN nodes **1311***a* and **1311***b* (collectively referred to as "RAN nodes **1311**" or "RAN node **1311**") that enable the connections **1303** and **1304**. As used herein, the terms "access node," "access point," or the like may describe equipment that provides the radio baseband functions for data and/or voice connectivity between a network and one or more users. These access nodes can be referred to as BS, gNBs, RAN nodes, eNBs, NodeBs, RSUs, TRxPs or TRPs, and so forth, and can comprise ground stations (e.g., terrestrial access points) or satellite stations providing coverage within a geographic area (e.g., a cell). As used herein, the term "NG RAN node" or the like may refer to a RAN node **1311** that operates in an NR or 5G system **1300** (for example, a gNB), and the term "E-UTRAN node" or the like may refer to a RAN node **1311** that operates in an LTE or 4G system **1300** (e.g., an eNB). According to various embodiments, the RAN nodes **1311** may be implemented as one or more of a dedicated physical device such as a macrocell base station, and/or a low power (LP) base station for providing femtocells, picocells or other like cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells.

In some embodiments, all or parts of the RAN nodes **1311** may be implemented as one or more software entities running on server computers as part of a virtual network, which may be referred to as a CRAN and/or a virtual baseband unit pool (vBBUP). In these embodiments, the CRAN or vBBUP may implement a RAN function split, such as a PDCP split wherein RRC and PDCP layers are operated by the CRAN/vBBUP and other L2 protocol entities are operated by individual RAN nodes **1311**; a MAC/PHY split wherein RRC, PDCP, RLC, and MAC layers are operated by the CRAN/vBBUP and the PHY layer is operated by individual RAN nodes **1311**; or a "lower PHY" split wherein RRC, PDCP, RLC, MAC layers and upper portions of the PHY layer are operated by the CRAN/vBBUP and lower portions of the PHY layer are operated by individual RAN nodes **1311**. This virtualized framework allows the freed-up processor cores of the RAN nodes **1311** to perform other virtualized applications. In some implementations, an individual RAN node **1311** may represent individual gNB-DUs that are connected to a gNB-CU via individual F1 interfaces (not shown by FIG. **13**). In these implementations, the gNB-DUs may include one or more remote radio heads or RFEMs (see, e.g., FIG. **16**), and the gNB-CU may be operated by a server that is located in the RAN **1310** (not shown) or by a server pool in a similar manner as the CRAN/vBBUP. Additionally or alternatively, one or more of the RAN nodes **1311** may be next generation eNBs (ng-eNBs), which are RAN nodes that provide E-UTRA user plane and control plane protocol terminations toward the UEs **1301**, and are connected to a 5GC (e.g., CN **1520** of FIG. **15**) via an NG interface (discussed infra).

In V2X scenarios one or more of the RAN nodes **1311** may be or act as RSUs. The term "Road Side Unit" or "RSU" may refer to any transportation infrastructure entity used for V2X communications. An RSU may be implemented in or by a suitable RAN node or a stationary (or relatively stationary) UE, where an RSU implemented in or by a UE may be referred to as a "UE-type RSU," an RSU implemented in or by an eNB may be referred to as an "eNB-type RSU," an RSU implemented in or by a gNB may be referred to as a "gNB-type RSU," and the like. In one example, an RSU is a computing device coupled with radio frequency circuitry located on a roadside that provides connectivity support to passing vehicle UEs **1301** (VUEs **1301**). The RSU may also include internal data storage circuitry to store intersection map geometry, traffic statistics, media, as well as applications/software to sense and control ongoing vehicular and pedestrian traffic. The RSU may operate on the 5.9 GHz Direct Short Range Communications (DSRC) band to provide very low latency communications required for high speed events, such as crash avoidance, traffic warnings, and the like. Additionally or alternatively, the RSU may operate on the cellular V2X band to provide the aforementioned low latency communications, as well as other cellular communications services. Additionally or alternatively, the RSU may operate as a Wi-Fi hotspot (2.4 GHz band) and/or provide connectivity to one or more cellular networks to provide uplink and downlink communications. The computing device(s) and some or all of the radiofrequency circuitry of the RSU may be packaged in a weatherproof enclosure suitable for outdoor installation, and may include a network interface controller to provide a wired connection (e.g., Ethernet) to a traffic signal controller and/or a backhaul network.

Any of the RAN nodes **1311** can terminate the air interface protocol and can be the first point of contact for the UEs **1301**. In some embodiments, any of the RAN nodes **1311** can fulfill various logical functions for the RAN **1310** including, but not limited to, radio network controller (RNC) functions such as radio bearer management, uplink and downlink dynamic radio resource management and data packet scheduling, and mobility management.

In embodiments, the UEs **1301** can be configured to communicate using OFDM communication signals with each other or with any of the RAN nodes **1311** over a multicarrier communication channel in accordance with various communication techniques, such as, but not limited to, an OFDMA communication technique (e.g., for downlink communications) or a SC-FDMA communication technique (e.g., for uplink and ProSe or sidelink communications), although the scope of the embodiments is not limited in this respect. The OFDM signals can comprise a plurality of orthogonal subcarriers.

In some embodiments, a downlink resource grid can be used for downlink transmissions from any of the RAN nodes **1311** to the UEs **1301**, while uplink transmissions can utilize similar techniques. The grid can be a time-frequency grid, called a resource grid or time-frequency resource grid, which is the physical resource in the downlink in each slot. Such a time-frequency plane representation is a common practice for OFDM systems, which makes it intuitive for radio resource allocation. Each column and each row of the resource grid corresponds to one OFDM symbol and one OFDM subcarrier, respectively. The duration of the resource grid in the time domain corresponds to one slot in a radio frame. The smallest time-frequency unit in a resource grid is denoted as a resource element. Each resource grid comprises a number of resource blocks, which describe the mapping of certain physical channels to resource elements. Each resource block comprises a collection of resource elements; in the frequency domain, this may represent the smallest quantity of resources that currently can be allocated. There are several different physical downlink channels that are conveyed using such resource blocks.

According to various embodiments, the UEs **1301** and the RAN nodes **1311** communicate data (for example, transmit and receive) data over a licensed medium (also referred to as the "licensed spectrum" and/or the "licensed band") and an unlicensed shared medium (also referred to as the "unlicensed spectrum" and/or the "unlicensed band"). The licensed spectrum may include channels that operate in the frequency range of approximately 400 MHz to approximately 3.8 GHz, whereas the unlicensed spectrum may include the 5 GHz band.

To operate in the unlicensed spectrum, the UEs **1301** and the RAN nodes **1311** may operate using LAA, eLAA, and/or feLAA mechanisms. In these implementations, the UEs **1301** and the RAN nodes **1311** may perform one or more known medium-sensing operations and/or carrier-sensing operations in order to determine whether one or more channels in the unlicensed spectrum is unavailable or otherwise occupied prior to transmitting in the unlicensed spectrum. The medium/carrier sensing operations may be performed according to a listen-before-talk (LBT) protocol.

LBT is a mechanism whereby equipment (for example, UEs **1301**, RAN nodes **1311**, etc.) senses a medium (for example, a channel or carrier frequency) and transmits when the medium is sensed to be idle (or when a specific channel in the medium is sensed to be unoccupied). The medium sensing operation may include CCA, which utilizes at least ED to determine the presence or absence of other signals on a channel in order to determine if a channel is occupied or clear. This LBT mechanism allows cellular/LAA networks to coexist with incumbent systems in the unlicensed spectrum and with other LAA networks. ED may include sensing RF energy across an intended transmission band for a period of time and comparing the sensed RF energy to a predefined or configured threshold

Typically, the incumbent systems in the 5 GHz band are WLANs based on IEEE 802.11 technologies. WLAN employs a contention-based channel access mechanism, called CSMA/CA. Here, when a WLAN node (e.g., a mobile station (MS) such as UE **1301**, AP **1306**, or the like) intends to transmit, the WLAN node may first perform CCA before transmission. Additionally, a backoff mechanism is used to avoid collisions in situations where more than one WLAN node senses the channel as idle and transmits at the same time. The backoff mechanism may be a counter that is drawn randomly within the CWS, which is increased exponentially upon the occurrence of collision and reset to a minimum value when the transmission succeeds. The LBT mechanism designed for LAA is somewhat similar to the CSMA/CA of WLAN. In some implementations, the LBT procedure for DL or UL transmission bursts including PDSCH or PUSCH transmissions, respectively, may have an LAA contention window that is variable in length between X and Y ECCA slots, where X and Y are minimum and maximum values for the CWSs for LAA. In one example, the minimum CWS for an LAA transmission may be 9 microseconds (μs); however, the size of the CWS and a MCOT (for example, a transmission burst) may be based on governmental regulatory requirements.

The LAA mechanisms are built upon CA technologies of LTE-Advanced systems. In CA, each aggregated carrier is referred to as a CC. A CC may have a bandwidth of 1.4, 3, 5, 10, 15 or 20 MHz and a maximum of five CCs can be aggregated, and therefore, a maximum aggregated bandwidth is 100 MHz. In FDD systems, the number of aggregated carriers can be different for DL and UL, where the number of UL CCs is equal to or lower than the number of DL component carriers. In some cases, individual CCs can have a different bandwidth than other CCs. In TDD systems, the number of CCs as well as the bandwidths of each CC is usually the same for DL and UL.

CA also comprises individual serving cells to provide individual CCs. The coverage of the serving cells may differ, for example, because CCs on different frequency bands will experience different pathloss. A primary service cell or PCell may provide a PCC for both UL and DL, and may handle RRC and NAS related activities. The other serving cells are referred to as SCells, and each SCell may provide an individual SCC for both UL and DL. The SCCs may be added and removed as required, while changing the PCC may require the UE **1301** to undergo a handover. In LAA, eLAA, and feLAA, some or all of the SCells may operate in the unlicensed spectrum (referred to as "LAA SCells"), and the LAA SCells are assisted by a PCell operating in the licensed spectrum. When a UE is configured with more than one LAA SCell, the UE may receive UL grants on the configured LAA SCells indicating different PUSCH starting positions within a same subframe.

The PDSCH carries user data and higher-layer signaling to the UEs **1301**. The PDCCH carries information about the transport format and resource allocations related to the PDSCH channel, among other things. It may also inform the UEs **1301** about the transport format, resource allocation, and HARQ information related to the uplink shared channel. Typically, downlink scheduling (assigning control and shared channel resource blocks to the UE **1301***b* within a cell) may be performed at any of the RAN nodes **1311** based on channel quality information fed back from any of the UEs **1301**. The downlink resource assignment information may be sent on the PDCCH used for (e.g., assigned to) each of the UEs **1301**.

The PDCCH uses CCEs to convey the control information. Before being mapped to resource elements, the PDCCH complex-valued symbols may first be organized into quadruplets, which may then be permuted using a sub-block interleaver for rate matching. Each PDCCH may be transmitted using one or more of these CCEs, where each CCE may correspond to nine sets of four physical resource elements known as REGs. Four Quadrature Phase Shift Keying (QPSK) symbols may be mapped to each REG. The PDCCH can be transmitted using one or more CCEs, depending on the size of the DCI and the channel condition. There can be four or more different PDCCH formats defined in LTE with different numbers of CCEs (e.g., aggregation level, L=1, 2, 4, or 8).

Some embodiments may use concepts for resource allocation for control channel information that are an extension of the above-described concepts. For example, some embodiments may utilize an EPDCCH that uses PDSCH resources for control information transmission. The EPDCCH may be transmitted using one or more ECCEs. Similar to above, each ECCE may correspond to nine sets of four physical resource elements known as an EREGs. An ECCE may have other numbers of EREGs in some situations.

The RAN nodes **1311** may be configured to communicate with one another via interface **1312**. In embodiments where the system **1300** is an LTE system (e.g., when CN **1320** is an EPC **1420** as in FIG. **14**), the interface **1312** may be an X2 interface **1312**. The X2 interface may be defined between two or more RAN nodes **1311** (e.g., two or more eNBs and the like) that connect to EPC **1320**, and/or between two eNBs connecting to EPC **1320**. In some implementations, the X2 interface may include an X2 user plane interface (X2-U) and an X2 control plane interface (X2-C). The X2-U may provide flow control mechanisms for user data packets transferred over the X2 interface, and may be used to communicate information about the delivery of user data between eNBs. For example, the X2-U may provide specific sequence number information for user data transferred from a MeNB to an SeNB; information about successful in sequence delivery of PDCP PDUs to a UE **1301** from an SeNB for user data; information of PDCP PDUs that were not delivered to a UE **1301**; information about a current minimum desired buffer size at the SeNB for transmitting to the UE user data; and the like. The X2-C may provide intra-LTE access mobility functionality, including context transfers from source to target eNBs, user plane transport control, etc.; load management functionality; as well as inter-cell interference coordination functionality.

In embodiments where the system **1300** is a 5G or NR system (e.g., when CN **1320** is an 5GC **1520** as in FIG. **15**), the interface **1312** may be an Xn interface **1312**. The Xn interface is defined between two or more RAN nodes **1311** (e.g., two or more gNBs and the like) that connect to 5GC **1320**, between a RAN node **1311** (e.g., a gNB) connecting to 5GC **1320** and an eNB, and/or between two eNBs connecting to 5GC **1320**. In some implementations, the Xn interface may include an Xn user plane (Xn-U) interface and an Xn control plane (Xn-C) interface. The Xn-U may provide non-guaranteed delivery of user plane PDUs and support/provide data forwarding and flow control functionality. The Xn-C may provide management and error handling functionality, functionality to manage the Xn-C interface; mobility support for UE **1301** in a connected mode (e.g., CM-CONNECTED) including functionality to manage the UE mobility for connected mode between one or more RAN nodes **1311**. The mobility support may include context transfer from an old (source) serving RAN node **1311** to new (target) serving RAN node **1311**; and control of user plane tunnels between old (source) serving RAN node **1311** to new (target) serving RAN node **1311**. A protocol stack of the Xn-U may include a transport network layer built on Internet Protocol (IP) transport layer, and a GTP-U layer on top of a UDP and/or IP layer(s) to carry user plane PDUs. The Xn-C protocol stack may include an application layer signaling protocol (referred to as Xn Application Protocol (Xn-AP)) and a transport network layer that is built on SCTP. The SCTP may be on top of an IP layer, and may provide the guaranteed delivery of application layer messages. In the transport IP layer, point-to-point transmission is used to deliver the signaling PDUs. In other implementations, the Xn-U protocol stack and/or the Xn-C protocol stack may be same or similar to the user plane and/or control plane protocol stack(s) shown and described herein.

The RAN **1310** is shown to be communicatively coupled to a core network—in this embodiment, core network (CN) **1320**. The CN **1320** may comprise a plurality of network elements **1322**, which are configured to offer various data and telecommunications services to customers/subscribers (e.g., users of UEs **1301**) who are connected to the CN **1320** via the RAN **1310**. The components of the CN **1320** may be implemented in one physical node or separate physical nodes including components to read and execute instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium). In some embodiments, NFV may be utilized to virtualize any or all of the above-described network node functions via executable instructions stored in one or more computer-readable storage mediums (described in further detail below). A logical instantiation of the CN **1320** may be referred to as a network slice, and a logical instantiation of a portion of the CN **1320** may be referred to as a network sub-slice. NFV architectures and infrastructures may be used to virtualize one or more network functions, alternatively performed by proprietary hardware, onto physical resources comprising a combination of industry-standard server hardware, storage hardware, or switches. In other words, NFV systems can be used to execute virtual or reconfigurable implementations of one or more EPC components/functions.

Generally, the application server **1330** may be an element offering applications that use IP bearer resources with the core network (e.g., UMTS PS domain, LTE PS data services, etc.). The application server **1330** can also be configured to support one or more communication services (e.g., VoIP sessions, PTT sessions, group communication sessions, social networking services, etc.) for the UEs **1301** via the EPC **1320**.

In embodiments, the CN **1320** may be a 5GC (referred to as "5GC **1320**" or the like), and the RAN **1310** may be connected with the CN **1320** via an NG interface **1313**. In embodiments, the NG interface **1313** may be split into two parts, an NG user plane (NG-U) interface **1314**, which carries traffic data between the RAN nodes **1311** and a UPF, and the S1 control plane (NG-C) interface **1315**, which is a signaling interface between the RAN nodes **1311** and AMFs. Embodiments where the CN **1320** is a 5GC **1320** are discussed in more detail with regard to FIG. **15**.

In embodiments, the CN **1320** may be a 5G CN(referred to as "5GC **1320**" or the like), while in other embodiments, the CN **1320** may be an EPC). Where CN **1320** is an EPC (referred to as "EPC **1320**" or the like), the RAN **1310** may be connected with the CN **1320** via an S1 interface **1313**. In embodiments, the S1 interface **1313** may be split into two parts, an S1 user plane (S1-U) interface **1314**, which carries traffic data between the RAN nodes **1311** and the S-GW, and the S1-MME interface **1315**, which is a signaling interface between the RAN nodes **1311** and MMEs. An example architecture wherein the CN **1320** is an EPC **1320** is shown by FIG. **14**.

FIG. **14** illustrates an example architecture of a system **1400** including a first CN **1420**, in accordance with various embodiments. In this example, system **1400** may implement the LTE standard wherein the CN **1420** is an EPC **1420** that corresponds with CN **1320** of FIG. **13**. Additionally, the UE **1401** may be the same or similar as the UEs **1301** of FIG. **13**, and the E-UTRAN **1410** may be a RAN that is the same or similar to the RAN **1310** of FIG. **13**, and which may include RAN nodes **1311** discussed previously. The CN **1420** may comprise MMEs **1421**, an S-GW **1422**, a P-GW **1423**, a HSS **1424**, and a SGSN **1425**.

The MMEs **1421** may be similar in function to the control plane of legacy SGSN, and may implement MM functions to keep track of the current location of a UE **1401**.

The MMEs **1421** may perform various MM procedures to manage mobility aspects in access such as gateway selection and tracking area list management. MM (also referred to as "EPS MM" or "EMM" in E-UTRAN systems) may refer to all applicable procedures, methods, data storage, etc. that are used to maintain knowledge about a present location of the UE **1401**, provide user identity confidentiality, and/or perform other like services to users/subscribers. Each UE **1401** and the MME **1421** may include an MM or EMM sublayer, and an MM context may be established in the UE **1401** and the MME **1421** when an attach procedure is successfully completed. The MM context may be a data structure or database object that stores MM-related information of the UE **1401**. The MMEs **1421** may be coupled with the HSS **1424** via an S6a reference point, coupled with the SGSN **1425** via an S3 reference point, and coupled with the S-GW **1422** via an S11 reference point.

The SGSN **1425** may be a node that serves the UE **1401** by tracking the location of an individual UE **1401** and performing security functions. In addition, the SGSN **1425** may perform Inter-EPC node signaling for mobility between 2G/3G and E-UTRAN 3GPP access networks; PDN and S-GW selection as specified by the MMEs **1421**; handling of UE **1401** time zone functions as specified by the MMEs **1421**; and MME selection for handovers to E-UTRAN 3GPP access network. The S3 reference point between the MMEs **1421** and the SGSN **1425** may enable user and bearer information exchange for inter-3GPP access network mobility in idle and/or active states.

The HSS **1424** may comprise a database for network users, including subscription-related information to support the network entities' handling of communication sessions. The EPC **1420** may comprise one or several HSSs **1424**, depending on the number of mobile subscribers, on the capacity of the equipment, on the organization of the network, etc. For example, the HSS **1424** can provide support for routing/roaming, authentication, authorization, naming/addressing resolution, location dependencies, etc. An S6a reference point between the HSS **1424** and the MMEs **1421** may enable transfer of subscription and authentication data for authenticating/authorizing user access to the EPC **1420** between HSS **1424** and the MMEs **1421**.

The S-GW **1422** may terminate the S1 interface **1313** ("S1-U" in FIG. **14**) toward the RAN **1410**, and routes data packets between the RAN **1410** and the EPC **1420**. In addition, the S-GW **1422** may be a local mobility anchor point for inter-RAN node handovers and also may provide an anchor for inter-3GPP mobility. Other responsibilities may include lawful intercept, charging, and some policy enforcement. The S11 reference point between the S-GW **1422** and the MMEs **1421** may provide a control plane between the MMEs **1421** and the S-GW **1422**. The S-GW **1422** may be coupled with the P-GW **1423** via an S5 reference point.

The P-GW **1423** may terminate an SGi interface toward a PDN **1430**. The P-GW **1423** may route data packets between the EPC **1420** and external networks such as a network including the application server **1330** (alternatively referred to as an "AF") via an IP interface **1325** (see e.g., FIG. **13**). In embodiments, the P-GW **1423** may be communicatively coupled to an application server (application server **1330** of FIG. **13** or PDN **1430** in FIG. **14**) via an IP communications interface **1325** (see, e.g., FIG. **13**). The S5 reference point between the P-GW **1423** and the S-GW **1422** may provide user plane tunneling and tunnel management between the P-GW **1423** and the S-GW **1422**. The S5 reference point may also be used for S-GW **1422** relocation due to UE **1401** mobility and if the S-GW **1422** needs to connect to a non-collocated P-GW **1423** for the required PDN connectivity. The P-GW **1423** may further include a node for policy enforcement and charging data collection (e.g., PCEF (not shown)). Additionally, the SGi reference point between the P-GW **1423** and the packet data network (PDN) **1430** may be an operator external public, a private PDN, or an intra operator packet data network, for example, for provision of IMS services. The P-GW **1423** may be coupled with a PCRF **1426** via a Gx reference point.

PCRF **1426** is the policy and charging control element of the EPC **1420**. In a non-roaming scenario, there may be a single PCRF **1426** in the Home Public Land Mobile Network (HPLMN) associated with a UE **1401**'s Internet Protocol Connectivity Access Network (IP-CAN) session. In a roaming scenario with local breakout of traffic, there may be two PCRFs associated with a UE **1401**'s IP-CAN session, a Home PCRF (H-PCRF) within an HPLMN and a Visited PCRF (V-PCRF) within a Visited Public Land Mobile Network (VPLMN). The PCRF **1426** may be communicatively coupled to the application server **1430** via the P-GW **1423**. The application server **1430** may signal the PCRF **1426** to indicate a new service flow and select the appropriate QoS and charging parameters. The PCRF **1426** may provision this rule into a PCEF (not shown) with the appropriate TFT and QCI, which commences the QoS and charging as specified by the application server **1430**. The Gx reference point between the PCRF **1426** and the P-GW **1423** may allow for the transfer of QoS policy and charging rules from the PCRF **1426** to PCEF in the P-GW **1423**. An Rx reference point may reside between the PDN **1430** (or "AF **1430**") and the PCRF **1426**.

FIG. **15** illustrates an architecture of a system **1500** including a second CN **1520** in accordance with various embodiments. The system **1500** is shown to include a UE **1501**, which may be the same or similar to the UEs **1301** and UE **1401** discussed previously; a (R) AN **1510**, which may be the same or similar to the RAN **1310** and RAN **1410** discussed previously, and which may include RAN nodes **1311** discussed previously; and a DN **1503**, which may be, for example, operator services, Internet access or 3rd party services; and a 5GC **1520**. The 5GC **1520** may include an AUSF **1522**; an AMF **1521**; a SMF **1524**; a NEF **1523**; a PCF **1526**; a NRF **1525**; a UDM **1527**; an AF **1528**; a UPF **1502**; and a NSSF **1529**.

The UPF **1502** may act as an anchor point for intra-RAT and inter-RAT mobility, an external PDU session point of interconnect to DN **1503**, and a branching point to support multi-homed PDU session. The UPF **1502** may also perform packet routing and forwarding, perform packet inspection, enforce the user plane part of policy rules, lawfully intercept packets (UP collection), perform traffic usage reporting, perform QoS handling for a user plane (e.g., packet filtering, gating, UL/DL rate enforcement), perform Uplink Traffic verification (e.g., SDF to QoS flow mapping), transport level packet marking in the uplink and downlink, and perform downlink packet buffering and downlink data notification triggering. UPF **1502** may include an uplink classifier to support routing traffic flows to a data network. The DN **1503** may represent various network operator services, Internet access, or third party services. DN **1503** may include, or be similar to, application server **1330** discussed previously. The UPF **1502** may interact with the SMF **1524** via an N4 reference point between the SMF **1524** and the UPF **1502**.

The AUSF **1522** may store data for authentication of UE **1501** and handle authentication-related functionality. The AUSF **1522** may facilitate a common authentication framework for various access types. The AUSF **1522** may communicate with the AMF **1521** via an N12 reference point between the AMF **1521** and the AUSF **1522**; and may communicate with the UDM **1527** via an N13 reference point between the UDM **1527** and the AUSF **1522**. Additionally, the AUSF **1522** may exhibit an Nausf service-based interface.

The AMF **1521** may be responsible for registration management (e.g., for registering UE **1501**, etc.), connection management, reachability management, mobility management, and lawful interception of AMF-related events, and access authentication and authorization. The AMF **1521** may be a termination point for the an N11 reference point between the AMF **1521** and the SMF **1524**. The AMF **1521** may provide transport for SM messages between the UE **1501** and the SMF **1524**, and act as a transparent pro21 for routing SM messages. AMF **1521** may also provide transport for SMS messages between UE **1501** and an SMSF (not shown by FIG. **15**). AMF **1521** may act as SEAF, which may include interaction with the AUSF **1522** and the UE **1501**, receipt of an intermediate key that was established as a result of the UE **1501** authentication process. Where USIM based authentication is used, the AMF **1521** may retrieve the security material from the AUSF **1522**. AMF **1521** may also include a SCM function, which receives a key from the SEA that it uses to derive access-network specific keys. Furthermore, AMF **1521** may be a termination point of a RAN CP interface, which may include or be an N2 reference point between the (R) AN **1510** and the AMF **1521**; and the AMF **1521** may be a termination point of NAS (N1) signalling, and perform NAS ciphering and integrity protection.

AMF **1521** may also support NAS signalling with a UE **1501** over an N3 IWF interface. The N3IWF may be used to provide access to untrusted entities. N3IWF may be a termination point for the N2 interface between the (R) AN **1510** and the AMF **1521** for the control plane, and may be a termination point for the N3 reference point between the (R) AN **1510** and the UPF **1502** for the user plane. As such, the AMF **1521** may handle N2 signalling from the SMF **1524** and the AMF **1521** for PDU sessions and QoS, encapsulate/de-encapsulate packets for IPSec and N3 tunnelling, mark N3 user-plane packets in the uplink, and enforce QoS corresponding to N3 packet marking taking into account QoS requirements associated with such marking received over N2. N3IWF may also relay uplink and downlink control-plane NAS signalling between the UE **1501** and AMF **1521** via an N1 reference point between the UE **1501** and the AMF **1521**, and relay uplink and downlink user-plane packets between the UE **1501** and UPF **1502**. The N3IWF also provides mechanisms for IPsec tunnel establishment with the UE **1501**. The AMF **1521** may exhibit an Namf service-based interface, and may be a termination point for an N14 reference point between two AMFs **1521** and an N17 reference point between the AMF **1521** and a 5G-EIR (not shown by FIG. **15**).

The UE **1501** may need to register with the AMF **1521** in order to receive network services. RM is used to register or deregister the UE **1501** with the network (e.g., AMF **1521**), and establish a UE context in the network (e.g., AMF **1521**). The UE **1501** may operate in an RM-REGISTERED state or an RM-DEREGISTERED state. In the RM-DEREGISTERED state, the UE **1501** is not registered with the network, and the UE context in AMF **1521** holds no valid location or routing information for the UE **1501** so the UE **1501** is not reachable by the AMF **1521**. In the RM-REGISTERED state, the UE **1501** is registered with the network, and the UE context in AMF **1521** may hold a valid location or routing information for the UE **1501** so the UE **1501** is reachable by the AMF **1521**. In the RM-REGISTERED state, the UE **1501** may perform mobility Registration Update procedures, perform periodic Registration Update procedures triggered by expiration of the periodic update timer (e.g., to notify the network that the UE **1501** is still active), and perform a Registration Update procedure to update UE capability information or to re-negotiate protocol parameters with the network, among others.

The AMF **1521** may store one or more RM contexts for the UE **1501**, where each RM context is associated with a specific access to the network. The RM context may be a data structure, database object, etc. that indicates or stores, inter alia, a registration state per access type and the periodic update timer. The AMF **1521** may also store a 5GC MM context that may be the same or similar to the (E) MM context discussed previously. In various embodiments, the AMF **1521** may store a CE mode B Restriction parameter of the UE **1501** in an associated MM context or RM context. The AMF **1521** may also derive the value, when needed, from the UE's usage setting parameter already stored in the UE context (and/or MM/RM context).

CM may be used to establish and release a signaling connection between the UE **1501** and the AMF **1521** over the N1 interface. The signaling connection is used to enable NAS signaling exchange between the UE **1501** and the CN **1520**, and comprises both the signaling connection between the UE and the AN(e.g., RRC connection or UE-N3IWF connection for non-3GPP access) and the N2 connection for the UE **1501** between the AN (e.g., RAN **1510**) and the AMF **1521**. The UE **1501** may operate in one of two CM states, CM-IDLE mode or CM-CONNECTED mode. When the UE **1501** is operating in the CM-IDLE state/mode, the UE **1501** may have no NAS signaling connection established with the AMF **1521** over the N1 interface, and there may be (R) AN **1510** signaling connection (e.g., N2 and/or N3 connections) for the UE **1501**. When the UE **1501** is operating in the CM-CONNECTED state/mode, the UE **1501** may have an established NAS signaling connection with the AMF **1521** over the N1 interface, and there may be a (R) AN **1510** signaling connection (e.g., N2 and/or N3 connections) for the UE **1501**. Establishment of an N2 connection between the (R) AN **1510** and the AMF **1521** may cause the UE **1501** to transition from CM-IDLE mode to CM-CONNECTED mode, and the UE **1501** may transition from the CM-CONNECTED mode to the CM-IDLE mode when N2 signaling between the (R) AN **1510** and the AMF **1521** is released.

The SMF **1524** may be responsible for SM (e.g., session establishment, modify and release, including tunnel maintain between UPF and AN node); UE IP address allocation and management (including optional authorization); selection and control of UP function; configuring traffic steering at UPF to route traffic to proper destination; termination of interfaces toward policy control functions; controlling part of policy enforcement and QoS; lawful intercept (for SM events and interface to LI system); termination of SM parts of NAS messages; downlink data notification; initiating AN specific SM information, sent via AMF over N2 to AN; and determining SSC mode of a session. SM may refer to management of a PDU session, and a PDU session or "session" may refer to a PDU connectivity service that provides or enables the exchange of PDUs between a UE **1501** and a data network (DN) **1503** identified by a Data Network Name (DNN). PDU sessions may be established upon UE **1501** request, modified upon UE **1501** and 5GC **1520** request, and released upon UE **1501** and 5GC **1520** request using NAS SM signaling exchanged over the N1 reference point between the UE **1501** and the SMF **1524**. Upon request from an application server, the 5GC **1520** may trigger a specific application in the UE **1501**. In response to receipt of the trigger message, the UE **1501** may pass the trigger message (or relevant parts/information of the trigger message) to one or more identified applications in the UE **1501**. The identified application(s) in the UE **1501** may establish a PDU session to a specific DNN. The SMF **1524** may check whether the UE **1501** requests are compliant with user subscription information associated with the UE **1501**. In this regard, the SMF **1524** may retrieve and/or request to receive update notifications on SMF **1524** level subscription data from the UDM **1527**.

The SMF **1524** may include the following roaming functionality: handling local enforcement to apply QOS SLAs (VPLMN); charging data collection and charging interface (VPLMN); lawful intercept (in VPLMN for SM events and interface to L1 system); and support for interaction with external DN for transport of signalling for PDU session authorization/authentication by external DN. An N16 reference point between two SMFs **1524** may be included in the system **1500**, which may be between another SMF **1524** in a visited network and the SMF **1524** in the home network in roaming scenarios. Additionally, the SMF **1524** may exhibit the Nsmf service-based interface.

The NEF **1523** may provide means for securely exposing the services and capabilities provided by 3GPP network functions for third party, internal exposure/re-exposure, Application Functions (e.g., AF **1528**), edge computing or fog computing systems, etc. In such embodiments, the NEF **1523** may authenticate, authorize, and/or throttle the AFs. NEF **1523** may also translate information exchanged with the AF **1528** and information exchanged with internal network functions. For example, the NEF **1523** may translate between an AF-Service-Identifier and an internal 5GC information. NEF **1523** may also receive information from other network functions (NFs) based on exposed capabilities of other network functions. This information may be stored at the NEF **1523** as structured data, or at a data storage NF using standardized interfaces. The stored information can then be re-exposed by the NEF **1523** to other NFs and AFs, and/or used for other purposes such as analytics. Additionally, the NEF **1523** may exhibit an Nnef service-based interface.

The NRF **1525** may support service discovery functions, receive NF discovery requests from NF instances, and provide the information of the discovered NF instances to the NF instances. NRF **1525** also maintains information of available NF instances and their supported services. As used herein, the terms "instantiate," "instantiation," and the like may refer to the creation of an instance, and an "instance" may refer to a concrete occurrence of an object, which may occur, for example, during execution of program code. Additionally, the NRF **1525** may exhibit the Nnrf service-based interface.

The PCF **1526** may provide policy rules to control plane function(s) to enforce them, and may also support unified policy framework to govern network behaviour. The PCF **1526** may also implement an FE to access subscription information relevant for policy decisions in a UDR of the UDM **1527**. The PCF **1526** may communicate with the AMF **1521** via an N15 reference point between the PCF **1526** and the AMF **1521**, which may include a PCF **1526** in a visited network and the AMF **1521** in case of roaming scenarios. The PCF **1526** may communicate with the AF **1528** via an N5 reference point between the PCF **1526** and the AF **1528**; and with the SMF **1524** via an N7 reference point between the PCF **1526** and the SMF **1524**. The system **1500** and/or CN **1520** may also include an N24 reference point between the PCF **1526** (in the home network) and a PCF **1526** in a visited network. Additionally, the PCF **1526** may exhibit an Npcf service-based interface.

The UDM **1527** may handle subscription-related information to support the network entities' handling of communication sessions, and may store subscription data of UE **1501**. For example, subscription data may be communicated between the UDM **1527** and the AMF **1521** via an N8 reference point between the UDM **1527** and the AMF. The UDM **1527** may include two parts, an application FE and a UDR (the FE and UDR are not shown by FIG. **15**). The UDR may store subscription data and policy data for the UDM **1527** and the PCF **1526**, and/or structured data for exposure and application data (including PFDs for application detection, application request information for multiple UEs **1501**) for the NEF **1523**. The Nudr service-based interface may be exhibited by the UDR **221** to allow the UDM **1527**, PCF **1526**, and NEF **1523** to access a particular set of the stored data, as well as to read, update (e.g., add, modify), delete, and subscribe to notification of relevant data changes in the UDR. The UDM may include a UDM-FE, which is in charge of processing credentials, location management, subscription management and so on. Several different front ends may serve the same user in different transactions. The UDM-FE accesses subscription information stored in the UDR and performs authentication credential processing, user identification handling, access authorization, registration/mobility management, and subscription management. The UDR may interact with the SMF **1524** via an N10 reference point between the UDM **1527** and the SMF **1524**. UDM **1527** may also support SMS management, wherein an SMS-FE implements the similar application logic as discussed previously. Additionally, the UDM **1527** may exhibit the Nudm service-based interface.

The AF **1528** may provide application influence on traffic routing, provide access to the NCE, and interact with the policy framework for policy control. The NCE may be a mechanism that allows the 5GC **1520** and AF **1528** to provide information to each other via NEF **1523**, which may be used for edge computing implementations. In such implementations, the network operator and third party services may be hosted close to the UE **1501** access point of attachment to achieve an efficient service delivery through the reduced end-to-end latency and load on the transport network. For edge computing implementations, the 5GC may select a UPF **1502** close to the UE **1501** and execute traffic steering from the UPF **1502** to DN **1503** via the N6 interface. This may be based on the UE subscription data, UE location, and information provided by the AF **1528**. In this way, the AF **1528** may influence UPF (re) selection and traffic routing. Based on operator deployment, when AF **1528** is considered to be a trusted entity, the network operator may permit AF **1528** to interact directly with relevant NFs. Additionally, the AF **1528** may exhibit an Naf service-based interface.

The NSSF **1529** may select a set of network slice instances serving the UE **1501**. The NSSF **1529** may also determine allowed NSSAI and the mapping to the subscribed S-NSSAIs, if needed. The NSSF **1529** may also determine the AMF set to be used to serve the UE **1501**, or a list of candidate AMF(s) **1521** based on a suitable configuration and possibly by querying the NRF **1525**. The selection of a set of network slice instances for the UE **1501** may be triggered by the AMF **1521** with which the UE **1501** is registered by interacting with the NSSF **1529**, which may lead to a change of AMF **1521**. The NSSF **1529** may interact with the AMF **1521** via an N22 reference point between AMF **1521** and NSSF **1529**; and may communicate with another NSSF **1529** in a visited network via an N31 reference point (not shown by FIG. **15**). Additionally, the NSSF **1529** may exhibit an Nnssf service-based interface.

As discussed previously, the CN **1520** may include an SMSF, which may be responsible for SMS subscription checking and verification, and relaying SM messages to/from the UE **1501** to/from other entities, such as an SMS-GMSC/IWMSC/SMS-router. The SMS may also interact with AMF **1521** and UDM **1527** for a notification procedure that the UE **1501** is available for SMS transfer (e.g., set a UE not reachable flag, and notifying UDM **1527** when UE **1501** is available for SMS).

The CN **120** may also include other elements that are not shown by FIG. **15**, such as a Data Storage system/architecture, a 5G-EIR, a SEPP, and the like. The Data Storage system may include a SDSF, an UDSF, and/or the like. Any NF may store and retrieve unstructured data into/from the UDSF (e.g., UE contexts), via N18 reference point between any NF and the UDSF (not shown by FIG. **15**). Individual NFs may share a UDSF for storing their respective unstructured data or individual NFs may each have their own UDSF located at or near the individual NFs. Additionally, the UDSF may exhibit an Nudsf service-based interface (not shown by FIG. **15**). The 5G-EIR may be an NF that checks the status of PEI for determining whether particular equipment/entities are blacklisted from the network; and the SEPP may be a non-transparent pro21 that performs topology hiding, message filtering, and policing on inter-PLMN control plane interfaces.

Additionally, there may be many more reference points and/or service-based interfaces between the NF services in the NFs; however, these interfaces and reference points have been omitted from FIG. **15** for clarity. In one example, the CN **1520** may include an Nx interface, which is an inter-CN interface between the MME (e.g., MME **1421**) and the AMF **1521** in order to enable interworking between CN **1520** and CN **1420**. Other example interfaces/reference points may include an N5g-EIR service-based interface exhibited by a 5G-EIR, an N27 reference point between the NRF in the visited network and the NRF in the home network; and an N31 reference point between the NSSF in the visited network and the NSSF in the home network.

FIG. **16** illustrates an example of infrastructure equipment **1600** in accordance with various embodiments. The infrastructure equipment **1600** (or "system **1600**") may be implemented as a base station, radio head, RAN node such as the RAN nodes **1311** and/or AP **1306** shown and described previously, application server(s) **1330**, and/or any other element/device discussed herein. In other examples, the system **1600** could be implemented in or by a UE.

The system **1600** includes application circuitry **1605**, baseband circuitry **1610**, one or more radio front end modules (RFEMs) **1615**, memory circuitry **1620**, power management integrated circuitry (PMIC) **1625**, power tee circuitry **1630**, network controller circuitry **1635**, network interface connector **1640**, satellite positioning circuitry **1645**, and user interface **1650**. In some embodiments, the device **1600** may include additional elements such as, for example, memory/storage, display, camera, sensor, or input/output (I/O) interface. In other embodiments, the components described below may be included in more than one device. For example, said circuitries may be separately included in more than one device for CRAN, vBBU, or other like implementations.

Application circuitry **1605** includes circuitry such as, but not limited to one or more processors (or processor cores), cache memory, and one or more of low drop-out voltage regulators (LDOs), interrupt controllers, serial interfaces such as SPI, I2C or universal programmable serial interface module, real time clock (RTC), timer-counters including interval and watchdog timers, general purpose input/output (I/O or IO), memory card controllers such as Secure Digital (SD) MultiMediaCard (MMC) or similar, Universal Serial Bus (USB) interfaces, Mobile Industry Processor Interface (MIPI) interfaces and Joint Test Access Group (JTAG) test access ports. The processors (or cores) of the application circuitry **1605** may be coupled with or may include memory/storage elements and may be configured to execute instructions stored in the memory/storage to enable various applications or operating systems to run on the system **1600**. In some implementations, the memory/storage elements may be on-chip memory circuitry, which may include any suitable volatile and/or non-volatile memory, such as DRAM, SRAM, EPROM, EEPROM, Flash memory, solid-state memory, and/or any other type of memory device technology, such as those discussed herein.

The processor(s) of application circuitry **1605** may include, for example, one or more processor cores (CPUs), one or more application processors, one or more graphics processing units (GPUs), one or more reduced instruction set computing (RISC) processors, one or more Acorn RISC Machine (ARM) processors, one or more complex instruction set computing (CISC) processors, one or more digital signal processors (DSP), one or more FPGAs, one or more PLDs, one or more ASICs, one or more microprocessors or controllers, or any suitable combination thereof. In some embodiments, the application circuitry **1605** may comprise, or may be, a special-purpose processor/controller to operate according to the various embodiments herein. As examples, the processor(s) of application circuitry **1605** may include one or more Intel Pentium®, Core®, or Xeon® processor(s); Advanced Micro Devices (AMD) Ryzen® processor(s), Accelerated Processing Units (APUs), or Epyc® processors; ARM-based processor(s) licensed from ARM Holdings, Ltd. such as the ARM Cortex-A family of processors and the ThunderX2® provided by Cavium™, Inc.; a MIPS-based design from MIPS Technologies, Inc. such as MIPS Warrior P-class processors; and/or the like. In some embodiments, the system **1600** may not utilize application circuitry **1605**, and instead may include a special-purpose processor/controller to process IP data received from an EPC or 5GC, for example.

In some implementations, the application circuitry **1605** may include one or more hardware accelerators, which may be microprocessors, programmable processing devices, or the like. The one or more hardware accelerators may include, for example, computer vision (CV) and/or deep learning (DL) accelerators. As examples, the programmable processing devices may be one or more a field-programmable devices (FPDs) such as field-programmable gate arrays (FPGAs) and the like; programmable logic devices (PLDs) such as complex PLDs (CPLDs), high-capacity PLDs (HCPLDs), and the like; ASICs such as structured ASICs and the like; programmable SoCs (PSoCs); and the like. In such implementations, the circuitry of application circuitry **1605** may comprise logic blocks or logic fabric, and other interconnected resources that may be programmed to perform various functions, such as the procedures, methods, functions, etc. of the various embodiments discussed herein. In such embodiments, the circuitry of application circuitry **1605** may include memory cells (e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), flash memory, static memory (e.g., static random access memory (SRAM), anti-fuses, etc.)) used to store logic blocks, logic fabric, data, etc. in look-up-tables (LUTs) and the like.

The baseband circuitry **1610** may be implemented, for example, as a solder-down substrate including one or more integrated circuits, a single packaged integrated circuit soldered to a main circuit board or a multi-chip module containing two or more integrated circuits. The various hardware electronic elements of baseband circuitry **1610** are discussed infra with regard to FIG. **18**.

User interface circuitry **1650** may include one or more user interfaces designed to enable user interaction with the system **1600** or peripheral component interfaces designed to enable peripheral component interaction with the system **1600**. User interfaces may include, but are not limited to, one or more physical or virtual buttons (e.g., a reset button), one or more indicators (e.g., light emitting diodes (LEDs)), a physical keyboard or keypad, a mouse, a touchpad, a touchscreen, speakers or other audio emitting devices, microphones, a printer, a scanner, a headset, a display screen or display device, etc. Peripheral component interfaces may include, but are not limited to, a nonvolatile memory port, a universal serial bus (USB) port, an audio jack, a power supply interface, etc.

The radio front end modules (RFEMs) **1615** may comprise a millimeter wave (mmWave) RFEM and one or more sub-mmWave radio frequency integrated circuits (RFICs). In some implementations, the one or more sub-mmWave RFICs may be physically separated from the mmWave RFEM. The RFICs may include connections to one or more antennas or antenna arrays (see e.g., antenna array **1811** of FIG. **18** infra), and the RFEM may be connected to multiple antennas. In alternative implementations, both mm Wave and sub-mm Wave radio functions may be implemented in the same physical RFEM **1615**, which incorporates both mmWave antennas and sub-mmWave.

The memory circuitry **1620** may include one or more of volatile memory including dynamic random access memory (DRAM) and/or synchronous dynamic random access memory (SDRAM), and nonvolatile memory (NVM) including high-speed electrically erasable memory (commonly referred to as Flash memory), phase change random access memory (PRAM), magnetoresistive random access memory (MRAM), etc., and may incorporate the three-dimensional (3D) cross-point (XPOINT) memories from Intel® and Micron®. Memory circuitry **1620** may be implemented as one or more of solder down packaged integrated circuits, socketed memory modules and plug-in memory cards.

The PMIC **1625** may include voltage regulators, surge protectors, power alarm detection circuitry, and one or more backup power sources such as a battery or capacitor. The power alarm detection circuitry may detect one or more of brown out (under-voltage) and surge (over-voltage) conditions. The power tee circuitry **1630** may provide for electrical power drawn from a network cable to provide both power supply and data connectivity to the infrastructure equipment **1600** using a single cable.

The network controller circuitry **1635** may provide connectivity to a network using a standard network interface protocol such as Ethernet, Ethernet over GRE Tunnels, Ethernet over Multiprotocol Label Switching (MPLS), or some other suitable protocol. Network connectivity may be provided to/from the infrastructure equipment **1600** via network interface connector **1640** using a physical connection, which may be electrical (commonly referred to as a "copper interconnect"), optical, or wireless. The network controller circuitry **1635** may include one or more dedicated processors and/or FPGAs to communicate using one or more of the aforementioned protocols. In some implementations, the network controller circuitry **1635** may include multiple controllers to provide connectivity to other networks using the same or different protocols.

The positioning circuitry **1645** includes circuitry to receive and decode signals transmitted/broadcasted by a positioning network of a global navigation satellite system (GNSS). Examples of navigation satellite constellations (or GNSS) include United States' Global Positioning System (GPS), Russia's Global Navigation System (GLONASS), the European Union's Galileo system, China's BeiDou Navigation Satellite System, a regional navigation system or GNSS augmentation system (e.g., Navigation with Indian Constellation (NAVIC), Japan's Quasi-Zenith Satellite System (QZSS), France's Doppler Orbitography and Radio-positioning Integrated by Satellite (DORIS), etc.), or the like. The positioning circuitry **1645** comprises various hardware elements (e.g., including hardware devices such as switches, filters, amplifiers, antenna elements, and the like to facilitate OTA communications) to communicate with components of a positioning network, such as navigation satellite constellation nodes. In some embodiments, the positioning circuitry **1645** may include a Micro-Technology for Positioning, Navigation, and Timing (Micro-PNT) IC that uses a master timing clock to perform position tracking/estimation without GNSS assistance. The positioning circuitry **1645** may also be part of, or interact with, the baseband circuitry **1610** and/or RFEMs **1615** to communicate with the nodes and components of the positioning network. The positioning circuitry **1645** may also provide position data and/or time data to the application circuitry **1605**, which may use the data to synchronize operations with various infrastructure (e.g., RAN nodes **1311**, etc.), or the like.

The components shown by FIG. **16** may communicate with one another using interface circuitry, which may include any number of bus and/or interconnect (IX) technologies such as industry standard architecture (ISA), extended ISA (EISA), peripheral component interconnect (PCI), peripheral component interconnect extended (PCIx), PCI express (PCIe), or any number of other technologies. The bus/IX may be a proprietary bus, for example, used in a SoC based system. Other bus/IX systems may be included, such as an I2C interface, an SPI interface, point to point interfaces, and a power bus, among others.

FIG. **17** illustrates an example of a platform **1700** (or "device **1700**") in accordance with various embodiments. In embodiments, the computer platform **1700** may be suitable for use as UEs **1301**, **1401**, **1502**, application servers **1330**, and/or any other element/device discussed herein. The platform **1700** may include any combinations of the components shown in the example. The components of platform **1700** may be implemented as integrated circuits (ICs), portions thereof, discrete electronic devices, or other modules, logic, hardware, software, firmware, or a combination thereof adapted in the computer platform **1700**, or as components otherwise incorporated within a chassis of a larger system. The block diagram of FIG. **17** is intended to show a high level view of components of the computer platform **1700**. However, some of the components shown may be omitted, additional components may be present, and different arrangement of the components shown may occur in other implementations.

Application circuitry **1705** includes circuitry such as, but not limited to one or more processors (or processor cores), cache memory, and one or more of LDOs, interrupt controllers, serial interfaces such as SPI, I2C or universal programmable serial interface module, RTC, timer-counters including interval and watchdog timers, general purpose I/O, memory card controllers such as SD MMC or similar, USB interfaces, MIPI interfaces, and JTAG test access ports. The processors (or cores) of the application circuitry **1705** may be coupled with or may include memory/storage elements and may be configured to execute instructions stored in the memory/storage to enable various applications or operating systems to run on the system **1700**. In some implementations, the memory/storage elements may be on-chip memory circuitry, which may include any suitable volatile and/or non-volatile memory, such as DRAM, SRAM, EPROM, EEPROM, Flash memory, solid-state memory, and/or any other type of memory device technology, such as those discussed herein.

The processor(s) of application circuitry **1605** may include, for example, one or more processor cores, one or more application processors, one or more GPUs, one or more RISC processors, one or more ARM processors, one or more CISC processors, one or more DSP, one or more FPGAs, one or more PLDs, one or more ASICs, one or more microprocessors or controllers, a multithreaded processor, an ultra-low voltage processor, an embedded processor, some other known processing element, or any suitable combination thereof. In some embodiments, the application circuitry **1605** may comprise, or may be, a special-purpose processor/controller to operate according to the various embodiments herein.

As examples, the processor(s) of application circuitry **1705** may include an Intel® Architecture Core™ based processor, such as a Quark™, an Atom™, an i3, an i5, an i7, or an MCU-class processor, or another such processor available from Intel® Corporation, Santa Clara, CA. The processors of the application circuitry **1705** may also be one or more of Advanced Micro Devices (AMD) Ryzen® processor(s) or Accelerated Processing Units (APUs); A5-A9 processor(s) from Apple® Inc., Snapdragon™ processor(s) from Qualcomm® Technologies, Inc., Texas Instruments, Inc.® Open Multimedia Applications Platform (OMAP)™ processor(s); a MIPS-based design from MIPS Technologies, Inc. such as MIPS Warrior M-class, Warrior I-class, and Warrior P-class processors; an ARM-based design licensed from ARM Holdings, Ltd., such as the ARM Cortex-A, Cortex-R, and Cortex-M family of processors; or the like. In some implementations, the application circuitry **1705** may be a part of a system on a chip (SoC) in which the application circuitry **1705** and other components are formed into a single integrated circuit, or a single package, such as the Edison™ or Galileo™ SoC boards from Intel® Corporation.

Additionally or alternatively, application circuitry **1705** may include circuitry such as, but not limited to, one or more a field-programmable devices (FPDs) such as FPGAs and the like; programmable logic devices (PLDs) such as complex PLDs (CPLDs), high-capacity PLDs (HCPLDs), and the like; ASICs such as structured ASICs and the like; programmable SoCs (PSoCs); and the like. In such embodiments, the circuitry of application circuitry **1705** may comprise logic blocks or logic fabric, and other interconnected resources that may be programmed to perform various functions, such as the procedures, methods, functions, etc. of the various embodiments discussed herein. In such embodiments, the circuitry of application circuitry **1705** may include memory cells (e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), flash memory, static memory (e.g., static random access memory (SRAM), anti-fuses, etc.)) used to store logic blocks, logic fabric, data, etc. in look-up tables (LUTs) and the like.

The baseband circuitry **1710** may be implemented, for example, as a solder-down substrate including one or more integrated circuits, a single packaged integrated circuit soldered to a main circuit board or a multi-chip module containing two or more integrated circuits. The various hardware electronic elements of baseband circuitry **1710** are discussed infra with regard to FIG. **18**.

The RFEMs **1715** may comprise a millimeter wave (mmWave) RFEM and one or more sub-mmWave radio frequency integrated circuits (RFICs). In some implementations, the one or more sub-mmWave RFICs may be physically separated from the mmWave RFEM. The RFICs may include connections to one or more antennas or antenna arrays (see e.g., antenna array **1811** of FIG. **18** infra), and the RFEM may be connected to multiple antennas. In alternative implementations, both mm Wave and sub-mm Wave radio functions may be implemented in the same physical RFEM **1715**, which incorporates both mm Wave antennas and sub-mmWave.

The memory circuitry **1720** may include any number and type of memory devices used to provide for a given amount of system memory. As examples, the memory circuitry **1720** may include one or more of volatile memory including random access memory (RAM), dynamic RAM (DRAM) and/or synchronous dynamic RAM (SDRAM), and nonvolatile memory (NVM) including high-speed electrically erasable memory (commonly referred to as Flash memory), phase change random access memory (PRAM), magnetoresistive random access memory (MRAM), etc. The memory circuitry **1720** may be developed in accordance with a Joint Electron Devices Engineering Council (JEDEC) low power double data rate (LPDDR)-based design, such as LPDDR2, LPDDR3, LPDDR4, or the like. Memory circuitry **1720** may be implemented as one or more of solder down packaged integrated circuits, single die package (SDP), dual die package (DDP) or quad die package (Q17P), socketed memory modules, dual inline memory modules (DIMMs) including microDIMMs or MiniDIMMs, and/or soldered onto a motherboard via a ball grid array (BGA). In low power implementations, the memory circuitry **1720** may be on-die memory or registers associated with the application circuitry **1705**. To provide for persistent storage of information such as data, applications, operating systems and so forth, memory circuitry **1720** may include one or more mass storage devices, which may include, inter alia, a solid state disk drive (SSDD), hard disk drive (HDD), a micro HDD, resistance change memories, phase change memories, holographic memories, or chemical memories, among others. For example, the computer platform **1700** may incorporate the three-dimensional (3D) cross-point (XPOINT) memories from Intel® and Micron®.

Removable memory circuitry **1723** may include devices, circuitry, enclosures/housings, ports or receptacles, etc. used to couple portable data storage devices with the platform **1700**. These portable data storage devices may be used for mass storage purposes, and may include, for example, flash memory cards (e.g., Secure Digital (SD) cards, microSD cards, XD picture cards, and the like), and USB flash drives, optical discs, external HDDs, and the like.

The platform **1700** may also include interface circuitry (not shown) that is used to connect external devices with the platform **1700**. The external devices connected to the platform **1700** via the interface circuitry include sensor circuitry **1721** and electro-mechanical components (EMCs) **1722**, as well as removable memory devices coupled to removable memory circuitry **1723**.

The sensor circuitry **1721** include devices, modules, or subsystems whose purpose is to detect events or changes in its environment and send the information (sensor data) about the detected events to some other a device, module, subsystem, etc. Examples of such sensors include, inter alia, inertia measurement units (IMUs) comprising accelerometers, gyroscopes, and/or magnetometers; microelectromechanical systems (MEMS) or nanoelectromechanical systems (NEMS) comprising 3-axis accelerometers, 3-axis gyroscopes, and/or magnetometers; level sensors; flow sensors; temperature sensors (e.g., thermistors); pressure sensors; barometric pressure sensors; gravimeters; altimeters; image capture devices (e.g., cameras or lensless apertures); light detection and ranging (LiDAR) sensors; proximity sensors (e.g., infrared radiation detector and the like), depth sensors, ambient light sensors, ultrasonic transceivers; microphones or other like audio capture devices; etc.

EMCs **1722** include devices, modules, or subsystems whose purpose is to enable platform **1700** to change its state, position, and/or orientation, or move or control a mechanism or (sub) system. Additionally, EMCs **1722** may be configured to generate and send messages/signalling to other components of the platform **1700** to indicate a current state of the EMCs **1722**. Examples of the EMCs **1722** include one or more power switches, relays including electromechanical relays (EMRs) and/or solid state relays (SSRs), actuators (e.g., valve actuators, etc.), an audible sound generator, a visual warning device, motors (e.g., DC motors, stepper motors, etc.), wheels, thrusters, propellers, claws, clamps, hooks, and/or other like electro-mechanical components. In embodiments, platform **1700** is configured to operate one or more EMCs **1722** based on one or more captured events and/or instructions or control signals received from a service provider and/or various clients.

In some implementations, the interface circuitry may connect the platform **1700** with positioning circuitry **1745**. The positioning circuitry **1745** includes circuitry to receive and decode signals transmitted/broadcasted by a positioning network of a GNSS. Examples of navigation satellite constellations (or GNSS) include United States' GPS, Russia's GLONASS, the European Union's Galileo system, China's BeiDou Navigation Satellite System, a regional navigation system or GNSS augmentation system (e.g., NAVIC), Japan's QZSS, France's DORIS, etc.), or the like. The positioning circuitry **1745** comprises various hardware elements (e.g., including hardware devices such as switches, filters, amplifiers, antenna elements, and the like to facilitate OTA communications) to communicate with components of a positioning network, such as navigation satellite constellation nodes. In some embodiments, the positioning circuitry **1745** may include a Micro-PNT IC that uses a master timing clock to perform position tracking/estimation without GNSS assistance. The positioning circuitry **1745** may also be part of, or interact with, the baseband circuitry **1610** and/or RFEMs **1715** to communicate with the nodes and components of the positioning network. The positioning circuitry **1745** may also provide position data and/or time data to the application circuitry **1705**, which may use the data to synchronize operations with various infrastructure (e.g., radio base stations), for turn-by-turn navigation applications, or the like

In some implementations, the interface circuitry may connect the platform **1700** with Near-Field Communication (NFC) circuitry **1740**. NFC circuitry **1740** is configured to provide contactless, short-range communications based on radio frequency identification (RFID) standards, wherein magnetic field induction is used to enable communication between NFC circuitry **1740** and NFC-enabled devices external to the platform **1700** (e.g., an "NFC touchpoint"). NFC circuitry **1740** comprises an NFC controller coupled with an antenna element and a processor coupled with the NFC controller. The NFC controller may be a chip/IC providing NFC functionalities to the NFC circuitry **1740** by executing NFC controller firmware and an NFC stack. The NFC stack may be executed by the processor to control the NFC controller, and the NFC controller firmware may be executed by the NFC controller to control the antenna element to emit short-range RF signals. The RF signals may power a passive NFC tag (e.g., a microchip embedded in a sticker or wristband) to transmit stored data to the NFC circuitry **1740**, or initiate data transfer between the NFC circuitry **1740** and another active NFC device (e.g., a smartphone or an NFC-enabled POS terminal) that is proximate to the platform **1700**.

The driver circuitry **1746** may include software and hardware elements that operate to control particular devices that are embedded in the platform **1700**, attached to the platform **1700**, or otherwise communicatively coupled with the platform **1700**. The driver circuitry **1746** may include individual drivers allowing other components of the platform **1700** to interact with or control various input/output (I/O) devices that may be present within, or connected to, the platform **1700**. For example, driver circuitry **1746** may include a display driver to control and allow access to a display device, a touchscreen driver to control and allow access to a touchscreen interface of the platform **1700**, sensor drivers to obtain sensor readings of sensor circuitry **1721** and control and allow access to sensor circuitry **1721**, EMC drivers to obtain actuator positions of the EMCs **1722** and/or control and allow access to the EMCs **1722**, a camera driver to control and allow access to an embedded image capture device, audio drivers to control and allow access to one or more audio devices.

The power management integrated circuitry (PMIC) **1725** (also referred to as "power management circuitry **1725**") may manage power provided to various components of the platform **1700**. In particular, with respect to the baseband circuitry **1710**, the PMIC **1725** may control power-source selection, voltage scaling, battery charging, or DC-to-DC conversion. The PMIC **1725** may often be included when the platform **1700** is capable of being powered by a battery **1730**, for example, when the device is included in a UE **1301**, **1401**, **1502**.

In some embodiments, the PMIC **1725** may control, or otherwise be part of, various power saving mechanisms of the platform **1700**. For example, if the platform **1700** is in an RRC_Connected state, where it is still connected to the RAN node as it expects to receive traffic shortly, then it may enter a state known as Discontinuous Reception Mode (DRX) after a period of inactivity. During this state, the platform **1700** may power down for brief intervals of time and thus save power. If there is no data traffic activity for an extended period of time, then the platform **1700** may transition off to an RRC_Idle state, where it disconnects from the network and does not perform operations such as channel quality feedback, handover, etc. The platform **1700** goes into a very low power state and it performs paging where again it periodically wakes up to listen to the network and then powers down again. The platform **1700** may not receive data in this state; in order to receive data, it must transition back to RRC_Connected state. An additional power saving mode may allow a device to be unavailable to the network for periods longer than a paging interval (ranging from seconds to a few hours). During this time, the device is totally unreachable to the network and may power down completely. Any data sent during this time incurs a large delay and it is assumed the delay is acceptable.

A battery **1730** may power the platform **1700**, although in some examples the platform **1700** may be mounted deployed in a fixed location, and may have a power supply coupled to an electrical grid. The battery **1730** may be a lithium ion battery, a metal-air battery, such as a zinc-air battery, an aluminum-air battery, a lithium-air battery, and the like. In some implementations, such as in V2X applications, the battery **1730** may be a typical lead-acid automotive battery.

In some implementations, the battery **1730** may be a "smart battery," which includes or is coupled with a Battery Management System (BMS) or battery monitoring integrated circuitry. The BMS may be included in the platform **1700** to track the state of charge (SoCh) of the battery **1730**. The BMS may be used to monitor other parameters of the battery **1730** to provide failure predictions, such as the state of health (SoH) and the state of function (SoF) of the battery **1730**. The BMS may communicate the information of the battery **1730** to the application circuitry **1705** or other components of the platform **1700**. The BMS may also include an analog-to-digital (ADC) convertor that allows the application circuitry **1705** to directly monitor the voltage of the battery **1730** or the current flow from the battery **1730**. The battery parameters may be used to determine actions that the platform **1700** may perform, such as transmission frequency, network operation, sensing frequency, and the like.

A power block, or other power supply coupled to an electrical grid may be coupled with the BMS to charge the battery **1730**. In some examples, the power block **1730** may be replaced with a wireless power receiver to obtain the power wirelessly, for example, through a loop antenna in the computer platform **1700**. In these examples, a wireless battery charging circuit may be included in the BMS. The specific charging circuits chosen may depend on the size of the battery **1730**, and thus, the current required. The charging may be performed using the Airfuel standard promulgated by the Airfuel Alliance, the Qi wireless charging standard promulgated by the Wireless Power Consortium, or the Rezence charging standard promulgated by the Alliance for Wireless Power, among others.

User interface circuitry **1750** includes various input/output (I/O) devices present within, or connected to, the platform **1700**, and includes one or more user interfaces designed to enable user interaction with the platform **1700** and/or peripheral component interfaces designed to enable peripheral component interaction with the platform **1700**. The user interface circuitry **1750** includes input device circuitry and output device circuitry. Input device circuitry includes any physical or virtual means for accepting an input including, inter alia, one or more physical or virtual buttons (e.g., a reset button), a physical keyboard, keypad, mouse, touchpad, touchscreen, microphones, scanner, headset, and/or the like. The output device circuitry includes any physical or virtual means for showing information or otherwise conveying information, such as sensor readings, actuator position(s), or other like information. Output device circuitry may include any number and/or combinations of audio or visual display, including, inter alia, one or more simple visual outputs/indicators (e.g., binary status indicators (e.g., light emitting diodes (LEDs)) and multi-character visual outputs, or more complex outputs such as display devices or touchscreens (e.g., Liquid Chrystal Displays (LCD), LED displays, quantum dot displays, projectors, etc.), with the output of characters, graphics, multimedia objects, and the like being generated or produced from the operation of the platform **1700**. The output device circuitry may also include speakers or other audio emitting devices, printer(s), and/or the like. In some embodiments, the sensor circuitry **1721** may be used as the input device circuitry (e.g., an image capture device, motion capture device, or the like) and one or more EMCs may be used as the output device circuitry (e.g., an actuator to provide haptic feedback or the like). In another example, NFC circuitry comprising an NFC controller coupled with an antenna element and a processing device may be included to read electronic tags and/or connect with another NFC-enabled device. Peripheral component interfaces may include, but are not limited to, a non-volatile memory port, a USB port, an audio jack, a power supply interface, etc.

Although not shown, the components of platform **1700** may communicate with one another using a suitable bus or interconnect (IX) technology, which may include any number of technologies, including ISA, EISA, PCI, PCIx, PCIe, a Time-Trigger Protocol (TTP) system, a FlexRay system, or any number of other technologies. The bus/IX may be a proprietary bus/IX, for example, used in a SoC based system. Other bus/IX systems may be included, such as an I2C interface, an SPI interface, point-to-point interfaces, and a power bus, among others.

FIG. **18** illustrates example components of baseband circuitry **1810** and radio front end modules (RFEM) **1815** in accordance with various embodiments. The baseband circuitry **1810** corresponds to the baseband circuitry **1610** and **1710** of FIGS. **16** and **17**, respectively. The RFEM **1815** corresponds to the RFEM **1615** and **1715** of FIGS. **16** and **17**, respectively. As shown, the RFEMs **1815** may include Radio Frequency (RF) circuitry **1806**, front-end module (FEM) circuitry **1808**, antenna array **1811** coupled together at least as shown.

The baseband circuitry **1810** includes circuitry and/or control logic configured to carry out various radio/network protocol and radio control functions that enable communication with one or more radio networks via the RF circuitry **1806**. The radio control functions may include, but are not limited to, signal modulation/demodulation, encoding/decoding, radio frequency shifting, etc. In some embodiments, modulation/demodulation circuitry of the baseband circuitry **1810** may include Fast-Fourier Transform (FFT), precoding, or constellation mapping/demapping functionality. In some embodiments, encoding/decoding circuitry of the baseband circuitry **1810** may include convolution, tail-biting convolution, turbo, Viterbi, or Low Density Parity Check (LDPC) encoder/decoder functionality. Embodiments of modulation/demodulation and encoder/decoder functionality are not limited to these examples and may include other suitable functionality in other embodiments. The baseband circuitry **1810** is configured to process baseband signals received from a receive signal path of the RF circuitry **1806** and to generate baseband signals for a transmit signal path of the RF circuitry **1806**. The baseband circuitry **1810** is configured to interface with application circuitry **1605/1705** (see FIGS. **16** and **17**) for generation and processing of the baseband signals and for controlling operations of the RF circuitry **1806**. The baseband circuitry **1810** may handle various radio control functions.

The aforementioned circuitry and/or control logic of the baseband circuitry **1810** may include one or more single or multi-core processors. For example, the one or more processors may include a 3G baseband processor **1804**A, a 4G/LTE baseband processor **1804**B, a 5G/NR baseband processor **1804**C, or some other baseband processor(s) **1804**D for other existing generations, generations in development or to be developed in the future (e.g., sixth generation (6G), etc.). In other embodiments, some or all of the functionality of baseband processors **1804**A-D may be included in modules stored in the memory **1804**G and executed via a Central Processing Unit (CPU) **1804**E. In other embodiments, some or all of the functionality of baseband processors **1804**A-D may be provided as hardware accelerators (e.g., FPGAs, ASICs, etc.) loaded with the appropriate bit streams or logic blocks stored in respective memory cells. In various embodiments, the memory **1804**G may store program code of a real-time OS (RTOS), which when executed by the CPU **1804**E (or other baseband processor), is to cause the CPU **1804**E (or other baseband processor) to manage resources of the baseband circuitry **1810**, schedule tasks, etc. Examples of the RTOS may include Operating System Embedded (OSE)™ provided by Enea®, Nucleus RTOS™ provided by Mentor Graphics®, Versatile Real-Time Executive (VRTX) provided by Mentor Graphics®, ThreadX™ provided by Express Logic®, FreeRTOS, REX OS provided by Qualcomm®, OKL4 provided by Open Kernel (OK) Labs®, or any other suitable RTOS, such as those discussed herein. In addition, the baseband circuitry **1810** includes one or more audio digital signal processor(s) (DSP) **1804**F. The audio DSP(s) **1804**F include elements for compression/decompression and echo cancellation and may include other suitable processing elements in other embodiments.

In some embodiments, each of the processors **1804**A-**1804**E include respective memory interfaces to send/receive data to/from the memory **1804**G. The baseband circuitry **1810** may further include one or more interfaces to communicatively couple to other circuitries/devices, such as an interface to send/receive data to/from memory external to the baseband circuitry **1810**; an application circuitry interface to send/receive data to/from the application circuitry **1605/1705** of FIGS. **16-18**); an RF circuitry interface to send/receive data to/from RF circuitry **1806** of FIG. **18**; a wireless hardware connectivity interface to send/receive data to/from one or more wireless hardware elements (e.g., Near Field Communication (NFC) components, Bluetooth®/Bluetooth® Low Energy components, Wi-Fi® components, and/or the like); and a power management interface to send/receive power or control signals to/from the PMIC **1725**. In alternate embodiments (which may be combined with the above described embodiments), baseband circuitry **1810** comprises one or more digital baseband systems, which are coupled with one another via an interconnect subsystem and to a CPU subsystem, an audio subsystem, and an interface subsystem. The digital baseband subsystems may also be coupled to a digital baseband interface and a mixed-signal baseband subsystem via another interconnect subsystem. Each of the interconnect subsystems may include a bus system, point-to-point connections, network-on-chip (NOC) structures, and/or some other suitable bus or interconnect technology, such as those discussed herein. The audio subsystem may include DSP circuitry, buffer memory, program memory, speech processing accelerator circuitry, data converter circuitry such as analog-to-digital and digital-to-analog converter circuitry, analog circuitry including one or more of amplifiers and filters, and/or other like components. In an aspect of the present disclosure, baseband circuitry **1810** may include protocol processing circuitry with one or more instances of control circuitry (not shown) to provide control functions for the digital baseband circuitry and/or radio frequency circuitry (e.g., the radio front end modules **1815**).

Although not shown by FIG. **18**, in some embodiments, the baseband circuitry **1810** includes individual processing device(s) to operate one or more wireless communication protocols (e.g., a "multi-protocol baseband processor" or "protocol processing circuitry") and individual processing device(s) to implement PHY layer functions. In these embodiments, the PHY layer functions include the aforementioned radio control functions. In these embodiments, the protocol processing circuitry operates or implements various protocol layers/entities of one or more wireless communication protocols. In a first example, the protocol processing circuitry may operate LTE protocol entities and/or 5G/NR protocol entities when the baseband circuitry **1810** and/or RF circuitry **1806** are part of mm Wave communication circuitry or some other suitable cellular communication circuitry. In the first example, the protocol processing circuitry would operate MAC, RLC, PDCP, SDAP, RRC, and NAS functions. In a second example, the protocol processing circuitry may operate one or more IEEE-based protocols when the baseband circuitry **1810** and/or RF circuitry **1806** are part of a Wi-Fi communication system. In the second example, the protocol processing circuitry would operate Wi-Fi MAC and logical link control (LLC) functions. The protocol processing circuitry may include one or more memory structures (e.g., **1804**G) to store program code and data for operating the protocol functions, as well as one or more processing cores to execute the program code and perform various operations using the data. The baseband circuitry **1810** may also support radio communications for more than one wireless protocol.

The various hardware elements of the baseband circuitry **1810** discussed herein may be implemented, for example, as a solder-down substrate including one or more integrated circuits (ICs), a single packaged IC soldered to a main circuit board or a multi-chip module containing two or more ICs. In one example, the components of the baseband circuitry **1810** may be suitably combined in a single chip or chipset, or disposed on a same circuit board. In another example, some or all of the constituent components of the baseband circuitry **1810** and RF circuitry **1806** may be implemented together such as, for example, a system on a chip (SoC) or System-in-Package (SiP). In another example, some or all of the constituent components of the baseband circuitry **1810** may be implemented as a separate SoC that is communicatively coupled with and RF circuitry **1806** (or multiple instances of RF circuitry **1806**). In yet another example, some or all of the constituent components of the baseband circuitry **1810** and the application circuitry **1605/1705** may be implemented together as individual SoCs mounted to a same circuit board (e.g., a "multi-chip package").

In some embodiments, the baseband circuitry **1810** may provide for communication compatible with one or more radio technologies. For example, in some embodiments, the baseband circuitry **1810** may support communication with an E-UTRAN or other WMAN, a WLAN, a WPAN. Embodiments in which the baseband circuitry **1810** is configured to support radio communications of more than one wireless protocol may be referred to as multi-mode baseband circuitry.

RF circuitry **1806** may enable communication with wireless networks using modulated electromagnetic radiation through a non-solid medium. In various embodiments, the RF circuitry **1806** may include switches, filters, amplifiers, etc. to facilitate the communication with the wireless network. RF circuitry **1806** may include a receive signal path, which may include circuitry to down-convert RF signals received from the FEM circuitry **1808** and provide baseband signals to the baseband circuitry **1810**. RF circuitry **1806** may also include a transmit signal path, which may include circuitry to up-convert baseband signals provided by the baseband circuitry **1810** and provide RF output signals to the FEM circuitry **1808** for transmission.

In some embodiments, the receive signal path of the RF circuitry **1806** may include mixer circuitry **1806***a*, amplifier circuitry **1806***b* and filter circuitry **1806***c*. In some embodiments, the transmit signal path of the RF circuitry **1806** may include filter circuitry **1806***c* and mixer circuitry **1806***a*. RF circuitry **1806** may also include synthesizer circuitry **1806***d* for synthesizing a frequency for use by the mixer circuitry **1806***a* of the receive signal path and the transmit signal path. In some embodiments, the mixer circuitry **1806***a* of the receive signal path may be configured to down-convert RF signals received from the FEM circuitry **1808** based on the synthesized frequency provided by synthesizer circuitry **1806***d*. The amplifier circuitry **1806***b* may be configured to amplify the down-converted signals and the filter circuitry **1806***c* may be a low-pass filter (LPF) or band-pass filter (BPF) configured to remove unwanted signals from the down-converted signals to generate output baseband signals. Output baseband signals may be provided to the baseband circuitry **1810** for further processing. In some embodiments, the output baseband signals may be zero-frequency baseband signals, although this is not a requirement. In some embodiments, mixer circuitry **1806***a* of the receive signal path may comprise passive mixers, although the scope of the embodiments is not limited in this respect.

In some embodiments, the mixer circuitry **1806***a* of the transmit signal path may be configured to up-convert input baseband signals based on the synthesized frequency provided by the synthesizer circuitry **1806***d* to generate RF output signals for the FEM circuitry **1808**. The baseband signals may be provided by the baseband circuitry **1810** and may be filtered by filter circuitry **1806***c*.

In some embodiments, the mixer circuitry **1806***a* of the receive signal path and the mixer circuitry **1806***a* of the transmit signal path may include two or more mixers and may be arranged for quadrature downconversion and upconversion, respectively. In some embodiments, the mixer circuitry **1806***a* of the receive signal path and the mixer circuitry **1806***a* of the transmit signal path may include two or more mixers and may be arranged for image rejection (e.g., Hartley image rejection). In some embodiments, the mixer circuitry **1806***a* of the receive signal path and the mixer circuitry **1806***a* of the transmit signal path may be arranged for direct downconversion and direct upconversion, respectively. In some embodiments, the mixer circuitry **1806***a* of the receive signal path and the mixer circuitry **1806***a* of the transmit signal path may be configured for super-heterodyne operation.

In some embodiments, the output baseband signals and the input baseband signals may be analog baseband signals, although the scope of the embodiments is not limited in this respect. In some alternate embodiments, the output baseband signals and the input baseband signals may be digital baseband signals. In these alternate embodiments, the RF circuitry **1806** may include analog-to-digital converter (ADC) and digital-to-analog converter (DAC) circuitry and the baseband circuitry **1810** may include a digital baseband interface to communicate with the RF circuitry **1806**.

In some dual-mode embodiments, a separate radio IC circuitry may be provided for processing signals for each spectrum, although the scope of the embodiments is not limited in this respect.

In some embodiments, the synthesizer circuitry **1806***d* may be a fractional-N synthesizer or a fractional N/N+1 synthesizer, although the scope of the embodiments is not limited in this respect as other types of frequency synthesizers may be suitable. For example, synthesizer circuitry **1806***d* may be a delta-sigma synthesizer, a frequency multiplier, or a synthesizer comprising a phase-locked loop with a frequency divider.

The synthesizer circuitry **1806***d* may be configured to synthesize an output frequency for use by the mixer circuitry **1806***a* of the RF circuitry **1806** based on a frequency input and a divider control input. In some embodiments, the synthesizer circuitry **1806***d* may be a fractional N/N+1 synthesizer.

In some embodiments, frequency input may be provided by a voltage controlled oscillator (VCO), although that is not a requirement. Divider control input may be provided by either the baseband circuitry **1810** or the application circuitry **1605/1705** depending on the desired output frequency. In some embodiments, a divider control input (e.g.,

N) may be determined from a look-up table based on a channel indicated by the application circuitry **1605/1705**.

Synthesizer circuitry **1806***d* of the RF circuitry **1806** may include a divider, a delay-locked loop (DLL), a multiplexer and a phase accumulator. In some embodiments, the divider may be a dual modulus divider (DMD) and the phase accumulator may be a digital phase accumulator (DPA). In some embodiments, the DMD may be configured to divide the input signal by either N or N+1 (e.g., based on a carry out) to provide a fractional division ratio. In some example embodiments, the DLL may include a set of cascaded, tunable, delay elements, a phase detector, a charge pump and a D-type flip-flop. In these embodiments, the delay elements may be configured to break a VCO period up into Nd equal packets of phase, where Nd is the number of delay elements in the delay line. In this way, the DLL provides negative feedback to help ensure that the total delay through the delay line is one VCO cycle.

In some embodiments, synthesizer circuitry **1806***d* may be configured to generate a carrier frequency as the output frequency, while in other embodiments, the output frequency may be a multiple of the carrier frequency (e.g., twice the carrier frequency, four times the carrier frequency) and used in conjunction with quadrature generator and divider circuitry to generate multiple signals at the carrier frequency with multiple different phases with respect to each other. In some embodiments, the output frequency may be a LO frequency (fLO). In some embodiments, the RF circuitry **1806** may include an IQ/polar converter.

FEM circuitry **1808** may include a receive signal path, which may include circuitry configured to operate on RF signals received from antenna array **1811**, amplify the received signals and provide the amplified versions of the received signals to the RF circuitry **1806** for further processing. FEM circuitry **1808** may also include a transmit signal path, which may include circuitry configured to amplify signals for transmission provided by the RF circuitry **1806** for transmission by one or more of antenna elements of antenna array **1811**. In various embodiments, the amplification through the transmit or receive signal paths may be done solely in the RF circuitry **1806**, solely in the FEM circuitry **1808**, or in both the RF circuitry **1806** and the FEM circuitry **1808**.

In some embodiments, the FEM circuitry **1808** may include a TX/RX switch to switch between transmit mode and receive mode operation. The FEM circuitry **1808** may include a receive signal path and a transmit signal path. The receive signal path of the FEM circuitry **1808** may include an LNA to amplify received RF signals and provide the amplified received RF signals as an output (e.g., to the RF circuitry **1806**). The transmit signal path of the FEM circuitry **1808** may include a power amplifier (PA) to amplify input RF signals (e.g., provided by RF circuitry **1806**), and one or more filters to generate RF signals for subsequent transmission by one or more antenna elements of the antenna array **1811**.

The antenna array **1811** comprises one or more antenna elements, each of which is configured convert electrical signals into radio waves to travel through the air and to convert received radio waves into electrical signals. For example, digital baseband signals provided by the baseband circuitry **1810** is converted into analog RF signals (e.g., modulated waveform) that will be amplified and transmitted via the antenna elements of the antenna array **1811** including one or more antenna elements (not shown). The antenna elements may be omnidirectional, direction, or a combination thereof. The antenna elements may be formed in a multitude of arranges as are known and/or discussed herein. The antenna array **1811** may comprise microstrip antennas or printed antennas that are fabricated on the surface of one or more printed circuit boards. The antenna array **1811** may be formed in as a patch of metal foil (e.g., a patch antenna) in a variety of shapes, and may be coupled with the RF circuitry **1806** and/or FEM circuitry **1808** using metal transmission lines or the like.

Processors of the application circuitry **1605/1705** and processors of the baseband circuitry **1810** may be used to execute elements of one or more instances of a protocol stack. For example, processors of the baseband circuitry **1810**, alone or in combination, may be used execute Layer 3, Layer 2, or Layer 1 functionality, while processors of the application circuitry **1605/1705** may utilize data (e.g., packet data) received from these layers and further execute Layer 4 functionality (e.g., TCP and UDP layers). As referred to herein, Layer 3 may comprise a RRC layer, described in further detail below. As referred to herein, Layer 2 may comprise a MAC layer, an RLC layer, and a PDCP layer, described in further detail below. As referred to herein, Layer 1 may comprise a PHY layer of a UE/RAN node, described in further detail below.

FIG. **19** illustrates various protocol functions that may be implemented in a wireless communication device according to various embodiments. In particular, FIG. **19** includes an arrangement **1900** showing interconnections between various protocol layers/entities. The following description of FIG. **19** is provided for various protocol layers/entities that operate in conjunction with the 5G/NR system standards and LTE system standards, but some or all of the aspects of FIG. **19** may be applicable to other wireless communication network systems as well.

The protocol layers of arrangement **1900** may include one or more of PHY **1910**, MAC **1920**, RLC **1930**, PDCP **1940**, SDAP **1947**, RRC **1955**, and NAS layer **1957**, in addition to other higher layer functions not illustrated. The protocol layers may include one or more service access points (e.g., items **1959**, **1956**, **1950**, **1949**, **1945**, **1935**, **1925**, and **1915** in FIG. **19**) that may provide communication between two or more protocol layers.

The PHY **1910** may transmit and receive physical layer signals **1905** that may be received from or transmitted to one or more other communication devices. The physical layer signals **1905** may comprise one or more physical channels, such as those discussed herein. The PHY **1910** may further perform link adaptation or adaptive modulation and coding (AMC), power control, cell search (e.g., for initial synchronization and handover purposes), and other measurements used by higher layers, such as the RRC **1955**. The PHY **1910** may still further perform error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, modulation/demodulation of physical channels, interleaving, rate matching, mapping onto physical channels, and MIMO antenna processing. In embodiments, an instance of PHY **1910** may process requests from and provide indications to an instance of MAC **1920** via one or more PHY-SAP **1915**. According to some embodiments, requests and indications communicated via PHY-SAP **1915** may comprise one or more transport channels.

Instance(s) of MAC **1920** may process requests from, and provide indications to, an instance of RLC **1930** via one or more MAC-SAPs **1925**. These requests and indications communicated via the MAC-SAP **1925** may comprise one or more logical channels. The MAC **1920** may perform mapping between the logical channels and transport channels, multiplexing of MAC SDUs from one or more logical channels onto TBs to be delivered to PHY **1910** via the transport channels, de-multiplexing MAC SDUs to one or more logical channels from TBs delivered from the PHY **1910** via transport channels, multiplexing MAC SDUs onto TBs, scheduling information reporting, error correction through HARQ, and logical channel prioritization.

Instance(s) of RLC **1930** may process requests from and provide indications to an instance of PDCP **1940** via one or more radio link control service access points (RLC-SAP) **1935**. These requests and indications communicated via RLC-SAP **1935** may comprise one or more RLC channels. The RLC **1930** may operate in a plurality of modes of operation, including: Transparent Mode (TM), Unacknowledged Mode (UM), and Acknowledged Mode (AM). The RLC **1930** may execute transfer of upper layer protocol data units (PDUs), error correction through automatic repeat request (ARQ) for AM data transfers, and concatenation, segmentation and reassembly of RLC SDUs for UM and AM data transfers. The RLC **1930** may also execute re-segmentation of RLC data PDUs for AM data transfers, reorder RLC data PDUs for UM and AM data transfers, detect duplicate data for UM and AM data transfers, discard RLC SDUs for UM and AM data transfers, detect protocol errors for AM data transfers, and perform RLC re-establishment.

Instance(s) of PDCP **1940** may process requests from and provide indications to instance(s) of RRC **1955** and/or instance(s) of SDAP **1947** via one or more packet data convergence protocol service access points (PDCP-SAP) **1945**. These requests and indications communicated via PDCP-SAP **1945** may comprise one or more radio bearers. The PDCP **1940** may execute header compression and decompression of IP data, maintain PDCP Sequence Numbers (SNs), perform in-sequence delivery of upper layer PDUs at re-establishment of lower layers, eliminate duplicates of lower layer SDUs at re-establishment of lower layers for radio bearers mapped on RLC AM, cipher and decipher control plane data, perform integrity protection and integrity verification of control plane data, control timer-based discard of data, and perform security operations (e.g., ciphering, deciphering, integrity protection, integrity verification, etc.).

Instance(s) of SDAP **1947** may process requests from and provide indications to one or more higher layer protocol entities via one or more SDAP-SAP **1949**. These requests and indications communicated via SDAP-SAP **1949** may comprise one or more QoS flows. The SDAP **1947** may map QoS flows to DRBs, and vice versa, and may also mark QFIs in DL and UL packets. A single SDAP entity **1947** may be configured for an individual PDU session. In the UL direction, the NG-RAN **1310** may control the mapping of QoS Flows to DRB(s) in two different ways, reflective mapping or explicit mapping. For reflective mapping, the SDAP **1947** of a UE **1301** may monitor the QFIs of the DL packets for each DRB, and may apply the same mapping for packets flowing in the UL direction. For a DRB, the SDAP **1947** of the UE **1301** may map the UL packets belonging to the QoS flows(s) corresponding to the QoS flow ID(s) and PDU session observed in the DL packets for that DRB. To enable reflective mapping, the NG-RAN **1510** may mark DL packets over the Uu interface with a QoS flow ID. The explicit mapping may involve the RRC **1955** configuring the SDAP **1947** with an explicit QoS flow to DRB mapping rule, which may be stored and followed by the SDAP **1947**. In embodiments, the SDAP **1947** may only be used in NR implementations and may not be used in LTE implementations.

The RRC **1955** may configure, via one or more management service access points (M-SAP), aspects of one or more protocol layers, which may include one or more instances of PHY **1910**, MAC **1920**, RLC **1930**, PDCP **1940** and SDAP **1947**. In embodiments, an instance of RRC **1955** may process requests from and provide indications to one or more NAS entities **1957** via one or more RRC-SAPs **1956**. The main services and functions of the RRC **1955** may include broadcast of system information (e.g., included in MIBs or SIBs related to the NAS), broadcast of system information related to the access stratum (AS), paging, establishment, maintenance and release of an RRC connection between the UE **1301** and RAN **1310** (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), establishment, configuration, maintenance and release of point to point Radio Bearers, security functions including key management, inter-RAT mobility, and measurement configuration for UE measurement reporting. The MIBs and SIBs may comprise one or more IEs, which may each comprise individual data fields or data structures.

The NAS **1957** may form the highest stratum of the control plane between the UE **1301** and the AMF **1521**. The NAS **1957** may support the mobility of the UEs **1301** and the session management procedures to establish and maintain IP connectivity between the UE **1301** and a P-GW in LTE systems.

According to various embodiments, one or more protocol entities of arrangement **1900** may be implemented in UEs **1301**, RAN nodes **1311**, AMF **1521** in NR implementations or MME **1421** in LTE implementations, UPF **1502** in NR implementations or S-GW **1422** and P-GW **1423** in LTE implementations, or the like to be used for control plane or user plane communications protocol stack between the aforementioned devices. In such embodiments, one or more protocol entities that may be implemented in one or more of UE **1301**, gNB **1311**, AMF **1521**, etc. may communicate with a respective peer protocol entity that may be implemented in or on another device using the services of respective lower layer protocol entities to perform such communication. In some embodiments, a gNB-CU of the gNB **1311** may host the RRC **1955**, SDAP **1947**, and PDCP **1940** of the gNB that controls the operation of one or more gNB-DUs, and the gNB-DUs of the gNB **1311** may each host the RLC **1930**, MAC **1920**, and PHY **1910** of the gNB **1311**.

In a first example, a control plane protocol stack may comprise, in order from highest layer to lowest layer, NAS **1957**, RRC **1955**, PDCP **1940**, RLC **1930**, MAC **1920**, and PHY **1910**. In this example, upper layers **1960** may be built on top of the NAS **1957**, which includes an IP layer **1961**, an SCTP **1962**, and an application layer signaling protocol (AP) **1963**.

In NR implementations, the AP **1963** may be an NG application protocol layer (NGAP or NG-AP) **1963** for the NG interface **1313** defined between the NG-RAN node **1311** and the AMF **1521**, or the AP **1963** may be an Xn application protocol layer (XnAP or Xn-AP) **1963** for the Xn interface **1312** that is defined between two or more RAN nodes **1311**.

The NG-AP **1963** may support the functions of the NG interface **1313** and may comprise Elementary Procedures (EPs). An NG-AP EP may be a unit of interaction between the NG-RAN node **1311** and the AMF **1521**. The NG-AP **1963** services may comprise two groups: UE-associated services (e.g., services related to a UE **1301**) and non-UE-associated services (e.g., services related to the whole NG interface instance between the NG-RAN node **1311** and AMF **1521**). These services may include functions including, but not limited to: a paging function for the sending of paging requests to NG-RAN nodes **1311** involved in a particular paging area; a UE context management function for allowing the AMF **1521** to establish, modify, and/or release a UE context in the AMF **1521** and the NG-RAN node **1311**; a mobility function for UEs **1301** in ECM-CONNECTED mode for intra-system HOs to support mobility within NG-RAN and inter-system HOs to support mobility from/to EPS systems; a NAS Signaling Transport function for transporting or rerouting NAS messages between UE **1301** and AMF **1521**; a NAS node selection function for determining an association between the AMF **1521** and the UE **1301**; NG interface management function (s) for setting up the NG interface and monitoring for errors over the NG interface; a warning message transmission function for providing means to transfer warning messages via NG interface or cancel ongoing broadcast of warning messages; a Configuration Transfer function for requesting and transferring of RAN configuration information (e.g., SON information, performance measurement (PM) data, etc.) between two RAN nodes **1311** via CN **1320**; and/or other like functions.

The XnAP **1963** may support the functions of the Xn interface **1312** and may comprise XnAP basic mobility procedures and XnAP global procedures. The XnAP basic mobility procedures may comprise procedures used to handle UE mobility within the NG RAN **1311** (or E-UTRAN **1410**), such as handover preparation and cancellation procedures, SN Status Transfer procedures, UE context retrieval and UE context release procedures, RAN paging procedures, dual connectivity related procedures, and the like. The XnAP global procedures may comprise procedures that are not related to a specific UE **1301**, such as Xn interface setup and reset procedures, NG-RAN update procedures, cell activation procedures, and the like.

In LTE implementations, the AP **1963** may be an S1 Application Protocol layer (S1-AP) **1963** for the S1 interface **1313** defined between an E-UTRAN node **1311** and an MME, or the AP **1963** may be an X2 application protocol layer (X2AP or X2-AP) **1963** for the X2 interface **1312** that is defined between two or more E-UTRAN nodes **1311**.

The S1 Application Protocol layer (S1-AP) **1963** may support the functions of the S1 interface, and similar to the NG-AP discussed previously, the S1-AP may comprise S1-AP EPs. An S1-AP EP may be a unit of interaction between the E-UTRAN node **1311** and an MME **1421** within an LTE CN **1320**. The S1-AP **1963** services may comprise two groups: UE-associated services and non UE-associated services. These services perform functions including, but not limited to: E-UTRAN Radio Access Bearer (E-RAB) management, UE capability indication, mobility, NAS signaling transport, RAN Information Management (RIM), and configuration transfer.

The X2AP **1963** may support the functions of the X2 interface **1312** and may comprise X2AP basic mobility procedures and X2AP global procedures. The X2AP basic mobility procedures may comprise procedures used to handle UE mobility within the E-UTRAN **1320**, such as handover preparation and cancellation procedures, SN Status Transfer procedures, UE context retrieval and UE context release procedures, RAN paging procedures, dual connectivity related procedures, and the like. The X2AP global procedures may comprise procedures that are not related to a specific UE **1301**, such as X2 interface setup and reset procedures, load indication procedures, error indication procedures, cell activation procedures, and the like.

The SCTP layer (alternatively referred to as the SCTP/IP layer) **1962** may provide guaranteed delivery of application layer messages (e.g., NGAP or XnAP messages in NR implementations, or S1-AP or X2AP messages in LTE implementations). The SCTP **1962** may ensure reliable delivery of signaling messages between the RAN node **1311** and the AMF **1521**/MME **1421** based, in part, on the IP protocol, supported by the IP **1961**. The Internet Protocol layer (IP) **1961** may be used to perform packet addressing and routing functionality. In some implementations the IP layer **1961** may use point-to-point transmission to deliver and convey PDUs. In this regard, the RAN node **1311** may comprise L2 and L1 layer communication links (e.g., wired or wireless) with the MME/AMF to exchange information.

In a second example, a user plane protocol stack may comprise, in order from highest layer to lowest layer, SDAP **1947**, PDCP **1940**, RLC **1930**, MAC **1920**, and PHY **1910**. The user plane protocol stack may be used for communication between the UE **1301**, the RAN node **1311**, and UPF **1502** in NR implementations or an S-GW **1422** and P-GW **1423** in LTE implementations. In this example, upper layers **1951** may be built on top of the SDAP **1947**, and may include a user datagram protocol (UDP) and IP security layer (UDP/IP) **1952**, a General Packet Radio Service (GPRS) Tunneling Protocol for the user plane layer (GTP-U) **1953**, and a User Plane PDU layer (UP PDU) **1963**.

The transport network layer **1954** (also referred to as a "transport layer") may be built on IP transport, and the GTP-U **1953** may be used on top of the UDP/IP layer **1952** (comprising a UDP layer and IP layer) to carry user plane PDUs (UP-PDUs). The IP layer (also referred to as the "Internet layer") may be used to perform packet addressing and routing functionality. The IP layer may assign IP addresses to user data packets in any of IPv4, IPv6, or PPP formats, for example.

The GTP-U **1953** may be used for carrying user data within the GPRS core network and between the radio access network and the core network. The user data transported can be packets in any of IPV4, IPV6, or PPP formats, for example. The UDP/IP **1952** may provide checksums for data integrity, port numbers for addressing different functions at the source and destination, and encryption and authentication on the selected data flows. The RAN node **1311** and the S-GW **1422** may utilize an S1-U interface to exchange user plane data via a protocol stack comprising an L1 layer (e.g., PHY **1910**), an L2 layer (e.g., MAC **1920**, RLC **1930**, PDCP **1940**, and/or SDAP **1947**), the UDP/IP layer **1952**, and the GTP-U **1953**. The S-GW **1422** and the P-GW **1423** may utilize an S5/S8a interface to exchange user plane data via a protocol stack comprising an L1 layer, an L2 layer, the UDP/IP layer **1952**, and the GTP-U **1953**. As discussed previously, NAS protocols may support the mobility of the UE **1301** and the session management procedures to establish and maintain IP connectivity between the UE **1301** and the P-GW **1423**.

Moreover, although not shown by FIG. **19**, an application layer may be present above the AP **1963** and/or the transport network layer **1954**. The application layer may be a layer in which a user of the UE **1301**, RAN node **1311**, or other network element interacts with software applications being executed, for example, by application circuitry **1605** or application circuitry **1705**, respectively. The application layer may also provide one or more interfaces for software applications to interact with communications systems of the UE **1301** or RAN node **1311**, such as the baseband circuitry **1810**. In some implementations the IP layer and/or the application layer may provide the same or similar functionality as layers 5-7, or portions thereof, of the Open Systems Interconnection (OSI) model (e.g., OSI Layer 7—the application layer, OSI Layer 6—the presentation layer, and OSI Layer 5—the session layer).

FIG. **20** illustrates components of a core network in accordance with various embodiments. The components of the CN **1420** may be implemented in one physical node or separate physical nodes including components to read and execute instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium). In embodiments, the components of CN **1520** may be implemented in a same or similar manner as discussed herein with regard to the components of CN **1420**. In some embodiments, NFV is utilized to virtualize any or all of the above-described network node functions via executable instructions stored in one or more computer-readable storage mediums (described in further detail below). A logical instantiation of the CN **1420** may be referred to as a network slice **2001**, and individual logical instantiations of the CN **1420** may provide specific network capabilities and network characteristics. A logical instantiation of a portion of the CN **1420** may be referred to as a network sub-slice **2002** (e.g., the network sub-slice **2002** is shown to include the P-GW **1423** and the PCRF **1426**).

As used herein, the terms “instantiate,” “instantiation,” and the like may refer to the creation of an instance, and an “instance” may refer to a concrete occurrence of an object, which may occur, for example, during execution of program code. A network instance may refer to information identifying a domain, which may be used for traffic detection and routing in case of different IP domains or overlapping IP addresses. A network slice instance may refer to a set of network functions (NFs) instances and the resources (e.g., compute, storage, and networking resources) required to deploy the network slice.

With respect to 5G systems (see, e.g., FIG. **15**), a network slice always comprises a RAN part and a CN part. The support of network slicing relies on the principle that traffic for different slices is handled by different PDU sessions. The network can realize the different network slices by scheduling and also by providing different L1/L2 configurations. The UE **1501** provides assistance information for network slice selection in an appropriate RRC message, if it has been provided by NAS. While the network can support large number of slices, the UE need not support more than 8 slices simultaneously.

A network slice may include the CN **1520** control plane and user plane NFs, NG-RANs **1510** in a serving PLMN, and a N3IWF functions in the serving PLMN. Individual network slices may have different S-NSSAI and/or may have different SSTs. NSSAI includes one or more S-NSSAIs, and each network slice is uniquely identified by an S-NSSAI. Network slices may differ for supported features and network functions optimizations, and/or multiple network slice instances may deliver the same service/features but for different groups of UEs **1501** (e.g., enterprise users). For example, individual network slices may deliver different committed service(s) and/or may be dedicated to a particular customer or enterprise. In this example, each network slice may have different S-NSSAIs with the same SST but with different slice differentiators. Additionally, a single UE may be served with one or more network slice instances simultaneously via a 5G AN and associated with eight different S-NSSAIs. Moreover, an AMF **1521** instance serving an individual UE **1501** may belong to each of the network slice instances serving that UE.

Network Slicing in the NG-RAN **1510** involves RAN slice awareness. RAN slice awareness includes differentiated handling of traffic for different network slices, which have been pre-configured. Slice awareness in the NG-RAN **1510** is introduced at the PDU session level by indicating the S-NSSAI corresponding to a PDU session in all signaling that includes PDU session resource information. How the NG-RAN **1510** supports the slice enabling in terms of NG-RAN functions (e.g., the set of network functions that comprise each slice) is implementation dependent. The NG-RAN **1510** selects the RAN part of the network slice using assistance information provided by the UE **1501** or the 5GC **1520**, which unambiguously identifies one or more of the pre-configured network slices in the PLMN. The NG-RAN **1510** also supports resource management and policy enforcement between slices as per SLAs. A single NG-RAN node may support multiple slices, and the NG-RAN **1510** may also apply an appropriate RRM policy for the SLA in place to each supported slice. The NG-RAN **1510** may also support QoS differentiation within a slice.

The NG-RAN **1510** may also use the UE assistance information for the selection of an AMF **1521** during an initial attach, if available. The NG-RAN **1510** uses the assistance information for routing the initial NAS to an AMF **1521**. If the NG-RAN **1510** is unable to select an AMF **1521** using the assistance information, or the UE **1501** does not provide any such information, the NG-RAN **1510** sends the NAS signaling to a default AMF **1521**, which may be among a pool of AMFs **1521**. For subsequent accesses, the UE **1501** provides a temp ID, which is assigned to the UE **1501** by the 5GC **1520**, to enable the NG-RAN **1510** to route the NAS message to the appropriate AMF **1521** as long as the temp ID is valid. The NG-RAN **1510** is aware of, and can reach, the AMF **1521** that is associated with the temp ID. Otherwise, the method for initial attach applies.

The NG-RAN **1510** supports resource isolation between slices. NG-RAN **1510** resource isolation may be achieved by means of RRM policies and protection mechanisms that should avoid that shortage of shared resources if one slice breaks the service level agreement for another slice. In some implementations, it is possible to fully dedicate NG-RAN **1510** resources to a certain slice. How NG-RAN **1510** supports resource isolation is implementation dependent.

Some slices may be available only in part of the network. Awareness in the NG-RAN **1510** of the slices supported in the cells of its neighbors may be beneficial for inter-frequency mobility in connected mode. The slice availability may not change within the UE’s registration area. The NG-RAN **1510** and the 5GC **1520** are responsible to handle a service request for a slice that may or may not be available in a given area. Admission or rejection of access to a slice may depend on factors such as support for the slice, availability of resources, support of the requested service by NG-RAN **1510**.

The UE **1501** may be associated with multiple network slices simultaneously. In case the UE **1501** is associated with multiple slices simultaneously, only one signaling connection is maintained, and for intra-frequency cell reselection, the UE **1501** tries to camp on the best cell. For inter-frequency cell reselection, dedicated priorities can be used to control the frequency on which the UE **1501** camps. The 5GC **1520** is to validate that the UE **1501** has the rights to access a network slice. Prior to receiving an Initial Context Setup Request message, the NG-RAN **1510** may be allowed to apply some provisional/local policies, based on awareness of a particular slice that the UE **1501** is requesting to access.

During the initial context setup, the NG-RAN **1510** is informed of the slice for which resources are being requested.

NFV architectures and infrastructures may be used to virtualize one or more NFS, alternatively performed by proprietary hardware, onto physical resources comprising a combination of industry-standard server hardware, storage hardware, or switches. In other words, NFV systems can be used to execute virtual or reconfigurable implementations of one or more EPC components/functions.

FIG. **21** is a block diagram illustrating components, according to some example embodiments, of a system **2100** to support NFV. The system **2100** is illustrated as including a VIM **2102**, an NFVI **2104**, an VNFM **2106**, VNFs **2108**, an EM **2110**, an NFVO **2112**, and a NM **2114**.

The VIM **2102** manages the resources of the NFVI **2104**. The NFVI **2104** can include physical or virtual resources and applications (including hypervisors) used to execute the system **2100**. The VIM **2102** may manage the life cycle of virtual resources with the NFVI **2104** (e.g., creation, maintenance, and tear down of VMs associated with one or more physical resources), track VM instances, track performance, fault and security of VM instances and associated physical resources, and expose VM instances and associated physical resources to other management systems.

The VNFM **2106** may manage the VNFs **2108**. The VNFs **2108** may be used to execute EPC components/functions. The VNFM **2106** may manage the life cycle of the VNFs **2108** and track performance, fault and security of the virtual aspects of VNFs **2108**. The EM **2110** may track the performance, fault and security of the functional aspects of VNFs **2108**. The tracking data from the VNFM **2106** and the EM **2110** may comprise, for example, PM data used by the VIM **2102** or the NFVI **2104**. Both the VNFM **2106** and the EM **2110** can scale up/down the quantity of VNFs of the system **2100**.

The NFVO **2112** may coordinate, authorize, release and engage resources of the NFVI **2104** in order to provide the requested service (e.g., to execute an EPC function, component, or slice). The NM **2114** may provide a package of end-user functions with the responsibility for the management of a network, which may include network elements with VNFs, non-virtualized network functions, or both (management of the VNFs may occur via the EM **2110**).

FIG. **22** is a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **22** shows a diagrammatic representation of hardware resources **2200** including one or more processors (or processor cores) **2210**, one or more memory/storage devices **2220**, and one or more communication resources **2230**, each of which may be communicatively coupled via a bus **2240**. For embodiments where node virtualization (e.g., NFV) is utilized, a hypervisor **2202** may be executed to provide an execution environment for one or more network slices/sub-slices to utilize the hardware resources **2200**.

The processors **2210** may include, for example, a processor **2212** and a processor **2214**. The processor(s) **2210** may be, for example, a central processing unit (CPU), a reduced instruction set computing (RISC) processor, a complex instruction set computing (CISC) processor, a graphics processing unit (GPU), a DSP such as a baseband processor, an ASIC, an FPGA, a radio-frequency integrated circuit (RFIC), another processor (including those discussed herein), or any suitable combination thereof.

The memory/storage devices **2220** may include main memory, disk storage, or any suitable combination thereof. The memory/storage devices **2220** may include, but are not limited to, any type of volatile or nonvolatile memory such as dynamic random access memory (DRAM), static random access memory (SRAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), Flash memory, solid-state storage, etc.

The communication resources **2230** may include interconnection or network interface components or other suitable devices to communicate with one or more peripheral devices **2204** or one or more databases **2206** via a network **2208**. For example, the communication resources **2230** may include wired communication components (e.g., for coupling via USB), cellular communication components, NFC components, Bluetooth® (or Bluetooth® Low Energy) components, Wi-Fi® components, and other communication components . . .

Instructions **2250** may comprise software, a program, an application, an applet, an app, or other executable code for causing at least any of the processors **2210** to perform any one or more of the methodologies discussed herein. The instructions **2250** may reside, completely or partially, within at least one of the processors **2210** (e.g., within the processor's cache memory), the memory/storage devices **2220**, or any suitable combination thereof. Furthermore, any portion of the instructions **2250** may be transferred to the hardware resources **2200** from any combination of the peripheral devices **2204** or the databases **2206**. Accordingly, the memory of processors **2210**, the memory/storage devices **2220**, the peripheral devices **2204**, and the databases **2206** are examples of computer-readable and machine-readable media.

For one or more embodiments, at least one of the components set forth in one or more of the preceding figures may be configured to perform one or more operations, techniques, processes, and/or methods as set forth in the example section below. For example, the baseband circuitry as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below. For another example, circuitry associated with a UE, base station, network element, etc. as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below in the example section.

FIG. **23** illustrates a flowchart **2300** that describes an apparatus, such as an originator apparatus, performing dynamic port discovery. The apparatus can be a UE, such as UE **1301**, **1401**, **1700**, according to some embodiments of the disclosure. Additionally, or alternatively, the apparatus can be a base station, radio head, RAN node such as infrastructure equipment **1600**, the RAN nodes **1311** and/or AP **1306**, according to embodiments of the disclosure. In embodiments, the flowchart **2300** can be performed or controlled by a processor or processor circuitry described in the various embodiments herein, including the processor shown in FIG. **22**, the baseband circuitry **1610** shown in FIG. **16**, and/or the baseband circuitry **1710** shown in FIG. **17**.

At **2302**, a command frame is generated. For example, an originator device generates the command frame for dynamic port discovery for communicating with a receiver device using a Reliable Data Service (RDS) protocol. According to some examples, the command frame can include a source port field, a destination port field, and an application identifier field. At **2304**, the command frame is transmitted to the receiver device. For example, the originator device transmits the command frame to the receiver apparatus.

At **2306**, a response frame is received from the receiver device. For example, the originator device receives the response frame from the receiver device. The response frame can include a source port field, a destination port field, an application identifier field, and a status field, according to some examples. The originator device and the receiver device can use the command frame and the response frame to negotiate the parameters for their communications. These parameters can include a source port number and a destination port number for an application.

At **2308**, the source port number and the destination port number are assigned to an application identifier associated with an application on the originator device communicating with a corresponding application on the receiver device.

EXAMPLES

Example 1 includes a 5G system comprising: gNB, AMF, SMF, UPF, NEF, UDM, NSSF, AUSF, AF, AS and other elements as described in 3GPP TS 23.501 and 3GPP TS 24.501.

Example 2 includes the 5G system of example 1 and/or some other examples herein, further comprising a UE, wherein the UE performs Reliable Data Transfer between UE and NEF using the RDS protocol as described in TS 24.250, wherein the communication between UE and NEF is bidirectional and supports both MO/MT data delivery in roaming and non-roaming scenarios.

Example 3 includes the 5G system of example 2 and/or some other examples herein, wherein the communication between UE and NEF can be performed over both 3GPP and non-3GPP accesses.

Example 4 includes the 5G system of example 2 and/or some other examples herein, wherein the UE discovers support for RDS with NEF during the Registration procedure. The UE specifies "RDS Supported" indication over non-3GPP and 3GPP accesses in REGISTRATION_REQUEST message. The network responds with "RDS Supported" or no support indication in REGISTRATION_ACCEPT/REGISTRATION_REJECT message.

Example 5 includes the 5G system of example 4 and/or some other examples herein, wherein RDS packets are transmitted over NAS without the need to establish data radio bearers, for example, via NAS transport message, which can carry RDS payload. UE and Network supports RDS protocol as specified in TS 24.250.

Example 6 includes the 5G system of example 5 and/or some other examples herein, wherein originator uses SET_PARAMETERS command to discover the list of all port numbers and associated application on the receiver by setting the Type parameter to RDS_Query_Port_List. The originator sends a command frame as specified in this disclosure to the receiver. The receiver in turn sends a response frame as specified in this disclosure.

Example 7 includes the 5G system of example 5 and/or some other examples herein, wherein originator uses SET_PARAMETERS command to reserve a Source port number on the receiver by setting the Type parameter to RDS_Reserve Port List. The originator sends a command frame as specified in this disclosure to the receiver. The receiver in turn sends a response frame as specified in this disclosure.

Example 8 includes the 5G system of example 5 and/or some other examples herein, wherein originator uses SET_PARAMETERS command to release a port number on the receiver by setting the Type parameter to RDS_Release_Port_List. The originator sends a command frame as specified in this disclosure to the receiver. The receiver in turn sends a response frame as specified in this disclosure.

Example 9 includes the 5G system of example 5 and/or some other examples herein, wherein originator uses a new command RDS_PORT_MGMT for RDS port management operations. This command is used to discover the list of all port numbers and associated application on the receiver, to reserve a port on receiver and release a port on the receiver by setting the Port Mgmt Command parameter appropriately as specified in this disclosure. The originator sends a command frame as specified in this disclosure to the receiver. The receiver in turn sends a response frame as specified in this disclosure.

Example 10 includes the 5G system of examples 6, 7, 8, 9, and/or some other examples herein, wherein the Application ID is set as combination of OS id and OS App id wherein the OS Id contains a Universally Unique IDentifier (UUID) as specified in IETF RFC 4122 of 16 bytes and OS app id contains an OS specific application identifier of N bytes.

Example 11 includes the 5G system of examples 6, 7, 8, 9, and/or some other examples herein, wherein the Application ID is set as only "OS App Id" (example: URN) of 4 bytes that is more universally recognized by IoT applications without subscription to IP-based APNs.

Example Z01 may include an apparatus comprising means to perform one or more elements of a method described in or related to any of examples 1-11, or any other method or process described herein.

Example Z02 may include one or more non-transitory computer-readable media comprising instructions to cause an electronic device, upon execution of the instructions by one or more processors of the electronic device, to perform one or more elements of a method described in or related to any of examples 1-11, or any other method or process described herein.

Example Z03 may include an apparatus comprising logic, modules, or circuitry to perform one or more elements of a method described in or related to any of examples 1-11, or any other method or process described herein.

Example Z04 may include a method, technique, or process as described in or related to any of examples 1-11, or portions or parts thereof.

Example Z05 may include an apparatus comprising: one or more processors and one or more computer-readable media comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples 1-11, or portions thereof.

Example Z06 may include a signal as described in or related to any of examples 1-11, or portions or parts thereof.

Example Z07 may include a frame, packet, and/or protocol data unit (PDU) as described in or related to any of examples 1-11, or portions or parts thereof.

Example Z08 may include a method of communicating in a wireless network as shown and described herein.

Example Z09 may include a system for providing wireless communication as shown and described herein.

Example Z10 may include a device for providing wireless communication as shown and described herein.

Any of the above-described examples may be combined with any other example (or combination of examples), unless explicitly stated otherwise. The foregoing description of one or more implementations provides illustration and description, but is not intended to be exhaustive or to limit the scope of embodiments to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments.

Abbreviations

For the purposes of the present document, the following abbreviations may apply to the examples and embodiments discussed herein, but are not meant to be limiting.

3GPP Third Generation Partnership Project
4G Fourth Generation
5G Fifth Generation
5GC 5G Core network
ACK Acknowledgement
AF Application Function
AM Acknowledged Mode
AMBR Aggregate Maximum Bit Rate
AMF Access and Mobility Management Function
AN Access Network
ANR Automatic Neighbour Relation
AP Application Protocol, Antenna Port, Access Point
API Application Programming Interface
APN Access Point Name
ARP Allocation and Retention Priority
ARQ Automatic Repeat Request
AS Access Stratum
ASN.1 Abstract Syntax Notation One
AUSF Authentication Server Function
AWGN Additive White Gaussian Noise
BCH Broadcast Channel
BER Bit Error Ratio
BLER Block Error Rate
BPSK Binary Phase Shift Keying
BRAS Broadband Remote Access Server
BSS Business Support System
BS Base Station
BSR Buffer Status Report
BW Bandwidth
BWP Bandwidth Part
C-RNTI Cell Radio Network Temporary Identity
CA Carrier Aggregation, Certification Authority
CAPEX CAPital Expenditure
CBRA Contention Based Random Access
CC Component Carrier, Country Code, Cryptographic Checksum
CCA Clear Channel Assessment
CCE Control Channel Element
CCCH Common Control Channel
CE Coverage Enhancement
CDM Content Delivery Network
CDMA Code-Division Multiple Access
CFRA Contention Free Random Access
CG Cell Group
CI Cell Identity
CID Cell-ID (e.g., positioning method)
CIM Common Information Model
CIR Carrier to Interference Ratio
CK Cipher Key
CM Connection Management, Conditional Mandatory
CMAS Commercial Mobile Alert Service
CMD Command
CMS Cloud Management System
CO Conditional Optional
COMP Coordinated Multi-Point
CORESET Control Resource Set
COTS Commercial Off-The-Shelf
CP Control Plane, Cyclic Prefix, Connection Point
CPD Connection Point Descriptor
CPE Customer Premise Equipment
CPICH Common Pilot Channel
CQI Channel Quality Indicator
CPU CSI processing unit, Central Processing Unit
C/R Command/Response field bit
CRAN Cloud Radio Access Network, Cloud RAN
CRB Common Resource Block
CRC Cyclic Redundancy Check
CRI Channel-State Information Resource Indicator, CSI-RS Resource Indicator
C-RNTI Cell RNTI
CS Circuit Switched
CSAR Cloud Service Archive
CSI Channel-State Information
CSI-IM CSI Interference Measurement
CSI-RS CSI Reference Signal
CSI-RSRP CSI reference signal received power
CSI-RSRQ CSI reference signal received quality
CSI-SINR CSI signal-to-noise and interference ratio
CSMA Carrier Sense Multiple Access
CSMA/CA CSMA with collision avoidance
CSS Common Search Space, Cell-specific Search Space
CTS Clear-to-Send
CW Codeword
CWS Contention Window Size
D2D Device-to-Device
DC Dual Connectivity, Direct Current
DCI Downlink Control Information
DF Deployment Flavour
DL Downlink
DMTF Distributed Management Task Force
DPDK Data Plane Development Kit
DM-RS, DMRS Demodulation Reference Signal
DN Data network
DRB Data Radio Bearer
DRS Discovery Reference Signal
DRX Discontinuous Reception
DSL Domain Specific Language. Digital Subscriber Line
DSLAM DSL Access Multiplexer
DwPTS Downlink Pilot Time Slot
E-LAN Ethernet Local Area Network
E2E End-to-End
ECCA extended clear channel assessment, extended CCA
ECCE Enhanced Control Channel Element, Enhanced CCE
ED Energy Detection
EDGE Enhanced Datarates for GSM Evolution (GSM Evolution)
EGMF Exposure Governance Management Function
EGPRS Enhanced GPRS
EIR Equipment Identity Register
eLAA enhanced Licensed Assisted Access, enhanced LAA
EM Element Manager
eMBB enhanced Mobile Broadband
eMBMS Evolved MBMS
EMS Element Management System
eNB evolved NodeB, E-UTRAN Node B
EN-DC E-UTRA-NR Dual Connectivity
EPC Evolved Packet Core
EPDCCH enhanced PDCCH, enhanced Physical Downlink Control Cannel

EPRE Energy per resource element
EPS Evolved Packet System
EREG enhanced REG, enhanced resource element groups
ETSI European Telecommunications Standards Institute
ETWS Earthquake and Tsunami Warning System
eUICC embedded UICC, embedded Universal Integrated Circuit Card
E-UTRA Evolved UTRA
E-UTRAN Evolved UTRAN
F1AP F1 Application Protocol
F1-C F1 Control plane interface
F1-U F1 User plane interface
FACCH Fast Associated Control CHannel
FACCH/F Fast Associated Control Channel/Full rate
FACCH/H Fast Associated Control Channel/Half rate
FACH Forward Access Channel
FAUSCH Fast Uplink Signalling Channel
FB Functional Block
FBI Feedback Information
FCC Federal Communications Commission
FCCH Frequency Correction CHannel
FDD Frequency Division Duplex
FDM Frequency Division Multiplex
FDMA Frequency Division Multiple Access
FE Front End
FEC Forward Error Correction
FFS For Further Study
FFT Fast Fourier Transformation
feLAA further enhanced Licensed Assisted Access, further enhanced LAA
FN Frame Number
FPGA Field-Programmable Gate Array
FR Frequency Range
GERAN Radio Network Temporary Identity G-RNTI
GERAN GSM EDGE RAN, GSM EDGE Radio Access Network
GGSN Gateway GPRS Support Node
GLONASS GLObal'naya NAvigatsionnaya Sputnikovaya Sistema (Engl.: Global Navigation Satellite System)
gNB Next Generation NodeB
gNB-CU gNB-centralized unit, Next Generation NodeB centralized unit
gNB-DU gNB-distributed unit, Next Generation NodeB distributed unit
GNSS Global Navigation Satellite System
GPRS General Packet Radio Service
GSM Global System for Mobile Communications, Groupe Spécial Mobile
GTP GPRS Tunneling Protocol
GTP-U GPRS Tunnelling Protocol for User Plane
GUMMEI Globally Unique MME Identifier
GUTI Globally Unique Temporary UE Identity
HARQ Hybrid ARQ, Hybrid Automatic Repeat Request
HANDO, HO Handover
HFN HyperFrame Number
HHO Hard Handover
HLR Home Location Register
HN Home Network
HPLMN Home Public Land Mobile Network
HSDPA High Speed Downlink Packet Access
HSN Hopping Sequence Number
HSPA High Speed Packet Access
HSS Home Subscriber Server
HSUPA High Speed Uplink Packet Access
HTTP Hyper Text Transfer Protocol

HTTPS Hyper Text Transfer Protocol Secure (https is http/1.1 over SSL, i.e. port 443)
I-Block Information Block
ICCID Integrated Circuit Card Identification
ICIC Inter-Cell Interference Coordination
ID Identity, identifier
IDFT Inverse Discrete Fourier Transform
IE Information element
IEEE Institute of Electrical and Electronics Engineers
IEI Information Element Identifier
IEIDL Information Element Identifier Data Length
IETF Internet Engineering Task Force
IF Infrastructure
IM Interference Measurement, Intermodulation, IP Multimedia
IMC IMS Credentials
IMEI International Mobile Equipment Identity
IMGI International mobile group identity
IMPI IP Multimedia Private Identity
IMPU
IMS
IP Multimedia PUblic identity
IP Multimedia Subsystem
IMSI International Mobile Subscriber Identity
IoT Internet of Things
IP Internet Protocol
Ipsec IP Security, Internet Protocol Security
IP-CAN IP-Connectivity Access Network
IP-M IP Multicast
IPv4 Internet Protocol Version 4
IPv6 Internet Protocol Version 6
IR Infrared
IRP Integration Reference Point
ISDN Integrated Services Digital Network
ISIM IM Services Identity Module
ISO International Organisation for Standardisation
ISP Internet Service Provider
IWF Interworking-Function
I-WLAN Interworking WLAN
K Constraint length of the convolutional code, USIM Individual key
kB Kilobyte (1000 bytes)
kbps kilo-bits per second
Kc Ciphering key
Ki Individual subscriber authentication key
KPI Key Performance Indicator
KQI Key Quality Indicator
KSI Key Set Identifier
ksps kilo-symbols per second
KVM Kernel Virtual Machine
L1 Layer 1 (physical layer)
L1-RSRP Layer 1 reference signal received power
L2 Layer 2 (data link layer)
L3 Layer 3 (network layer)
LAA Licensed Assisted Access
LAN Local Area Network
LBT Listen Before Talk
LCM LifeCycle Management
LCR Low Chip Rate
LCS Location Services
LI Layer Indicator
LLC Logical Link Control, Low Layer Compatibility
LPLMN Local PLMN
LPP LTE Positioning Protocol
LSB Least Significant Bit
LTE Long Term Evolution
LWA LTE-WLAN aggregation

LWIP LTE/WLAN Radio Level Integration with IPsec Tunnel
LTE Long Term Evolution
M2M Machine-to-Machine
MAC Medium Access Control (protocol layering context)
MAC Message authentication code (security/encryption context)
MAC-A MAC used for authentication and key agreement (TSG T WG3 context)
MAC-I MAC used for data integrity of signalling messages (TSG T WG3 context)
MANO Management and Orchestration
MBMS Multimedia Broadcast and Multicast Service
MBSFN Multimedia Broadcast multicast service Single Frequency Network
MCC Mobile Country Code
MCG Master Cell Group
MCOT Maximum Channel Occupancy Time
MCS Modulation and coding scheme
MDAF Management Data Analytics Function
MDAS Management Data Analytics Service
MDT Minimization of Drive Tests
ME Mobile Equipment
MeNB master eNB
MER Message Error Ratio
MGL Measurement Gap Length
MGRP Measurement Gap Repetition Period
MIB Master Information Block, Management Information Base
MIMO Multiple Input Multiple Output
MLC Mobile Location Centre
MM Mobility Management
MME Mobility Management Entity
MN Master Node
MO Measurement Object, Mobile Originated
MPBCH MTC Physical Broadcast CHannel
MPDCCH MTC Physical Downlink Control CHannel
MPDSCH MTC Physical Downlink Shared CHannel
MPRACH MTC Physical Random Access CHannel
MPUSCH MTC Physical Uplink Shared Channel
MPLS MultiProtocol Label Switching
MS Mobile Station
MSB Most Significant Bit
MSC Mobile Switching Centre
MSI Minimum System Information, MCH Scheduling Information
MSID Mobile Station Identifier
MSIN Mobile Station Identification Number
MSISDN Mobile Subscriber ISDN Number
MT Mobile Terminated, Mobile Termination
MTC Machine-Type Communications
mMTC massive MTC, massive Machine-Type Communications
MU-MIMO Multi User MIMO
MWUS MTC wake-up signal, MTC WUS
NACK Negative Acknowledgement
NAI Network Access Identifier
NAS Non-Access Stratum, Non-Access Stratum layer
NCT Network Connectivity Topology
NEC Network Capability Exposure
NE-DC NR-E-UTRA Dual Connectivity
NEF Network Exposure Function
NF Network Function
NFP Network Forwarding Path
NFPD Network Forwarding Path Descriptor
NFV Network Functions Virtualization
NFVI NFV Infrastructure
NFVO NFV Orchestrator
NG Next Generation, Next Gen
NGEN-DC NG-RAN E-UTRA-NR Dual Connectivity
NM Network Manager
NMS Network Management System
N-POP Network Point of Presence
NMIB, N-MIB Narrowband MIB
NPBCH Narrowband Physical Broadcast CHannel
NPDCCH Narrowband Physical Downlink Control CHannel
NPDSCH Narrowband Physical Downlink Shared CHannel
NPRACH Narrowband Physical Random Access CHannel
NPUSCH Narrowband Physical Uplink Shared CHannel
NPSS Narrowband Primary Synchronization Signal
NSSS Narrowband Secondary Synchronization Signal
NR New Radio, Neighbour Relation
NRF NF Repository Function
NRS Narrowband Reference Signal
NS Network Service
NSA Non-Standalone operation mode
NSD Network Service Descriptor
NSR Network Service Record
NSSAI ‘Network Slice Selection Assistance Information
S-NNSAI Single-NSSAI
NSSF Network Slice Selection Function
NW Network
NWUS Narrowband wake-up signal, Narrowband WUS
NZP Non-Zero Power
O&M Operation and Maintenance
ODU2 Optical channel Data Unit—type 2
OFDM Orthogonal Frequency Division Multiplexing
OFDMA Orthogonal Frequency Division Multiple Access
OOB Out-of-band
OPEX OPerating EXpense
OSI Other System Information
OSS Operations Support System
OTA over-the-air
PAPR Peak-to-Average Power Ratio
PAR Peak to Average Ratio
PBCH Physical Broadcast Channel
PC Power Control, Personal Computer
PCC Primary Component Carrier, Primary CC
PCell Primary Cell
PCI Physical Cell ID, Physical Cell Identity
PCEF Policy and Charging Enforcement Function
PCF Policy Control Function
PCRF Policy Control and Charging Rules Function
PDCP Packet Data Convergence Protocol, Packet Data Convergence Protocol layer
PDCCH Physical Downlink Control Channel
PDCP Packet Data Convergence Protocol
PDN Packet Data Network, Public Data Network
PDSCH Physical Downlink Shared Channel
PDU Protocol Data Unit
PEI Permanent Equipment Identifiers
PFD Packet Flow Description
P-GW PDN Gateway
PHICH Physical hybrid-ARQ indicator channel
PHY Physical layer
PLMN Public Land Mobile Network
PIN Personal Identification Number
PM Performance Measurement
PMI Precoding Matrix Indicator
PNF Physical Network Function
PNFD Physical Network Function Descriptor

PNFR Physical Network Function Record
POC PTT over Cellular
PP, PTP Point-to-Point
PPP Point-to-Point Protocol
PRACH Physical RACH
PRB Physical resource block
PRG Physical resource block group
ProSe Proximity Services, Proximity-Based Service
PRS Positioning Reference Signal
PS Packet Services
PSBCH Physical Sidelink Broadcast Channel
PSDCH Physical Sidelink Downlink Channel
PSCCH Physical Sidelink Control Channel
PSSCH Physical Sidelink Shared Channel
PSCell Primary SCell
PSS Primary Synchronization Signal
PSTN Public Switched Telephone Network
PT-RS Phase-tracking reference signal
PTT Push-to-Talk
PUCCH Physical Uplink Control Channel
PUSCH Physical Uplink Shared Channel
QAM Quadrature Amplitude Modulation
QCI QoS class of identifier
QCL Quasi co-location
QFI QoS Flow ID, QoS Flow Identifier
QoS Quality of Service
QPSK Quadrature (Quaternary) Phase Shift Keying
QZSS Quasi-Zenith Satellite System
RA-RNTI Random Access RNTI
RAB Radio Access Bearer, Random Access Burst
RACH Random Access Channel
RADIUS Remote Authentication Dial In User Service
RAN Radio Access Network
RAND RANDom number (used for authentication)
RAR Random Access Response
RAT Radio Access Technology
RAU Routing Area Update
RB Resource block, Radio Bearer
RBG Resource block group
REG Resource Element Group
Re Release
REQ REQuest
RF Radio Frequency
RI Rank Indicator
RIV Resource indicator value
RL Radio Link
RLC Radio Link Control, Radio Link Control layer
RLF Radio Link Failure
RLM Radio Link Monitoring
RLM-RS Reference Signal for RLM
RM Registration Management
RMC Reference Measurement Channel
RMSI Remaining MSI, Remaining Minimum System Information
RN Relay Node
RNC Radio Network Controller
RNL Radio Network Layer
RNTI Radio Network Temporary Identifier
ROHC RObust Header Compression
RRC Radio Resource Control, Radio Resource Control layer
RRM Radio Resource Management
RS Reference Signal
RSRP Reference Signal Received Power
RSRQ Reference Signal Received Quality
RSSI Received Signal Strength Indicator
RSU Road Side Unit
RSTD Reference Signal Time difference
RTP Real Time Protocol
RTS Ready-To-Send
RTT Round Trip Time
Rx Reception, Receiving, Receiver
S1AP S1 Application Protocol
S1-MME S1 for the control plane
S1-U S1 for the user plane
S-GW Serving Gateway
S-RNTI SRNC Radio Network Temporary Identity
S-TMSI SAE Temporary Mobile Station Identifier
SA Standalone operation mode
SAE System Architecture Evolution
SAP Service Access Point
SAPD Service Access Point Descriptor
SAPI Service Access Point Identifier
SCC Secondary Component Carrier, Secondary CC
SCEF Service Capability Exposure Function
SCell Secondary Cell
SC-FDMA Single Carrier Frequency Division Multiple Access
SCG Secondary Cell Group
SCM Security Context Management
SCS Subcarrier Spacing
SCTP Stream Control Transmission Protocol
SDAP Service Data Adaptation Protocol, Service Data Adaptation Protocol layer
SDL Supplementary Downlink
SDNF Structured Data Storage Network Function
SDP Service Discovery Protocol (Bluetooth related)
SDSF Structured Data Storage Function
SDU Service Data Unit
SEAF Security Anchor Function
SeNB secondary eNB
SEPP Security Edge Protection Proxy
SFI Slot format indication
SFTD Space-Frequency Time Diversity, SFN and frame timing difference
SFN System Frame Number
SgNB Secondary gNB
SGSN Serving GPRS Support Node
S-GW Serving Gateway
SI System Information
SI-RNTI System Information RNTI
SIB System Information Block
SIM Subscriber Identity Module
SIP Session Initiated Protocol
SiP System in Package
SL Sidelink
SLA Service Level Agreement
SM Session Management
SMF Session Management Function
SMS Short Message Service
SMSF SMS Function
SMTC SSB-based Measurement Timing Configuration
SN Secondary Node, Sequence Number
SoC System on Chip
SON Self-Organizing Network
SpCell Special Cell
SP-CSI-RNTI Semi-Persistent CSI RNTI
SPS Semi-Persistent Scheduling
SQN Sequence number
SR Scheduling Request
SRB Signalling Radio Bearer
SRS Sounding Reference Signal
SS Synchronization Signal
SSB Synchronization Signal Block, SS/PBCH Block

SSBRI SS/PBCH Block Resource Indicator, Synchronization Signal Block Resource Indicator
SSC Session and Service Continuity
SS-RSRP Synchronization Signal based Reference Signal Received Power
SS-RSRQ Synchronization Signal based Reference Signal Received Quality
SS-SINR Synchronization Signal based Signal to Noise and Interference Ratio
SSS Secondary Synchronization Signal
SST Slice/Service Types
SU-MIMO Single User MIMO
SUL Supplementary Uplink
TA Timing Advance, Tracking Area
TAC Tracking Area Code
TAG Timing Advance Group
TAU Tracking Area Update
TB Transport Block
TBS Transport Block Size
TBD To Be Defined
TCI Transmission Configuration Indicator
TCP Transmission Communication Protocol
TDD Time Division Duplex
TDM Time Division Multiplexing
TDMA Time Division Multiple Access
TE Terminal Equipment
TEID Tunnel End Point Identifier
TFT Traffic Flow Template
TMSI Temporary Mobile Subscriber Identity
TNL Transport Network Layer
TPC Transmit Power Control
TPMI Transmitted Precoding Matrix Indicator
TR Technical Report
TRP, TRxP Transmission Reception Point
TRS Tracking Reference Signal
TRx Transceiver
TS Technical Specifications, Technical Standard
TTI Transmission Time Interval
Tx Transmission, Transmitting, Transmitter
U-RNTI UTRAN Radio Network Temporary Identity
UART Universal Asynchronous Receiver and Transmitter
UCI Uplink Control Information
UE User Equipment
UDM Unified Data Management
UDP User Datagram Protocol
UDSF Unstructured Data Storage Network Function
UICC Universal Integrated Circuit Card
UL Uplink
UM Unacknowledged Mode
UML Unified Modelling Language
UMTS Universal Mobile Telecommunications System
UP User Plane
UPF User Plane Function
URI Uniform Resource Identifier
URL Uniform Resource Locator
URLLC Ultra-Reliable and Low Latency
USB Universal Serial Bus
USIM Universal Subscriber Identity Module
USS UE-specific search space
UTRA UMTS Terrestrial Radio Access
UTRAN Universal Terrestrial Radio Access Network
UwPTS Uplink Pilot Time Slot
V2I Vehicle-to-Infrastruction
V2P Vehicle-to-Pedestrian
V2V Vehicle-to-Vehicle
V2X Vehicle-to-everything
VIM Virtualized Infrastructure Manager
VL Virtual Link,
VLAN Virtual LAN, Virtual Local Area Network
VM Virtual Machine
VNF Virtualized Network Function
VNFFG VNF Forwarding Graph
VNFFGD VNF Forwarding Graph Descriptor
VNFM VNF Manager
VOIP Voice-over-IP, Voice-over-Internet Protocol
VPLMN Visited Public Land Mobile Network
VPN Virtual Private Network
VRB Virtual Resource Block
WiMAX Worldwide Interoperability for Microwave Access
WLAN Wireless Local Area Network
WMAN Wireless Metropolitan Area Network
WPAN Wireless Personal Area Network
X2-C X2-Control plane
X2-U X2-User plane
XML extensible Markup Language
XRES Expected user RESponse
XOR exclusive OR
ZC Zadoff-Chu
ZP Zero Power

Terminology

For the purposes of the present document, the following terms and definitions are applicable to the examples and embodiments discussed herein, but are not meant to be limiting.

The term "circuitry" as used herein refers to, is part of, or includes hardware components such as an electronic circuit, a logic circuit, a processor (shared, dedicated, or group) and/or memory (shared, dedicated, or group), an Application Specific Integrated Circuit (ASIC), a field-programmable device (FPD) (e.g., a field-programmable gate array (FPGA), a programmable logic device (PLD), a complex PLD (CPLD), a high-capacity PLD (HCPLD), a structured ASIC, or a programmable SoC), digital signal processors (DSPs), etc., that are configured to provide the described functionality. In some embodiments, the circuitry may execute one or more software or firmware programs to provide at least some of the described functionality. The term "circuitry" may also refer to a combination of one or more hardware elements (or a combination of circuits used in an electrical or electronic system) with the program code used to carry out the functionality of that program code. In these embodiments, the combination of hardware elements and program code may be referred to as a particular type of circuitry.

The term "processor circuitry" as used herein refers to, is part of, or includes circuitry capable of sequentially and automatically carrying out a sequence of arithmetic or logical operations, or recording, storing, and/or transferring digital data. The term "processor circuitry" may refer to one or more application processors, one or more baseband processors, a physical central processing unit (CPU), a single-core processor, a dual-core processor, a triple-core processor, a quad-core processor, and/or any other device capable of executing or otherwise operating computer-executable instructions, such as program code, software modules, and/or functional processes. The terms "application circuitry" and/or "baseband circuitry" may be considered synonymous to, and may be referred to as, "processor circuitry."

The term "interface circuitry" as used herein refers to, is part of, or includes circuitry that enables the exchange of information between two or more components or devices. The term "interface circuitry" may refer to one or more hardware interfaces, for example, buses, I/O interfaces, peripheral component interfaces, network interface cards, and/or the like.

The term "user equipment" or "UE" as used herein refers to a device with radio communication capabilities and may describe a remote user of network resources in a communications network. The term "user equipment" or "UE" may be considered synonymous to, and may be referred to as, client, mobile, mobile device, mobile terminal, user terminal, mobile unit, mobile station, mobile user, subscriber, user, remote station, access agent, user agent, receiver, radio equipment, reconfigurable radio equipment, reconfigurable mobile device, etc. Furthermore, the term "user equipment" or "UE" may include any type of wireless/wired device or any computing device including a wireless communications interface.

The term "network element" as used herein refers to physical or virtualized equipment and/or infrastructure used to provide wired or wireless communication network services. The term "network element" may be considered synonymous to and/or referred to as a networked computer, networking hardware, network equipment, network node, router, switch, hub, bridge, radio network controller, RAN device, RAN node, gateway, server, virtualized VNF, NFVI, and/or the like.

The term "computer system" as used herein refers to any type interconnected electronic devices, computer devices, or components thereof. Additionally, the term "computer system" and/or "system" may refer to various components of a computer that are communicatively coupled with one another. Furthermore, the term "computer system" and/or "system" may refer to multiple computer devices and/or multiple computing systems that are communicatively coupled with one another and configured to share computing and/or networking resources.

The term "appliance," "computer appliance," or the like, as used herein refers to a computer device or computer system with program code (e.g., software or firmware) that is specifically designed to provide a specific computing resource. A "virtual appliance" is a virtual machine image to be implemented by a hypervisor-equipped device that virtualizes or emulates a computer appliance or otherwise is dedicated to provide a specific computing resource.

The term "resource" as used herein refers to a physical or virtual device, a physical or virtual component within a computing environment, and/or a physical or virtual component within a particular device, such as computer devices, mechanical devices, memory space, processor/CPU time, processor/CPU usage, processor and accelerator loads, hardware time or usage, electrical power, input/output operations, ports or network sockets, channel/link allocation, throughput, memory usage, storage, network, database and applications, workload units, and/or the like. A "hardware resource" may refer to compute, storage, and/or network resources provided by physical hardware element(s). A "virtualized resource" may refer to compute, storage, and/or network resources provided by virtualization infrastructure to an application, device, system, etc. The term "network resource" or "communication resource" may refer to resources that are accessible by computer devices/systems via a communications network. The term "system resources" may refer to any kind of shared entities to provide services, and may include computing and/or network resources. System resources may be considered as a set of coherent functions, network data objects or services, accessible through a server where such system resources reside on a single host or multiple hosts and are clearly identifiable.

The term "channel" as used herein refers to any transmission medium, either tangible or intangible, which is used to communicate data or a data stream. The term "channel" may be synonymous with and/or equivalent to "communications channel," "data communications channel," "transmission channel," "data transmission channel," "access channel," "data access channel," "link," "data link," "carrier," "radiofrequency carrier," and/or any other like term denoting a pathway or medium through which data is communicated. Additionally, the term "link" as used herein refers to a connection between two devices through a RAT for the purpose of transmitting and receiving information.

The terms "instantiate," "instantiation," and the like as used herein refers to the creation of an instance. An "instance" also refers to a concrete occurrence of an object, which may occur, for example, during execution of program code.

The terms "coupled," "communicatively coupled," along with derivatives thereof are used herein. The term "coupled" may mean two or more elements are in direct physical or electrical contact with one another, may mean that two or more elements indirectly contact each other but still cooperate or interact with each other, and/or may mean that one or more other elements are coupled or connected between the elements that are said to be coupled with each other. The term "directly coupled" may mean that two or more elements are in direct contact with one another. The term "communicatively coupled" may mean that two or more elements may be in contact with one another by a means of communication including through a wire or other interconnect connection, through a wireless communication channel or ink, and/or the like.

The term "information element" refers to a structural element containing one or more fields. The term "field" refers to individual contents of an information element, or a data element that contains content.

The term "SMTC" refers to an SSB-based measurement timing configuration configured by SSB-MeasurementTimingConfiguration.

The term "SSB" refers to an SS/PBCH block.

The term "a "Primary Cell" refers to the MCG cell, operating on the primary frequency, in which the UE either performs the initial connection establishment procedure or initiates the connection re-establishment procedure.

The term "Primary SCG Cell" refers to the SCG cell in which the UE performs random access when performing the Reconfiguration with Sync procedure for DC operation.

The term "Secondary Cell" refers to a cell providing additional radio resources on top of a Special Cell for a UE configured with CA.

The term "Secondary Cell Group" refers to the subset of serving cells comprising the PSCell and zero or more secondary cells for a UE configured with DC.

The term "Serving Cell" refers to the primary cell for a UE in RRC_CONNECTED not configured with CA/DC there is only one serving cell comprising of the primary cell.

The term "serving cell" or "serving cells" refers to the set of cells comprising the Special Cell(s) and all secondary cells for a UE in RRC_CONNECTED configured with CA/.

The term "Special Cell" refers to the PCell of the MCG or the PSCell of the SCG for DC operation, otherwise, the term "Special Cell" refers to the Pcell.

What is claimed is:

1. An originator device, comprising:

radio front end circuitry; and processor circuitry configured to:

generate a first command frame for dynamic port discovery for communicating with a receiver device using a Reliable Data Service (RDS) protocol, wherein the first command frame comprises a query for assigned destination ports on the receiver device and available destination ports on the receiver device;

transmit, using the radio front end circuitry, the first command frame to the receiver device;

receive, using the radio front end circuitry, a first response frame with a bitmap indicating which destination ports on the receiver device are the assigned destination ports and which of the destination ports on the receiver device are the available destination ports;

assign, based on the bitmap of the first response frame, a source port number and a destination port number to an application identifier, wherein the application identifier is associated with an application on the originator device communicating with a corresponding application on the receiver device;

generate a second command frame using the RDS protocol, wherein the second command frame comprises three separate fields comprising a first source port field for the source port number, a first destination port field for the destination port number, and a first application identifier field for the application identifier;

transmit, using the radio front end circuitry, the second command frame to the receiver device; and

receive, using the radio front end circuitry, a second response frame from the receiver device, wherein the second response frame comprises four separate fields comprising a second source port field for the source port number, a second destination port field for the destination port number, a second application identifier field for the application identifier, and a status field indicating a status of a response to a command associated with the second command frame,

wherein the status field indicates whether the receiver device was successful in assigning the source port number and the destination port number to the application identifier.

2. The originator device of claim 1, wherein the processor circuitry is further configured to transmit, using the radio front end circuitry and to the receiver device, a source bitmap associated with source ports on the originator device, wherein the source bitmap indicates which of the source ports are assigned and which of the source ports are available.

3. The originator device of claim 1, wherein to transmit the second command frame, the processor circuitry is configured to transmit, using the radio front end circuitry and to the receiver device, the source port number, the destination port number, and the application identifier to release the source port number and the destination port number from the application associated with the application identifier.

4. The originator device of claim 1, wherein the second command frame further comprises a port management command comprising a port list query, a reserve port command, or a release port command.

5. The originator device of claim 1, wherein the originator device comprises a user equipment (UE) and the receiver device comprises a public data network (PDN) gateway (P-GW).

6. The originator device of claim 1, wherein the originator device comprises a public data network (PDN) gateway (P-GW) and the receiver device comprises a user equipment (UE).

7. The originator device of claim 1, wherein the processor circuitry is further configured to use the second command frame and the second response frame for communicating with the receiver device.

8. A method comprising:

generating a first command frame for dynamic port discovery for communicating with a receiver device using a Reliable Data Service (RDS) protocol, wherein the first command frame comprises a query for assigned destination ports on the receiver device and available destination ports on the receiver device;

transmitting the first command frame to the receiver device;

receiving a first response frame with a bitmap indicating which destination ports on the receiver device are the assigned destination ports and which of the destination ports on the receiver device are the available destination ports;

assigning, based on the bitmap if the first response frame, at an originator device, a source port number and a destination port number to an application identifier, wherein the application identifier is associated with an application on the originator device communicating with a corresponding application on the receiver device;

generating, at the originator device, a second command frame using the RDS protocol, wherein the second command frame comprises three separate fields comprising a first source port field for the source port number, a first destination port field for the destination port number, and a first application identifier field for the application identifier;

transmitting the second command frame to the receiver device; and

receiving a second response frame from the receiver device, wherein the second response frame comprises four separate fields comprising a second source port field for the source port number, a second destination port field for the destination port number, a second application identifier field for the application identifier, and a status field indicating a status of a response to a command associated with the second command frame,

wherein the status field indicates whether the receiver device was successful in assigning the source port number and the destination port number to the application identifier.

9. The method of claim 8, further comprising:

transmitting, to the receiver device, a source bitmap associated with source ports on the originator device, wherein the source bitmap indicates which of the source ports are assigned and which of the source ports are available.

10. The method of claim 8, wherein transmitting the second command frame comprises:

transmitting, to the receiver device, the source port number, the destination port number, and the application identifier to release the source port number and the destination port number from the application associated with the application identifier.

11. The method of claim 8, further comprising:

using the second command frame and the second response frame for communicating with the receiver device.

12. An originator device, comprising:

a memory that stores instructions; and

a processor, upon executing the instructions, configured to:

generate a first command frame for dynamic port discovery for communicating with a receiver device using a Reliable Data Service (RDS) protocol, wherein the first command frame comprises a query for assigned destination ports on the receiver device and available destination ports on the receiver device;

transmit the first command frame to the receiver device;

receive a first response frame with a bitmap indicating which destination ports on the receiver device are the assigned destination ports and which of the destination ports on the receiver device are the available destination ports;

assign, based on the bitmap of the first response frame, a source port number and a destination port number to an application identifier, wherein the application identifier is associated with an application on the originator device communicating with a corresponding application on the receiver device;

generate a second command frame using the RDS protocol, wherein the second command frame comprises three separate fields comprising a first source port field for the source port number, a first destination port field for the destination port number, and a first application identifier field for the application identifier;

transmit the second command frame to the receiver device; and

receive a second response frame from the receiver device, wherein the second response frame comprises four separate fields comprising a second source port field for the source port number, a second destination port field for the destination port number, a second application identifier field for the application identifier, and a status field indicating a status of a response to a command associated with the second command frame,

wherein the status field indicates whether the receiver device was successful in assigning the source port number and the destination port number to the application identifier.

13. The originator device of claim 12, wherein the processor is further configured to transmit, to the receiver device, a source bitmap associated with source ports on the originator device, wherein the source bitmap indicates which of the source ports are assigned and which of the source ports are available.

14. The originator device of claim 12, wherein to transmit the second command frame, the processor is configured to transmit, to the receiver device, the source port number, the destination port number, and the application identifier to release the source port number and the destination port number from the application associated with the application identifier.

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